

Agentic OCR in 2026

An opinionated practitioner's field guide

Anuj Sadani

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Preface

i In a hurry?

This is a practitioner's field guide to **agentic OCR in 2026** — covering what it is, where it came from, the vendor landscape (open-source and paid), how to evaluate it, how it interacts with RAG, and where it still falls short. The book is structured for two audiences read in parallel: each chapter opens with an executive summary, then dives into the technical body.

Who this book is for

Two readers, sharing each chapter:

- **Technical implementers** — ML engineers, IDP architects, applied scientists building document-processing pipelines. You want architecture diagrams, eval rubrics, named tradeoffs, and a working mental model of how agentic loops actually behave.
- **Technical decision-makers** — CTOs, heads of AI, buyers evaluating vendors. You want the shift-left thesis explained without hand-waving, ROI math you can stress-test, and a vendor-selection framework that doesn't read like a Gartner brochure.

If you can't tell which one you are, read the executive callouts in every chapter; if any of them makes you want more depth, the technical body is right below it.

How to read this book

The book is roughly linear, but skippable:

- **Read straight through** if agentic OCR is new to you. Parts I and II build the vocabulary used everywhere else.
- **Skip to Part III** if you already know what agentic OCR is and want the 2026 vendor landscape.
- **Skip to Part IV** if your team has already chosen a vendor and you need to evaluate output and wire it into RAG.
- **Read Part V first** if you suspect you're being sold something. The anti-patterns chapter is where most of the field's pain is.

What this book is not

- Not a tutorial. There's no code-along section that has you implementing your own agent. You don't need one — the field has plenty.
- Not vendor-neutral. The book has opinions. They are labeled.
- Not stable forever. Quarter by quarter the leaderboard moves, pricing changes, and new entrants show up. The frameworks should age better than the player names.

A note on sources

This is a synthesis of original research, public benchmarks, vendor documentation as of mid-2026, and field experience. Citations are inline; the full bibliography is in the appendix.

I

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1. Introduction: The Agentic Turn

i Executive summary

- The 2026 bottleneck on agent quality is not reasoning. It is reading. Frontier agents that look brilliant in benchmark sandboxes routinely fail in production on enterprise documents, and most of that failure is happening at the parsing layer, not the reasoning layer.
- “Agentic OCR” is the load-bearing term of the year, and most of what is sold under that label does not qualify. This book gives you a five-attribute test to call it.
- 67% of enterprise document-processing initiatives are now actively evaluating agentic approaches — up from 23% two years ago. The vocabulary, vendor maturity, and benchmarks have crossed a usability threshold simultaneously.
- The book is layered: executive callouts at the top of each chapter, technical body underneath. Read both, or read either, depending on what you need this week.

1.1 What changed

A frontier agent reads a treasury bond document. The face value is \$10,000. The agent’s output says \$3,000. The agent’s reasoning layer — a model that can write a five-paragraph argument about credit risk and make tea — accepts the \$3,000 without complaint and proceeds to plan a downstream action on the wrong number.

This is a true 2026 failure mode, lifted almost verbatim from a Databricks blog [1], and it is the cleanest single illustration of what changed in this field over the last eighteen months. The agent did not get the reasoning wrong. It got the *reading* wrong. The reasoning was excellent reasoning on the wrong inputs. By the time the planner saw the data, the data was already poisoned.

This pattern repeats across every domain where enterprise agents touch documents. An insurance prior-authorization agent denies coverage because it misread a date on a referral letter. A financial-analysis agent produces a flawless quarterly summary based on numbers that were merged from two adjacent table cells the parser couldn’t disentangle. A legal-discovery agent flags the wrong clause for review because a footnote referenced “Section 4.2” and the parser dropped the footnote. In every case the reasoning layer is doing its job; the document layer is not.

The thesis of this book, stated in one sentence: **the ceiling on agent quality in 2026 is set by document understanding, not by reasoning.** Or, in the version that fits on a slide: *reading is the ceiling, not reasoning.*

The category of system that addresses this — that does multi-pass, self-correcting, tool-using, grounded document parsing in service of a structured downstream goal — is what the field has settled on calling “agentic OCR.” The term is doing a lot of work. Vendors who do nothing of the sort are using it freely. Chapter 3 will give you a five-attribute test to separate the real systems from the relabeled wrappers; the rest of the book will give you the architecture, the players, the benchmarks, the evaluation rubrics, and the integration patterns to act on the distinction.

1.2 Why 2026 matters

Three things happened in roughly the same eighteen-month window that turned agentic OCR from a research curiosity into a category enterprises buy.

Adoption crossed a threshold. As of mid-2026, **67% of enterprise document-processing initiatives are explicitly evaluating agentic approaches over traditional OCR-plus-rules stacks** [2]. Two years ago that number was 23%. This is the kind of jump that goes from “interesting niche pilot” to “you should probably know what this is when it shows up in a procurement deck.”

Regulation started forcing the issue. Beginning **March 31, 2026**, CMS requires U.S. health plans to publicly report their average turnaround times for prior authorizations, plus denial, appeal, and overturn

rates [3]. That single rule turned manual prior-auth workflows from a tolerable cost center into a board-level risk. The math now favors any system that can credibly cut review time without inflating denial rates. The healthcare vertical is moving first because regulation moved first; legal and finance are following.

Benchmarks finally caught up to the work. ParseBench, the first 2026 benchmark designed specifically for AI agents reading documents [4], publishes a public leaderboard with ~2,000 human-verified pages and 167,000 test rules. olmOCR-Bench produces reproducible unit-test-style scores for open-source models [5]. RD-TableBench gives you a thousand adversarial tables to humble vendors who claim table extraction is solved [6]. DABstep, the only benchmark that tests multi-step reasoning over real-world data and document packets, reports a top score of around **14.55% on hard tasks** — a number low enough to function as a permanent reality check on anyone selling magic [7].

Each of these on its own would be modest. Together they explain why the conversation has changed. The technology is no longer prospective. The vendors are no longer all startups. The benchmarks are no longer all gameable. The regulation is no longer hypothetical. Pick any quarter from 2024 and the same conversation about agentic OCR would have ended in “interesting, let’s revisit next year.” In 2026 it ends with a budget line item.

The flip side of that maturity is that the field is also more crowded, more confused, and more mis-marketed than at any previous point. “Agentic” is the word that lets people charge ten times more for a retry loop. The five-attribute test in Chapter 3 exists precisely so that you do not get fooled.

⚠ When this book may not be for you

Before you read further: agentic OCR is not for every workload. The most expensive mistake in this space is deploying it for the sake of it, on a use case that cannot pay for it. The 2026 numbers make the threshold concrete.

Realistic all-in cost-per-correct-extraction for **true agentic OCR** in 2026:

- Parsing API spend: \$0.005 – \$0.056 per page (Reducto, LlamaParse Agentic, LandingAI ADE).
- HITL amortized over a typical 5% escalation rate: \$0.05 – \$0.25 per page.
- Engineering, eval infrastructure, vendor management: real but case-dependent.

Total: roughly \$0.05 – \$0.30 per correctly-extracted page.

If your downstream business value per page is around **one cent**, premium agentic OCR’s parsing cost alone is 1.2× to 6× your entire per-page budget — before any engineering or HITL costs. The category structurally cannot pay for itself at that price point. The cheapest credible parser (Mistral OCR 3 Batch at \$0.001/page) still consumes about 10% of a one-cent ROI on parsing alone, leaving nothing for the surrounding work. Self-hosted open-source (OlmOCR-2 at ~\$0.00018/page on an H100) is the only configuration whose *parsing* cost fits — but those deployments still need an engineering team that the per-page ROI has to fund.

You probably should **not** build agentic OCR if:

- Your downstream per-page value is below roughly 5 cents (and certainly below 1 cent).
- Your documents are clean, native PDFs or HTML where parsing is not actually the bottleneck. Diagnose the chain (Chapter 12) before you upgrade.
- Your workload is narrow and stable — one or two vendor templates that change rarely. Classic template-driven IDP still wins on TCO here.
- Your stakes are low and your volume is enormous (10M+ pages/month of commodity invoices). Hyperscaler basic OCR plus a thin cleanup layer can be 50× cheaper for the same downstream outcome.
- Your real bottleneck is reasoning, retrieval, or workflow design, not document understanding. Agentic OCR will not fix those.

This book is most useful for readers whose answer to all five is “doesn’t apply” or “I’m not sure” — the second case is exactly what Section 9.5 will help you resolve. If the answer to one or more is a confident “yes,” save the budget and revisit when the numbers change.

1.3 Who this book is for

There are two readers, and the book is structured so they read it together.

The first reader is a **technical implementer** — an ML engineer, an applied scientist, an IDP architect. You want architecture diagrams that actually decompose into deployable components. You want named

tradeoffs between single-pass VLMs, hybrid CV-plus-VLM pipelines, and reflection-loop agents. You want evaluation rubrics you can paste into a doc and start running on Monday. You want to know which open-source model to start with for a self-hosted deployment, and what a calibrated confidence score actually looks like.

The second reader is a **technical decision-maker** — a CTO, a head of AI, a VP of engineering choosing among vendor pitches. You want the shift-left thesis explained without hand-waving and without becoming a slide deck. You want a vendor-selection framework that does not read like a Gartner brochure. You want to know what 99.24% accuracy on healthcare prior-auth actually represents, what it cost the buyer who got there, and whether the ROI replicates. You want to know what you can and cannot delegate to your team without sitting in on the weekly review.

These two readers are not the same person. They are, however, frequently in the same room making the same decision from different perspectives. The book is organized so they can read it together: each chapter opens with a callout-styled executive summary aimed at the second reader, then proceeds into the technical body aimed at the first reader. If you are the executive, the callout is yours; if a callout makes you want more detail, the body is right below it. If you are the implementer, the callouts are still worth a glance because they tell you what your management thinks the chapter is going to say.

A third reader exists in smaller numbers: the **skeptic** who suspects the whole category is overhyped. The book is friendly to that reader, too. Chapters 13 and 14, on anti-patterns and unsolved problems, were written with the skeptic specifically in mind. If your prior is that agentic OCR is the next round of NLP hype, start there; if the case for the skeptical position survives those chapters, your prior is well-founded.

1.4 How to read this book

The book has five parts and fifteen chapters. It is roughly linear but designed for skipping.

Part I (Chapters 1–3) — Foundations. What agentic OCR is, where it came from, why the definition matters.

Part II (Chapters 4–6) — Architecture. The three reference architectures, how self-correction actually works, and the shift-left thesis.

Part III (Chapters 7–9) — The 2026 landscape. Commercial vendors, open-source players, vendor selection.

Part IV (Chapters 10–12) — Evaluation and integration. The benchmark landscape, how to evaluate your own pipeline, and the (often-misunderstood) relationship between agentic OCR and retrieval-augmented generation.

Part V (Chapters 13–15) — Reality check. Anti-patterns, what the technology does not solve, and where 2027 is heading.

Read straight through if the field is new to you; Parts I and II build the vocabulary that everything else depends on. Skip to Part III if you already know what agentic OCR is and you want the vendor landscape. Skip to Part IV if you have already chosen a vendor and need to wire it into your stack. Read Part V first if you suspect you are being sold something — that part is calibrated to puncture vendor narratives, and if the case for agentic OCR survives that puncturing, the rest of the book is the field guide for acting on it.

1.5 What this book is not

A few honest disclaimers.

This is not a tutorial. There is no code-along chapter that walks you through implementing your own agent. There is plenty of code in the open-source ecosystem already; the gap this book fills is not “another tutorial.”

This is not vendor-neutral. The book has opinions. They are labeled — every chapter ends with at least one explicitly tagged **Named take** that you can quote, push back on, or steal. The opinions are derived from public benchmarks, vendor documentation, primary research, and field experience, all cited in the bibliography. You should disagree with some of them. The labels are there precisely so the disagreement is productive.

This is not stable forever. Mid-2026 is a snapshot. Quarter by quarter the leaderboard moves, pricing changes, models get released, vendors get acquired. The frameworks in this book — the five-attribute test, the three reference architectures, the vendor-selection decision tree, the anti-pattern checklist — should age better than the vendor names. When the names change, look up the new names; the categories should still hold.

1.6 A note on sources

This book is a synthesis of three things: original research collected during 2025–2026, public benchmarks and vendor documentation through May 2026, and field experience working on agentic OCR pipelines across healthcare, financial services, and enterprise SaaS. Where vendors publish numbers, those numbers are cited with the appropriate “vendor-published” caveat. Where benchmarks publish leaderboards, the dates are noted, because OCR benchmark leaderboards have an unusually short half-life. Where claims come from primary research, the paper is cited.

The bibliography lives in Appendix B. The glossary lives in Appendix A. There is a one-page cheatsheet in Appendix C that you can print and tape to a wall; it is the book in one page if you ever need to remember the framework at speed.

Named take

If your agent can read perfectly but reason imperfectly, you have a model problem. If your agent reasons brilliantly on the wrong inputs, you have a document problem — and 2026 says the document problem is bigger.

2. A Short History of Document Processing

i Executive summary

- Document processing has gone through four distinct eras: classic OCR (1990s–2010s), template-driven IDP (2010s), single-pass VLMs (2022–2024), and agentic loops (2024–present). Each generation eliminated the previous generation’s most expensive engineering — and replaced it with a new expensive engineering category.
- The reason agentic OCR works *now* is not that the underlying models suddenly got better. The models have been steadily improving for years. The change is that the *surrounding scaffolding* — tool use, schema validation, grounding, HITL queues, evaluation infrastructure — finally got good enough to wrap around the models in production.
- Anyone selling you the story that agentic OCR makes the engineering disappear is selling you a screenshot of the demo. The engineering moves; it does not vanish.

2.1 The classic OCR era (1990s–2010s)

Tesseract was open-sourced by Google in 2005 [8]. By that point its underlying work was already nearly a decade old, developed at HP Labs starting in the mid-1980s. The release matters because it codified an assumption that would shape the next fifteen years of enterprise document processing:

Text on a page is a stream of characters in a known layout.

Every architectural decision in the classic OCR era is downstream of that assumption. You feed an image to the system. The system returns text. Maybe — if you are lucky and the vendor is good — bounding boxes for each word. The layout has to be discovered or imposed by something *else*: a layout-analysis library, a deskewing pass, a language-specific tuning step, a forms-recognition module bolted on top. Each of those somethings was a cottage industry. ABBYY built one of the most successful businesses of the 2000s on the strength of its layout-analysis pipeline alone.

What the classic OCR era got right was the engineering culture. Pipelines were modular. Each stage had clear inputs and outputs. You could measure character-level accuracy on a fixed test set and improve it. Tesseract on clean printed English crossed 99% character accuracy by the early 2010s, and that number is real — it is also, in retrospect, almost completely useless for what enterprises actually wanted to do.

What the classic OCR era got wrong, eventually, is that **enterprise documents are not streams of characters in known layouts**. They are nested structures — invoices with line items, contracts with clauses, claims forms with attached supporting documents, medical records with handwritten margin notes — and the relationships between regions of the page carry as much information as the characters do. A 99% character-accurate Tesseract output of an invoice is useless if you cannot tell which line items belong to which header, or which total goes with which subtotal. The character was the wrong unit of analysis.

By the late 2010s, every team that took document processing seriously had built (or bought) a layer of structural logic on top of OCR. The teams that did this well became IDP vendors. The teams that did it poorly built fragile internal tooling that broke every time a vendor changed an invoice format.

2.2 ICR and template-driven IDP (2010s)

The 2010s answer to “OCR alone is not enough” was **intelligent character recognition (ICR)** and **template-driven intelligent document processing (IDP)**.

ICR was the modest version of the answer: extend OCR to handle handwriting and printed-form fields. Forms have known field locations; if you train a model on enough variants of a known form, you can read

the values out reliably. This worked extremely well for narrow domains — bank deposit slips, insurance claims forms, government tax returns — where the form was standardized and high-volume.

Template-driven IDP was the ambitious version. The idea: for each document type you cared about, build a template that mapped known fields to known regions of the page. Run OCR underneath; apply the template to the OCR output; pull out the structured fields. ABBYY Vantage [9], Hyperscience [10], Kofax (later UiPath Document Understanding), and a half-dozen other vendors built businesses on this pattern through the 2010s.

The template-driven approach has two virtues that should not be underestimated. The first is that **on a known document type, accuracy can be excellent** — well into the high 90s for fields that are reliably present. The second is that **the system is auditable**. You can point at the template, point at the OCR output, and explain exactly why the extractor produced the value it did. For regulated industries — banks, insurers, government — that auditability is a feature, not an afterthought.

The fatal weakness was operational. Every new vendor adds a new template. Every layout change breaks every template. The cost is not in initial accuracy; it is in maintenance. A bank deploying template-driven IDP for invoice processing might support fifty vendors at launch and find itself, two years later, maintaining six hundred templates and falling behind on the next two hundred. The TCO graph for template-driven IDP is the wrong shape: it goes up over time, not down.

By the late 2010s, the most successful classic-IDP vendors had already begun layering machine-learning classifiers on top of their templates — using the template as a starting point and ML to handle variance. This bought them another four or five years of relevance, but it did not solve the underlying problem. The template was still the load-bearing abstraction, and the template was still brittle.

2.3 The single-pass VLM era (2022–2024)

The vision-language model collapsed the OCR-plus-layout-plus-template stack into a single model call.

Donut [11] is one of the cleanest early examples — a transformer trained directly to map document images to structured outputs without an explicit OCR stage. LayoutLMv3 [12] is another, taking a different architectural route but landing in the same place: a single model that understands document structure as part of its native operation, not as a downstream cleanup step.

The arrival of GPT-4V in late 2023 [13] generalized the pattern to commercial frontier models. Suddenly any developer with an API key could feed a document image to a model and ask for structured output. For the first time, the OCR-plus-layout-plus-template scaffolding that had taken a decade to build could be replaced with a single prompt.

What the single-pass VLM era got right was generality. The classic IDP world had been a thousand narrow templates; the VLM world was one general capability. A model that worked at all worked across hundreds of document types. The friction of supporting a new vendor's invoice format dropped from "build a template" to "tweak the prompt."

What the single-pass VLM era got wrong became clear within eighteen months of widespread adoption. **Tables.** VLMs at this generation reliably and quietly garbled tables — merging cells, dropping rows, confusing rowspan and colspan. **Grounding.** A VLM might emit a plausible-looking JSON object with no traceable connection between any field and any region of the source document, making compliance audits effectively impossible. **Confidence.** Single-pass VLMs emit text with no calibrated probability that the text is correct. **Long documents.** The context windows of frontier VLMs in 2023 were not large enough for many enterprise documents, and the workarounds — chunking, sliding windows — re-introduced the same structural-loss problems that the VLM was supposed to eliminate.

The single-pass VLM era is best understood not as a finished destination but as a **stopover**. It proved that document understanding could be collapsed into a model. It did not prove that one collapse, in one pass, was enough.

2.4 The agentic turn (2024–2026)

Sometime around mid-2024, the field collectively realized that the single-pass VLM was the wrong architecture for high-stakes extraction. The model could read the page; the model could not, reliably, *check its own work*.

The OCR-Agent paper [14] — one of the canonical references for the agentic-OCR pattern — formalized the alternative. Plan an extraction. Execute it. Critique it. Re-prompt if necessary. Invoke a specialized tool (a table extractor, a layout detector, a schema validator) when the base model is the wrong instrument. Escalate to a human reviewer when the system cannot resolve its own uncertainty. The single-pass output became a multi-pass loop.

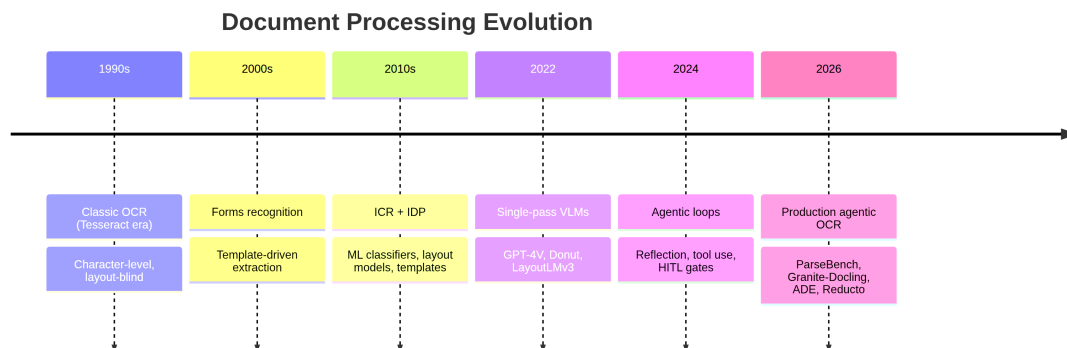
The architecture is older than the term. Software engineers had been wrapping retry logic around LLM calls since 2023. What changed in 2024–2026 is that the surrounding ecosystem caught up:

- **Tool calling** matured into a first-class capability across frontier models, making the “invoke a specialized tool” step reliable rather than fragile.
- **Schema validation** libraries (Pydantic, Zod, JSON Schema) became standard, giving the loop an external oracle to check outputs against.
- **Grounding** — the requirement that every output field be traceable to a region of the source — moved from research nice-to-have to compliance requirement, especially after early agentic deployments in healthcare and legal made their absence painful.
- **HITL queue infrastructure** at vendors like Hyperscience, Landing AI, and Reducto became sophisticated enough that “escalate to a human” was a real workflow, not a thrown exception.
- **Evaluation infrastructure** — golden sets, regression testing, vendor version pinning — became a serious discipline rather than an afterthought.

By 2026, production deployments have benchmarks to point at. Anterior AI, working with Reducto on healthcare prior-authorization workflows, reports 99.24% accuracy on a vertical-specific golden set [15]. Databricks Document Intelligence, launched in 2026, reports a **16% performance gain across every agent framework they tested** simply by inserting `ai_parse_document` upstream of the agent’s reasoning step [1]. IBM Granite-Docling-258M — a 258-million-parameter model — was released under Apache-2.0 in January 2026 and gets within five percentage points of human accuracy on page-element identification [16].

These are not hypothetical numbers. They are vendor-reported, and they carry the usual vendor-reported caveats, but the consistency across vendors and the public benchmarks underneath them (ParseBench, RD-TableBench, OmniDocBench, olmOCR-Bench) form a coherent picture: a field that crossed a maturity threshold in 2025–2026.

2.5 Timeline



Document-processing evolution, 1990–2026

2.6 The pattern, restated

Look at the four eras together and a pattern emerges.

Classic OCR eliminated the cost of manual transcription. It created the cost of layout-and-template engineering.

Template-driven IDP eliminated the cost of writing custom layout-extraction code for each form. It created the cost of maintaining a thousand templates.

Single-pass VLMs eliminated the cost of templates. They created the cost of tolerating opaque, ungrounded, table-mangling output.

Agentic loops eliminated the cost of accepting opaque output. They created the cost of orchestration, retry budgets, evaluation infrastructure, HITL queue management, and vendor version drift.

Each generation moved the engineering. None of them eliminated it. Anyone selling you the story that agentic OCR makes the engineering disappear is selling you a screenshot of the demo. The honest version: agentic OCR moves the engineering to places where it has higher leverage. Pipeline glue code, brittle templates, and post-hoc cleanup logic are bad places for engineering effort. Evaluation infrastructure,

prompt engineering, and HITL workflow design are good places — because they compound across document types instead of being rebuilt for each one.

That is the trade. It is a good trade. It is not the same trade as “the model figures it out.”

 Named take

Each generation of document processing eliminates the previous generation’s most expensive engineering — and creates a new expensive engineering category in its place. Anyone telling you agentic OCR eliminates pipeline work is selling you a screenshot of the demo.

3. What Agentic OCR Actually Is

i Executive summary

- Agentic OCR is a document-processing system that is **goal-driven, multi-pass, self-correcting, tool-using, and grounded**. All five attributes must be present. Three out of five is “VLM with retries,” not agentic OCR.
- The reason the definition matters is that the word “agentic” is the marketing differentiator that lets vendors charge ten times more for what may or may not be a retry loop. The five-attribute test gives you a defensible procurement filter.
- **Grounding is the line.** A system that cannot point at the pixels that produced a field is not doing agentic OCR. It is producing confident hallucinations with structure.
- This chapter is the load-bearing chapter for the rest of the book. Later chapters on architecture, vendor selection, anti-patterns, and evaluation all assume the definition established here.

3.1 Three demos walk into a procurement meeting

Picture three vendor demos, one after another.

Demo One is a frontier-model API. The salesperson uploads a scanned invoice; the model returns clean JSON; the audience claps. There is one model call. The system answers in a second and a half. The system has no idea whether its answer is right.

Demo Two is a CV-plus-VLM specialist. The salesperson uploads a much messier document — handwritten notes in margins, a nested table, a stamp partially obscuring a field. A pipeline appears: a layout detector segments the page into regions; a text extractor handles the prose blocks; a separate model handles the table; a schema validator checks the output; a confidence score below threshold triggers a re-extraction with revised instructions; a final assembler stitches the regions back together with provenance pointers. The audience claps because they recognize *engineering*.

Demo Three is a Tesseract pipeline with an LLM post-processor that “cleans up” the OCR output. The salesperson uploads a moderately messy invoice; the system extracts text; the LLM rewrites it into JSON. The salesperson calls this “agentic OCR.” The audience claps because, by 2026, “agentic” is the word that gets the room.

Only one of these is actually agentic OCR. This chapter will tell you which one, and why it matters. The cost difference between them is roughly an order of magnitude. The accuracy difference, on a sufficiently messy real-world workload, is also roughly an order of magnitude.

3.2 Definition

The working definition that the rest of this book uses:

Agentic OCR is a document-processing system that is goal-driven, multi-pass, self-correcting, tool-using, and grounded.

All five attributes must be present in a system for the term to apply. The five are not a wishlist. They are interlocking — each one fills a gap the others cannot fill on their own — and a system missing any one of them has a predictable failure mode that the marketing language often hides.

The five attributes, expanded:

3.3 The five attributes

3.3.1 Goal-driven

A goal-driven system operates against a **structured target**, not against the document. The target is the schema, the extraction goal, or the downstream contract that the output has to satisfy. The system plans its extraction against that target.

This matters because the alternative — “extract everything” — produces output that is structurally indistinguishable from a fancy `cat` command piped into a markdown formatter. The single-pass VLM era of 2022–2024 was full of such systems. They emitted markdown or HTML; they expected the downstream consumer to figure out what it cared about; they treated the document as the unit of analysis.

A goal-driven system treats the **schema** as the unit of analysis. Given a “vendor invoice” schema with fields like `invoice_number`, `total_amount`, `line_items[]`, the system plans an extraction that looks for those specific fields, succeeds or fails on each one independently, and reports per-field provenance. The same document fed to two different schemas produces two different extractions, and both are correct on their own terms.

The smell test: ask a vendor what their system does when you change the schema. If the answer involves changing a template, building a new model, or “let us run it through our pipeline and see,” it is not goal-driven. A goal-driven system takes the schema as input and adapts.

3.3.2 Multi-pass

A multi-pass system iterates rather than emits-and-exits. It runs at least one self-review pass, and ideally more than one when the first pass fails its own checks.

This is the attribute that most distinguishes agentic OCR from its single-pass VLM ancestor. A single-pass VLM extracts once. If the extraction is wrong, the system does not know. A multi-pass system extracts, checks (via a schema validator, a secondary model, a tool output, or a confidence calibration), and re-extracts if the check fails.

The retry budget is real engineering. Typical agentic OCR systems run two to four retries before escalating to a human. The choice of retry budget is a tradeoff: more retries means more cost and more latency, but also more chances to catch errors. Vendors that hide this number from you are not doing you a favor.

The smell test: ask a vendor how many model calls their system makes per document on average. If the answer is “one,” it is not multi-pass. If the answer is “it depends on the document complexity,” you are talking to someone who has thought about this seriously.

3.3.3 Self-correcting

Self-correction is the most over-loaded attribute in the vendor lexicon. The phrase covers three concrete subsystems, and most vendors who use the term have, at best, one of the three working.

The three subsystems are:

- **Detection** — knowing that an extraction is wrong. This requires either an external oracle (a schema validator, a secondary model, a tool output) or a calibrated confidence signal that actually correlates with accuracy on out-of-distribution data.
- **Repair** — fixing the detected error. This is usually a re-extraction with revised instructions, but it can also be a routing decision (send this region to a different tool) or a contextual lookup (resolve this entity against an external knowledge base).
- **Escalation** — failing safely when detection or repair cannot close the loop. Escalation routes the document to a human reviewer with the relevant context attached.

Chapter 5 unpacks each of these in depth. For the definition, the load-bearing fact is that **all three are required**. A system that detects but cannot repair is a half-built loop. A system that repairs but cannot escalate gracefully will, eventually, lose customers’ trust on the documents it could not fix. A system that escalates without detection is just a queue with extra steps.

Most production failures in agentic OCR are **detection failures**, not repair failures. The model never noticed the error. This is the deepest reason that confidence scores from VLMs need to be treated as ordinal signals at best — they are an opinion the model has about its own work, and opinions are not the same as ground truth.

The smell test: ask a vendor what their system does when its first extraction is wrong but the confidence score is high. A good answer involves an external oracle. A bad answer is “we re-prompt the model” — that is the model talking to itself.

3.3.4 Tool-using

A tool-using system invokes specialized tools as a routine part of its workflow. The tools are real software: a layout detector, a table extractor, a schema validator, a KB lookup, a fallback OCR engine, a redaction module, a classification model.

This is where the 2026 leaderboards have been most clarifying. **Specialization beats generality on adversarial workloads.** Reducto’s hybrid CV-plus-VLM architecture sits at the top of RD-TableBench at roughly 0.90 similarity — about twenty percentage points ahead of the hyperscaler APIs, all of which use general-purpose VLMs without specialized table tooling [6]. PP-StructureV3, at under 100 million parameters, matches Gemini-2.5-Pro on OmniDocBench precisely because it is a layout-specialized model and not a general one [17].

The thing tool-using is *not* is “the model has access to a function call.” It is the routine orchestration of specialized tools as part of the workflow. A model that *could* call a table tool but in practice almost never does is not tool-using; it is tool-capable. The distinction matters because the operational reliability of the system depends on the tools being invoked predictably, on the workflow being designed around tool use, and on the tool failures being handled gracefully.

The smell test: ask a vendor for an architecture diagram of their system. If it shows a single model in the middle with an arrow labeled “extraction,” it is not tool-using. If it shows a router that dispatches regions to specialized handlers, it is.

3.3.5 Grounded

A grounded system traces every output field back to a region of the source document.

In practice, this means each extracted field carries metadata: the page number, the bounding box, often the specific text span. The downstream consumer can render the output alongside the document with the cited region highlighted; the compliance auditor can verify the extraction; the human reviewer can correct it efficiently by pointing at the right place.

Grounding sounds like a compliance feature. It is also the structural prerequisite for everything else in this list. Without grounding:

- **Self-correction’s detection step is impaired.** You cannot reliably critique an answer you cannot localize. “Is this \$10,000 right?” is much easier to answer when you can point at the pixels that produced it than when you have only the text.
- **Tool-using suffers.** Routing a region to a specialized tool requires knowing where the region is. A system that does not track regions has nothing to route.
- **HITL escalation degrades.** A human reviewer presented with an ungrounded JSON object and a fifty-page PDF will spend most of their time scrolling. A reviewer presented with the same JSON and a highlighted source region resolves the case in seconds.

Grounding is the line. A system that cannot point at the pixels that produced a field is not doing agentic OCR. It is producing confident hallucinations with structure.

The smell test: ask a vendor to show you, in their UI, the source region for a specific output field. If the answer involves a screen-recording delay, a developer pulling up a debug log, or the phrase “we can add that,” it is not grounded.

3.4 The five-attribute scorecard

Use this as a procurement filter.

Attribute	Classic OCR	Single-pass VLM	True agentic OCR
Goal-driven	No	Partial	Yes
Multi-pass	No	No	Yes
Self-correcting	No	No	Yes
Tool-using	No	No	Yes

Attribute	Classic OCR	Single-pass VLM	True agentic OCR
Grounded	Partial (bboxes)	No	Yes

Classic OCR scores partial on grounding only because it does produce per-word bounding boxes. It scores zero on every other attribute because OCR was never designed to do those things; it was designed to convert images to text.

Single-pass VLMs score partial on goal-driven because most modern VLMs do accept structured-output prompts (a schema in the prompt, JSON mode, function calling). They score zero on the other attributes, in the strictest reading, because a single model call cannot be multi-pass, cannot self-correct, cannot route to tools, and emits ungrounded output by default. Adding a retry wrapper around a VLM call gets you partial credit on multi-pass and self-correction, but it does not get you to “true agentic.” That is wrapper-washing (Chapter 13).

True agentic OCR scores yes on all five. There is no half-credit row in this table for the systems this book is about. A system missing any one attribute has a predictable, named failure mode that the rest of the book will return to repeatedly.

3.5 What it is not

The definition is sharpest when you also say what falls outside it.

Classic OCR with a confidence threshold is not agentic OCR. Confidence gating is a feature of agentic OCR; it is not the definition. A Tesseract pipeline that re-runs on low-confidence regions has one of five attributes (a weak version of self-correction) and zero of the other four. Marketing such a system as “agentic” is the wrapper-washing anti-pattern.

A single-pass VLM is not agentic OCR. Even GPT-4V on a document, even with a beautifully crafted prompt that requests grounded structured output with confidence scores, is one extraction call. Single-pass VLMs are valuable. They are not agentic OCR. The category exists for systems that *iterate*.

Template-bound IDP with an LLM cleanup step is not agentic OCR. The template is still load-bearing; the LLM is decoration. If removing the LLM cleanup step would cause the system to fail gracefully on unchanged documents, the LLM is the agentic part. If the system breaks on unchanged documents without the LLM, the LLM is doing real work. Most systems sold as “agentic” by classic-IDP vendors are in the first category.

An LLM with one function-calling tool is not agentic OCR. Tool-using means routinely orchestrated tools, not technically-possible tools. A vendor whose architecture diagram has one box labeled “the LLM can call a table tool if needed” is showing you a feature, not an architecture.

These four anti-categories cover, in the author’s experience, well over half of all vendor pitches that use the word “agentic.” The five-attribute test is the defense.

3.6 Why the definition matters more than the word

A working definition is more important than a label, but the label is doing real work in 2026 procurement. The word “agentic” appears in vendor pricing pages where, three years ago, the same systems were sold as “intelligent” or “AI-powered” without a price premium. The premium is real; it averages roughly five to ten times the per-page cost of the prior generation. Some of that premium reflects genuinely new capability. Some of it is a relabeling.

The five-attribute test is a way to tell which is which. It is also a way to talk to vendors in a structured way. A vendor who can answer all five smell tests has thought seriously about the architecture. A vendor who deflects on one or two of them has gaps. A vendor who deflects on three or more is selling marketing.

The chapters that follow assume this definition. When Chapter 4 lays out the three reference architectures, every one of them satisfies all five attributes; that is what makes them reference architectures. When Chapter 7 ranks commercial vendors, the ranking implicitly weighs how convincingly each one satisfies the five. When Chapter 13 enumerates anti-patterns, every anti-pattern is a way of failing the five-attribute test.

Hold onto the test. It is the most useful procurement tool this book offers.

 Named takes

“Agentic” is the word that lets people charge 10× for a retry loop. The five-attribute test exists so you don’t get fooled.

Grounding is the line. If your system can’t point at the pixels that produced a field, it isn’t doing agentic OCR — it’s doing confident hallucination.

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4. Architecture Patterns

i Executive summary

- Three reference architectures dominate 2026 agentic OCR: **single-pass VLM** (Pattern A), **agentic loop with reflection** (Pattern B), and **hybrid CV+VLM** (Pattern C). Each is valid for a different intersection of document variability and stakes.
- The choice is not “which architecture is best.” The choice is “which architecture matches my workload.” Pattern A is right for low-stakes, low-variability work; Pattern B is the 2026 default for general agentic workloads; Pattern C is winning the high-stakes benchmarks and the high-stakes verticals (healthcare, legal, finance) by a non-trivial margin.
- Tool chains are not optional. A reflection loop with no specialized tools is a model talking to itself. The vendors leading 2026 benchmarks all route work to specialized extractors at the region level — that is what is winning.
- HITL gates are not a fallback bolted on the side. In production they are part of the architecture, and the rate of escalation (1–10% in healthy deployments) is one of the most honest signals you can read about a system.

4.1 A working architecture, walked through

Before the abstractions, a concrete system. Anterior AI runs healthcare prior-authorization workflows in production [15]. Their reported pipeline, simplified for the purpose of this walk-through:

A clinical packet arrives — a referral letter, an ICD-coded diagnosis sheet, a stack of supporting lab reports and prior office notes, often a fax cover page with a phone number that has been scribbled over and corrected. Total length, ten to forty pages. The packet enters a layout-first analysis: a CV model partitions every page into regions, classifies each region (header, prose paragraph, table, signature block, form field, handwritten margin note, image), and emits a region map. A router dispatches each region to the right extractor — printed text regions go through one path, hand-written notes through another, tables through a specialized table extractor, signature blocks to a stamp-and-signature classifier. The extracted regions assemble back into a structured document object whose schema is the prior-auth schema, not a generic markdown blob.

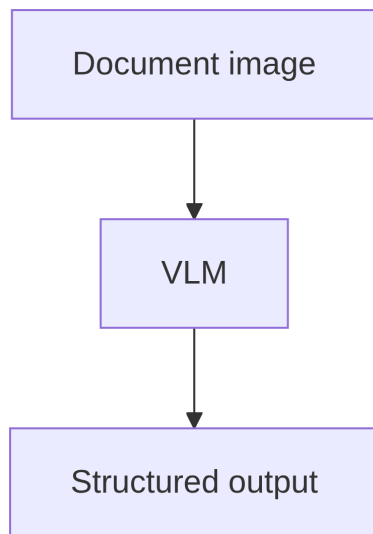
A reflection step then validates the assembled object against schema rules: dates are in expected ranges, ICD codes resolve, the patient ID matches across pages, the requested procedure is on the payer’s covered-procedures list. Anything that fails validation triggers either a re-extraction (with a revised prompt that names the specific issue) or, if re-extraction has burned its retry budget, an escalation to a human reviewer with the failing field highlighted on the source page.

The system reports about 99.24% field-level accuracy on its golden set. The HITL escalation rate sits in the low single digits. The architecture has all three of the things the rest of this chapter is going to abstract: a layout-first decomposition, a router that sends regions to specialized tools, a reflection loop with both retry and escalation paths. It is, in 2026 vocabulary, a Pattern C architecture.

This chapter unpacks the three architectures, gives you criteria for choosing between them, and is honest about the engineering each one costs.

4.2 Pattern A: single-pass VLM

The simplest architecture in modern agentic OCR is the one that, by the definition in Chapter 3, barely qualifies as “agentic” at all.



Pattern A — Single-pass VLM

A document image goes in. A vision-language model emits structured output — typically JSON conforming to a schema declared in the prompt or via the model’s structured-output API. There is one model call. There is no reflection, no validation loop, no specialized tool, no retry, no grounding by default. Latency is around 200ms to 2s per page depending on model and document complexity. Cost is the cost of one VLM call.

This architecture works, and it works well, on a specific kind of workload: documents whose layout is simple, whose tables are small or absent, whose fonts are clean, whose stakes for any single error are low, and whose schemas are stable. A consumer-facing app that lets a user photograph a receipt and pulls out total, merchant, and date is a textbook Pattern A workload. So is bulk indexing of email attachments where 95% accuracy is good enough because the only downstream consumer is a search box.

Pattern A’s failure modes are not edge cases; they are central tendencies of the workloads it gets misapplied to:

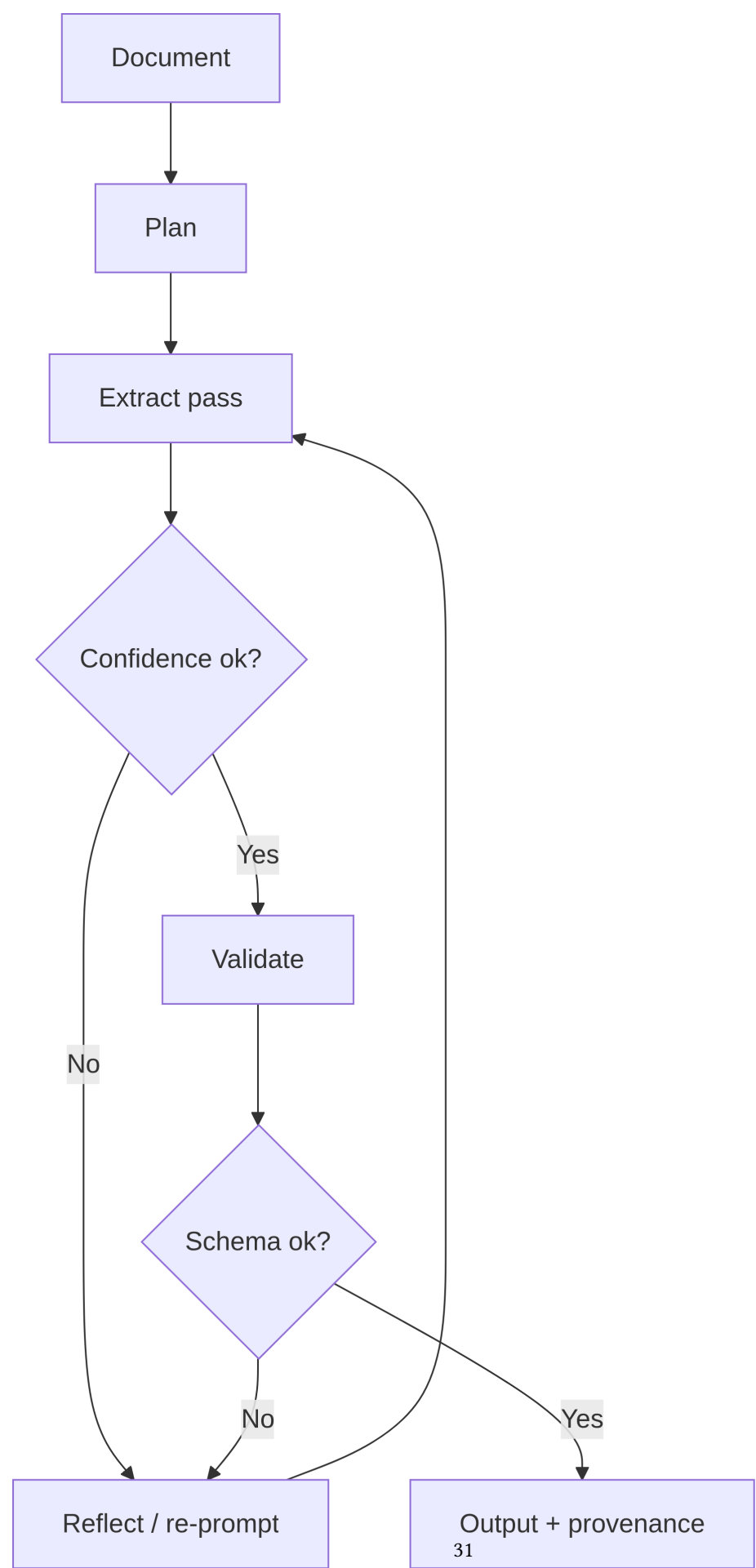
- **Tables.** A single-pass VLM in 2026 will still routinely merge adjacent table cells, drop rows when the table crosses a page break, and confuse rowspan/colspan in tables with merged headers. This is not a model-quality problem that another year of training will fix; it is a structural limitation of single-pass extraction. The Reducto vs. hyperscaler gap on RD-TableBench — about twenty percentage points of table similarity in Reducto’s favor — is the most reliable signal of this fact [6].
- **Long documents.** Even with current context windows, the practical accuracy of single-pass extraction degrades over multi-page documents. The model attends to early pages well and forgets the middle. This shows up in production as fields that come from page 1 being reliably extracted and fields that come from page 8 being silently dropped.
- **Grounding.** Pattern A emits structured output with no built-in connection to source regions. You can ask the model to include bounding-box coordinates and it will, but the coordinates are themselves model outputs — they are not ground truth, and they are wrong some non-trivial fraction of the time. Pattern A is the architecture that produces beautiful JSON that you cannot audit.
- **Confidence.** A single-pass VLM produces output with no calibrated probability that the output is right. Anything that claims to be a confidence score in Pattern A is, at best, a model’s opinion about its own work.

Pattern A is not bad. It is not “agentic OCR” in the strict five-attribute sense — it satisfies, at best, two of the five attributes — but it is the right architecture for workloads where the cost of an error is low and the document variability is modest. The mistake is using it for a workload it cannot serve and then being surprised when production accuracy falls short of the demo.

The 2026 representative implementations of Pattern A are the hyperscaler APIs (Google Document AI, AWS Textract, Azure Document Intelligence) running in their default modes, frontier VLM APIs called with a structured-output prompt (GPT-4V/4o, Claude, Gemini, Qwen3-VL), and the cheapest tier of specialist vendors (LlamaParse Fast, Mistral OCR 3 Standard).

4.3 Pattern B: agentic loop with reflection

Pattern B is the 2026 default for general-purpose agentic OCR. It is what the field talks about when it says “agentic.” It satisfies all five attributes in the Chapter 3 scorecard, and it does so without requiring the engineering investment that Pattern C demands.



Pattern B — Agentic loop with reflection

The shape is: plan, extract, critique, re-prompt if necessary, validate, output with provenance. Each step has a real engineering parameter behind it.

Plan. The system articulates, before extracting, what it is looking for. In simpler implementations this is the schema injected into the prompt. In more sophisticated implementations the system generates a per-document extraction plan that varies by document type — different field priorities, different validation rules, different retry strategies. The plan step is what makes the architecture *goal-driven* in the Chapter 3 sense.

Extract pass. A VLM call (or sequence of calls for long documents) produces a candidate extraction. This is structurally similar to Pattern A but happens inside a loop that knows the extraction might not be final.

Confidence check. This is where Pattern B’s engineering quality varies most between implementations. The naïve version compares a model-emitted confidence score against a threshold. The sophisticated version cross-checks the candidate extraction against external oracles — schema validators (does it parse? do dates lie in plausible ranges?), tool outputs (does the layout detector agree on where this field is?), secondary models (does a second VLM with a different prompt produce the same answer?). The difference between naïve and sophisticated confidence checks is, in practice, the difference between detection failures and detection successes.

Reflect / re-prompt. If the confidence check fails, the system re-prompts with a revised instruction. The revision typically references the specific failure — “the date you extracted as 2026-13-45 is not valid; please re-examine the date field” — rather than rerunning the same prompt. This is why retry-thrash is a real failure mode: a system that re-prompts identically each time learns nothing and converges on nothing.

Validate. A final structural check against the target schema. Schema validation is cheap and surprisingly diagnostic. Many extraction errors that confidence checks miss show up as schema failures (a required field is null; a numeric field contains “TBD”; an enum field contains an unexpected value).

Output with provenance. Every field in the final output carries a pointer to the source region. This is what makes Pattern B *grounded* in the Chapter 3 sense.

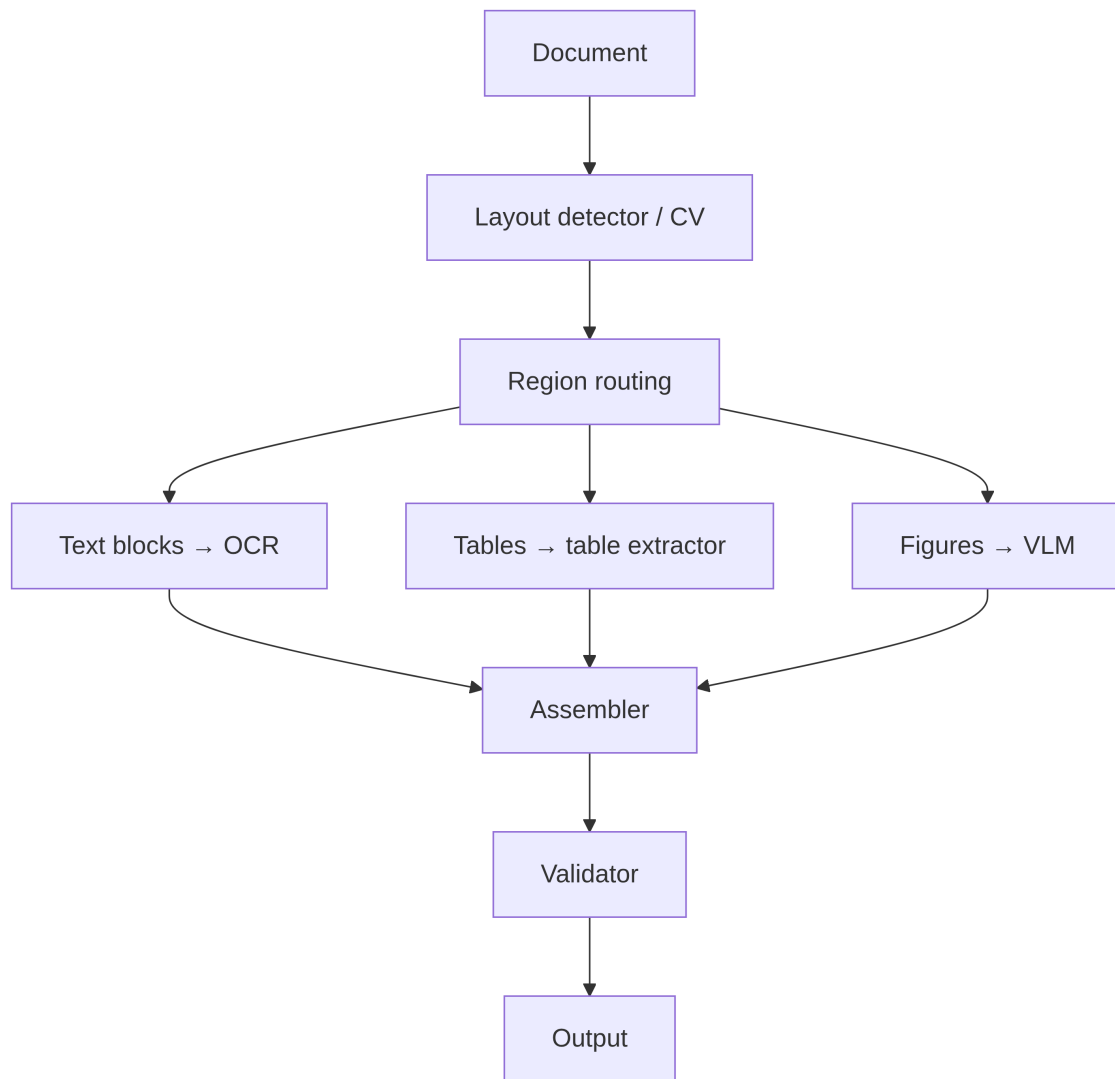
The retry budget — typically two to four attempts before escalation — is the most visible engineering parameter, and the one most worth asking vendors about. Too low and the system gives up on documents it could have handled. Too high and a single recalcitrant document can run up cost and latency without ever converging.

Pattern B is well-suited to general-purpose agentic OCR: messy enterprise documents, mixed layouts, moderate stakes. Most LlamaParse Agentic [4] and LandingAI ADE [18] deployments are Pattern B. The strengths are flexibility (one architecture handles a wide variety of document types) and engineering tractability (the loop is conceptually simple even if the implementation is not). The cost is, well, cost — multi-pass means multiple VLM calls, which means real money per page, plus the latency penalty of running through a loop.

Pattern B’s failure modes are subtler than Pattern A’s. The architecture detects more errors but can produce silent hallucinations when the reflection step approves a wrong answer (Chapter 5 unpacks this in detail). It can also produce retry-thrash on documents that the model is genuinely incapable of extracting correctly — burning the retry budget without improving the answer. Production deployments need observability into both failure modes, and most teams discover this fact the hard way.

4.4 Pattern C: hybrid CV+VLM

Pattern C is the architecture that is winning the high-stakes benchmarks and the high-stakes verticals. It is also the architecture that costs the most engineering to build and operate.



Pattern C — Hybrid CV+VLM

The core move is **layout-first decomposition**. Before any extraction happens, a computer-vision model partitions the page into regions and classifies each one. The classification is fine-grained enough to drive routing: text blocks go to a dedicated text-extraction path, tables go to a specialized table extractor, figures go to a VLM, signature blocks go to a signature classifier, handwritten regions go to a handwriting-tuned extractor.

The reason this architecture wins on adversarial workloads is that **specialization beats generality where the distribution is narrow and the stakes are high**. A table extractor that has been trained specifically on tables, with table-specific evaluation metrics (Needleman-Wunsch cell alignment, header-disambiguation accuracy), is better at tables than a general VLM that has been trained on everything including tables. The Reducto result on RD-TableBench — about 0.90 cell-alignment similarity on adversarial tables, roughly twenty points ahead of the hyperscaler APIs in Pattern A — is the clearest data point [6]. PP-StructureV3 making the same point at under 100 million parameters, matching Gemini-2.5-Pro on OmniDocBench at a tiny fraction of the compute cost, is the strongest argument that this is structural rather than incidental [17].

Pattern C also has the strongest grounding story by construction. Because the system tracks regions from the layout step onward, every extracted field carries a region pointer all the way through to the output. There is no “model emitted a bounding box and we hope it’s right” — the bounding box came from the layout detector, not the extractor.

The cost of Pattern C is the engineering. Every component is a real piece of software with its own training data, eval, and failure modes. The layout detector needs to be accurate or every downstream extractor receives garbage. The router needs to know which extractor to send each region to, which means the classification taxonomy needs to be maintained as document types evolve. The assembler needs

to handle cases where regions overlap, where the layout detector segments wrongly, where extractors disagree on overlapping regions. Each of these is solvable. None of them is free.

Pattern C is what's behind Reducto's hybrid architecture, IBM Granite-Docling and the Docling pipeline framework, OpenDataLab's MinerU, and the PaddleOCR / PP-Structure lineage. It is also implicit in any in-house implementation that takes high-stakes extraction seriously enough to refuse the "one model does everything" path.

It is worth naming that the **hyperscaler ecosystems also expose layout-detector primitives** that can serve the same architectural role inside a Pattern C pipeline, for buyers already committed to a cloud vendor:

- **Azure AI Document Intelligence Layout** (\$10/1k pages) — text, tables, paragraphs, section headings, selection marks, figure regions with bounding boxes. The Azure-native answer for the layout-extraction stage.
- **Google Document AI Layout Parser** (\$10/1k pages) — Gemini-powered layout extraction plus initial chunking in a single call; closest hyperscaler equivalent to LlamaParse Agentic for buyers on GCP.
- **AWS Textract AnalyzeDocument with Layout feature** (\$10–\$25/1k pages, stackable with Forms/Tables/Queries) — newer addition to the Textract feature set; less mature than Azure or Google Layout but functional.

These hyperscaler Layout APIs serve the same architectural role as PP-StructureV3 or Granite-Docling in the layout-detector slot of Pattern C — they emit structured regions that downstream extractors can route off. A pipeline that uses Azure DI Layout to segment, an LLM (frontier or Granite-Docling) to extract from text blocks, and a specialized table extractor for table regions, plus a schema validator and HITL queue, is a valid Pattern C architecture for organizations committed to Azure. The same shape works on GCP with Document AI Layout Parser and on AWS with Textract AnalyzeDocument Layout. Chapter 7 walks the hyperscaler sub-products in more detail.

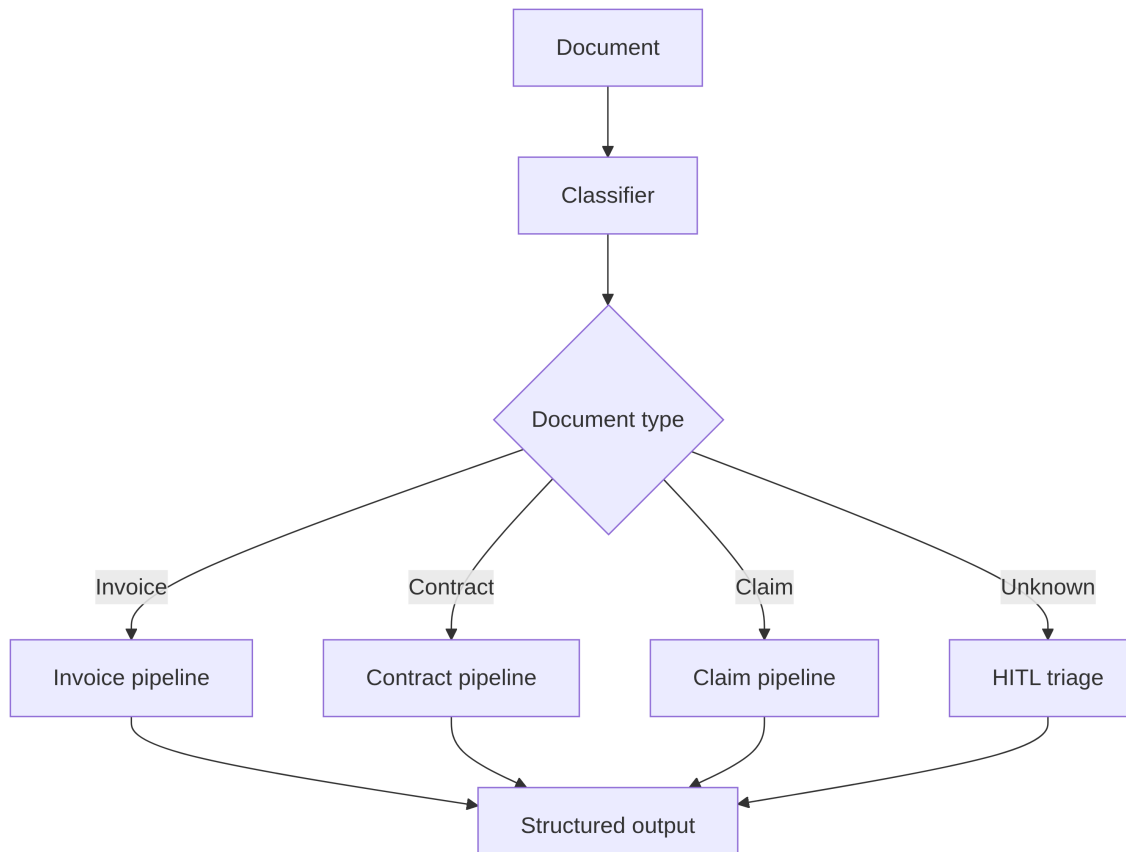
The honest framing: **Pattern C is the architecture for workloads where errors cost more than engineering does.** Healthcare prior-authorization fits because a denied claim costs hundreds to thousands of dollars in downstream rework, dwarfing the marginal engineering cost of a hybrid pipeline. Legal contract review fits because a missed clause can be a six-figure legal exposure. KYC document review fits because a wrongly approved high-risk account can be a regulatory event. Commodity invoice processing rarely fits, because the marginal error cost is too low to justify the engineering.

4.5 Document classification — the often-missed primitive

The three reference architectures above describe the *extraction* core. In production deployments handling more than one document type — which is most production deployments — there is a step that conceptually precedes extraction and meaningfully changes the architecture: **document classification**.

The classification step's job is to decide *what kind of document this is* before deciding how to extract from it. An incoming PDF could be an invoice, a contract, a claim form, a clinical note, a driver's license, or a fax cover sheet with the actual document on the back. The downstream extraction logic depends on the answer: an invoice goes through a vendor-specific schema; a contract goes through clause extraction; a claim form goes through a different pipeline with HITL gating tuned to claim severity.

The architectural shape is **classify-then-route**:



Classify-then-route — adding classification ahead of extraction

This is a Pattern C variation (sometimes integrated into a larger Pattern C; sometimes a stand-alone routing layer in front of Pattern B pipelines). Three common implementations in 2026:

- **Lightweight classifier model.** A small, fast specialized classifier — typically a fine-tuned BERT-class model or an embedding-plus-kNN approach — produces a document-type label in milliseconds. Cheap, accurate on in-distribution types, fails ungracefully on novel types. The right default for narrow-but-stable document distributions where the class set is small and known.
- **LLM-based zero-shot classification.** A VLM or LLM call with a prompt like “Classify this document as one of: invoice, contract, claim, clinical note, other.” Slower and more expensive per document than a dedicated classifier, but flexible — new document types can be added by editing the prompt rather than retraining a model. The right choice when the document distribution is wide and evolving, or when the classification needs to consider visual cues (a logo, a layout fingerprint).
- **Layout-based heuristics.** Cheap rule-based classifiers using bounding-box patterns from the layout-detector stage. (“If there are exactly four large text blocks in the top quadrant and a table below, it’s an invoice.”) Brittle but free; useful as a confidence-boosting cross-check for the LLM-based classifier rather than as the sole classifier.

The hyperscaler equivalents are first-class. AWS Bedrock Data Automation and Azure DI Prebuilt models effectively do classification + extraction in one call by routing to a specific blueprint or prebuilt model; Databricks’ `ai_classify` is the explicit primitive. The hyperscaler-platform tier has converged on classify-then-route as the default architecture, which is one indicator that it is the right default.

4.5.1 Why classification gets missed

The single largest 2026 procurement mistake involving classification is **assuming the parser also handles it**. Most parsers — including the agentic-OCR specialists — do not classify. They extract whatever they are pointed at against whatever schema they are given. If you point an invoice parser at a contract, you get an extraction that conforms to the invoice schema and is mostly nonsense. The system did its job; the system was given the wrong job.

The fix is to put classification *in front of* extraction as an explicit step, with its own evaluation, its own monitoring, and its own HITL escalation path for novel or low-confidence cases. Teams that build agentic

OCR pipelines without classification routinely encounter a class of production failures that look like “the extractor is wrong” but are actually “the extractor was given a document outside its design distribution and had no way to flag the mismatch.”

4.5.2 Failure modes when classification is wrong

A wrong classification fails worse than a missing extraction, because the system is now confidently doing the wrong thing. The two failure modes:

- **Misclassification followed by confident extraction.** The system classifies a contract as an invoice, routes to the invoice pipeline, and emits a structured “invoice” with fields scraped from the contract. The values are plausible (vendor name, dates, dollar amounts all exist in contracts) but wrong. Downstream consumers may not notice until reconciliation fails weeks later.
- **Novel-class blindness.** A document arrives that does not match any known class. A well-engineered classifier returns “unknown” with low confidence and routes to HITL triage. A poorly-engineered classifier returns its closest-match class with moderate confidence, sending the novel document into the wrong extraction pipeline. The defense is an explicit “unknown / out-of-distribution” class in the classifier’s label set, and routing on that class to HITL by default.

Both failures are catchable with the eval discipline of Chapter 11. — *if* classification is measured as a separate accuracy axis from extraction. Bundling them into a single “did we get the answer right” metric hides exactly the failures that originate in classification.

4.6 Tool chains, memory, and HITL gates

The three patterns above describe the extraction core. Production agentic OCR systems have additional components that the simplified diagrams elide.

Tool inventory. A production system typically has access to: a layout detector (Pattern C central; optional for Pattern B), a table extractor, a schema validator (always), an external knowledge-base lookup for entity resolution (vendor IDs, ICD codes, ISO country codes), a redaction module for PII, a classification model for document-type routing, and a fallback OCR for when the primary extractor fails. Tool-using in the Chapter 3 sense means these are routinely invoked, not just technically available.

Memory. Short-term memory carries context within a single document — heading hierarchy, units of measurement, vendor identity established on page 1 — into extractions on later pages. Long-term memory carries vendor-specific corrections from HITL queues back into prompts, document-type priors learned across deployments, and audit logs of model behavior for drift detection. Few public vendors talk much about their memory architectures, but the gap between vendors that have solid memory engineering and those that do not is sharply visible in production over a six-month window.

HITL gates. Every production architecture has a path to a human reviewer. The question is where in the pipeline the gate sits. The healthier patterns place HITL gates post-validation, before output, with the failing field highlighted on the source page and the candidate extraction pre-populated for the reviewer to accept, reject, or correct. The unhealthier patterns dump entire documents into a queue with no annotation, requiring the reviewer to find the issue manually. The healthy version cuts reviewer time per case from minutes to seconds.

The escalation rate is one of the most diagnostic production signals. A healthy 2026 agentic OCR deployment escalates somewhere between 1% and 10% of documents. Below 1% suggests the system is silently swallowing errors that should have escalated; above 10% suggests the extraction is too unreliable for the workload (or that the eval rules are too strict, but that’s the easier failure to diagnose).

4.7 Choosing between the three patterns

The decision reduces to two axes: **document variability** and **stakes**.

- **Narrow variability, low stakes** → Pattern A. Don’t pay for a loop you don’t need.
- **Wide variability, low or moderate stakes** → Pattern B. The 2026 sweet spot for most enterprise agentic OCR.
- **High stakes, any variability** → Pattern C with HITL gates. Engineering cost pays for itself fast when the alternative is downstream rework cost.

Volume modifies the decision but does not change its shape. At very low volume (under 10k pages/month), Pattern A via a hyperscaler API is almost always the right choice regardless of stakes — the math does

not support investing in Pattern B or C until volume justifies it. At very high volume (millions of pages/month), the open-source Pattern B or C path (Granite-Docling, PaddleOCR + Pattern C orchestration, OlmOCR-2 self-hosted) starts to pay for itself versus commercial APIs.

A practical heuristic: **build the simplest pattern that satisfies your error budget, not the most sophisticated pattern your engineering can support.** Sophistication has a maintenance cost. The vendors that look impressive at demo time but become operational burdens six months in are almost always the ones who deployed Pattern C when Pattern B would have done the job.

The chapters that follow assume these three patterns as vocabulary. When Chapter 7 ranks commercial vendors, the ranking implicitly is “which pattern do they implement well, and at what price.” When Chapter 11 talks about evaluation, the rubric varies by pattern (the metrics that matter for Pattern A are not all the same as the metrics that matter for Pattern C). When Chapter 13 enumerates anti-patterns, several of them are specific ways each of the three patterns fails when misapplied.

💡 Named takes

If your architecture has no place for a table-specific tool, you’re betting the model is uniformly good at everything. Twelve months of leaderboards say it isn’t.

Hybrid CV+VLM is winning the table benchmarks for the same reason ensembles win on tabular ML: specialization beats generality where the distribution is narrow and the stakes are high.

Build the simplest pattern that satisfies your error budget. Sophistication you do not need is technical debt with a long fuse.

5. Self-Correction: What It Means and What It Doesn't

i Executive summary

- Self-correction is the single most over-loaded phrase in 2026 agentic-OCR marketing. It is doing the work of three distinct subsystems — **detection**, **repair**, and **escalation** — and most vendors who use the term have, at best, one of the three working well.
- The dominant production failure mode is **detection failure**: the model is wrong, the confidence score is high, the loop never triggers, the error ships. The most discussed result in the 2026 literature (“Tiny Silent Hallucinations in Agentic AI”) is essentially a paper-length elaboration of this point.
- Confidence scores from VLMs are not calibrated probabilities. Treat them as an ordinal signal, cross-validate with external oracles (schema validators, secondary models, tool outputs), and never let an un-cross-checked confidence threshold be the only gate between extraction and output.
- Self-correction does not fix bad source documents, does not detect missing semantic context, and does not eliminate the need for HITL. It reduces HITL volume, sometimes by an order of magnitude. It does not replace it.

5.1 The most under-discussed paper of 2026

Sometime in early 2026, an unflashy short paper titled “*Tiny Silent Hallucinations in Agentic AI*” circulated on OpenReview [19]. The paper studied a class of failure mode in agentic systems where the model produced an output that was confidently wrong, and the surrounding loop — the reflection step, the confidence check, the schema validator — failed to flag it. The errors were small (a transposed digit, a misread date, a single field misclassified) but they were silent, in the precise sense that the system gave no indication anything had gone wrong. They shipped.

The paper got modest attention at publication and considerably more by mid-2026 as production teams started seeing the pattern in their own deployments. The finding is unsettling because it punctures the most reassuring story that agentic OCR vendors tell about their systems: that the loop catches errors the model misses. It often does. It also, with a surprising frequency, does not.

This chapter is about why. Self-correction is a real capability that real systems implement, and the field has made measurable progress on it. But the marketing has run far ahead of the engineering, and the production deployments that survive contact with reality are the ones whose teams understand what self-correction does and — equally important — what it does not do.

5.2 Three subsystems, not one capability

When a vendor says “the system self-corrects,” they could be describing any of three operationally distinct things. They are usually describing one of the three and implying the other two. The first move in evaluating a self-correction story is to insist on the disambiguation.

Detection — knowing that an extraction is wrong. This requires either (a) an external oracle that can verify the extraction independently or (b) a confidence signal that is genuinely calibrated against accuracy on the system’s actual production distribution. Detection is by far the hardest of the three subsystems to do well. It is also the one most production failures are downstream of.

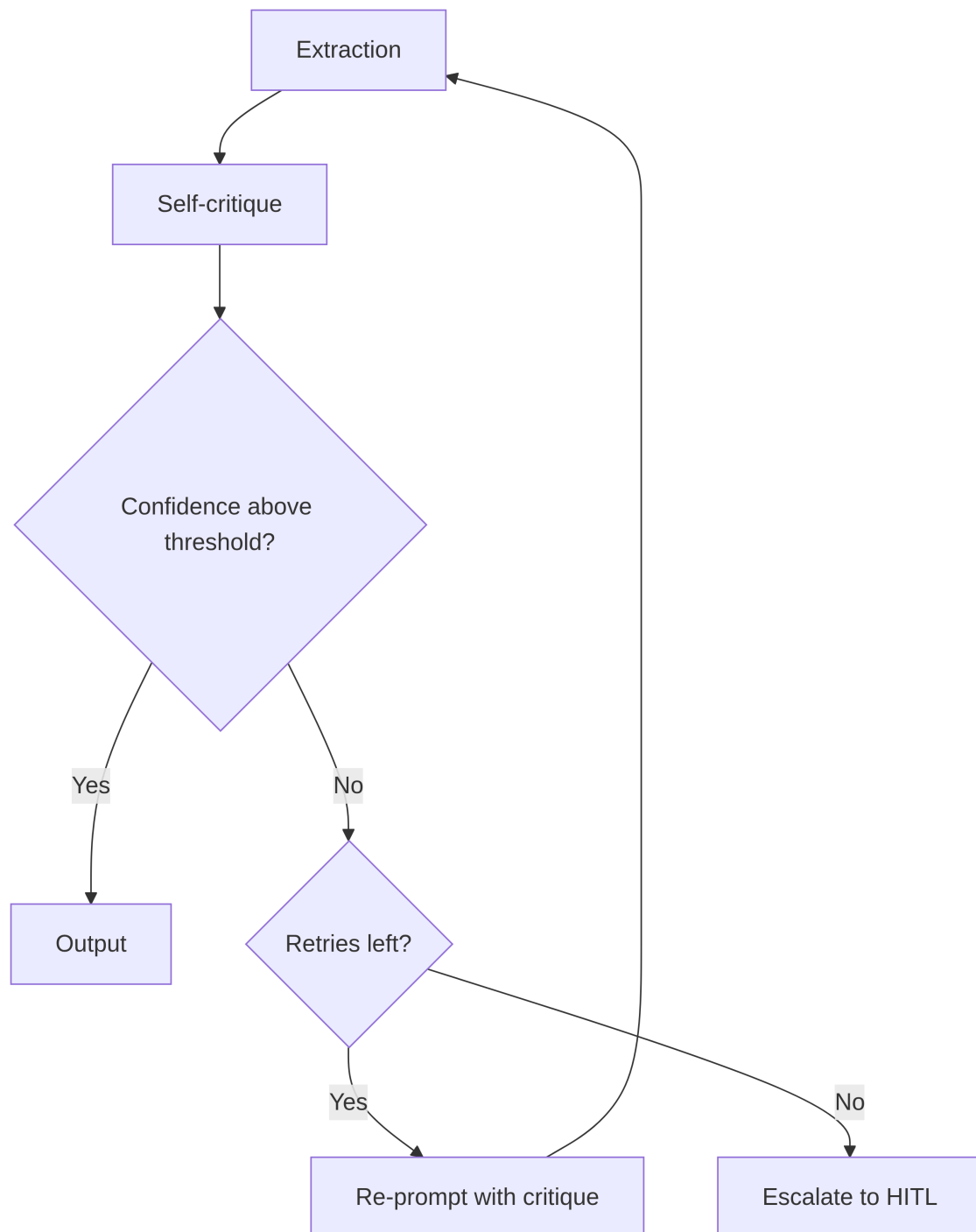
Repair — fixing a detected error. This is typically a re-extraction with revised instructions (“the date you produced is invalid; re-examine the date field”), but it can also be a routing decision (send this region to a specialized tool the system did not initially use) or a contextual lookup (resolve the entity against an external KB before deciding the extraction is wrong). Repair is the easiest of the three. Most vendors do it reasonably well, because the LLM behind it is good at responding to specific revision prompts.

Escalation — failing safely when detection or repair cannot close the loop. Escalation routes the document to a human reviewer, attaches the relevant context (the failing field, the candidate extraction, the source region highlighted), and pre-populates the reviewer’s interface for an efficient decision. Escalation is, in some ways, the most engineering-intensive of the three because it requires a HITL queue, a reviewer-facing UI, a feedback loop back into the system, and operational discipline around the queue’s response time.

A system that does all three well is a real agentic OCR system. A system that does two of three has predictable, named failure modes (Chapter 13 returns to these as anti-patterns). A system that does only one — typically repair — is what most vendor demos are showing you, and it is much less useful than it looks.

5.3 The reflection loop, mechanically

The canonical reflection loop, simplified to its essential structure, looks like this:



Reflection loop with confidence gating

A few things about this diagram are worth holding onto, because the failure modes hide in the details.

The self-critique step. What the model is critiquing here is its own output. That is the source of most detection failures: a model that produced a wrong answer with high confidence is often *also* the model that critiques that answer favorably. The loop has no external oracle in the simple version. Adding one — a schema validator, a secondary model with a different prompt, a tool output — is what turns a fragile loop into a robust one. Production-grade implementations almost always wire in at least one external oracle. Demo-grade implementations often do not.

The confidence threshold. This is a number. Typically somewhere in the 0.6 to 0.9 range for the model-emitted confidence scores. The right value is the value that, on your golden set, maximizes a defensible utility function — usually some combination of accuracy and HITL escalation rate. Choosing it by intuition is the sort of thing that looks fine in development and quietly destroys production. The threshold needs to be calibrated, and calibration on out-of-distribution documents (i.e., the documents

your system will see in production that are unlike anything it saw in development) is the live research problem.

The retry budget. Usually two to four. The lower bound matters because a single retry is often not enough to converge on the right answer; the upper bound matters because retries cost real money and a recalcitrant document can burn through retries without ever improving. Some sophisticated systems vary the retry budget by document complexity or by extraction confidence — burning more retries on the documents where the marginal probability of getting it right is high, and giving up faster on documents the model clearly cannot handle.

The escalation path. When the loop gives up, what happens? In production, escalation usually means “added to a HITL queue with the failing field highlighted.” In demos, escalation often means “returns an error to the API caller” — which is operationally useless because there is nobody on the other end of the API to do something about it.

5.4 Confidence gating: real versus theatre

Confidence scores deserve their own section because so much of the field’s self-correction story rides on them, and so much of that story is wrong.

A confidence score, in agentic OCR, is a number between zero and one that the model emits alongside an extraction. The intuitive story is that it represents the probability that the extraction is correct. The actual reality is more complicated.

For a *well-calibrated* model on its *training distribution*, the confidence score does approximately what the intuitive story claims. A confidence of 0.95 on an extraction means that, across many similar extractions, roughly 95% of them are correct. This is a useful signal. The threshold is meaningful. The reflection loop works.

For an *uncalibrated* model on an *out-of-distribution* document — which is what most production deployments see most of the time — the confidence score is at best an ordinal signal. The model is more confident on extraction A than on extraction B, but neither number means what the intuitive story claims. A confidence of 0.95 on an OOD document can be accompanied by an accuracy as low as 60%, and a confidence of 0.6 on an in-distribution document can be accompanied by an accuracy of 90%. The number is informative; the number is not a probability.

This matters for self-correction because the standard reflection loop gates on confidence as if it were a probability. If your gating logic is `if confidence < 0.85 then retry`, you are assuming the model knows when it does not know. On in-distribution data with calibrated models, this assumption is approximately true. On out-of-distribution data with realistic models, this assumption is wrong, and the gate fails closed (high-confidence wrong answers ship) much more often than fail open.

The Databricks “\$10,000 → \$3,000” hallucination is a textbook example [1]. The model misread the value. The model was confident in its misreading. The reflection loop, gating on confidence, approved the wrong answer. The error shipped. No external oracle was in the loop to verify against the actual digit on the page.

The fix is not “improve the confidence scores.” The fix is to stop treating confidence as the only gate. Real production self-correction wires in **external oracles**:

- **Schema validators.** Does the output parse? Are dates in valid ranges? Are required fields present? Are enums populated with allowed values? Schema validation is cheap, deterministic, and catches a surprising fraction of high-confidence wrong answers — particularly the kind that hallucinate plausible-looking values.
- **Secondary models.** A second VLM with a different prompt produces a candidate extraction. If the two extractions disagree, escalate or re-prompt with the disagreement as context. This is expensive (two model calls instead of one) but catches errors that single-model self-critique structurally cannot.
- **Tool cross-checks.** A layout detector says the date field lives in this bounding box. The VLM extracted the date from a different region. Disagreement → flag.
- **External KB lookups.** The extracted ICD code is not in the ICD-10 codebook. The extracted vendor name does not match any vendor in the system’s vendor list. These are deterministic checks that any agentic system handling structured domains should run by default.

Each external oracle has costs (compute, latency, integration effort), and a production system rarely uses all of them on every document. The healthy pattern is to use them adaptively: schema validation always (it is nearly free), KB lookups when the extracted entity falls in a domain the system knows about, secondary models when the primary extraction is below some confidence threshold or when the document is flagged as high-stakes.

5.5 A failure taxonomy

The most useful tool for evaluating a self-correction implementation — your own or a vendor's — is a taxonomy that names the specific failure modes so you can ask whether the system handles each one.

Detection failures are failures of subsystem one — the system did not notice the error.

- *Silent hallucination.* The model produces a wrong answer with high confidence. The loop sees high confidence, accepts the output, ships the error. The most common production failure mode. Mitigations: external oracles, secondary-model cross-checking, calibrated confidence thresholds tuned on a representative golden set.
- *Confidence miscalibration.* The model produces a borderline answer but reports either much higher or much lower confidence than the accuracy data would justify. Mitigations: empirical calibration on a golden set, treating confidence as ordinal rather than absolute.
- *Out-of-distribution blindness.* The document is sufficiently different from anything in the model's training distribution that the model has no real basis for opinion, but emits one anyway. Mitigations: OOD detection as a pre-extraction step; route OOD documents directly to HITL.

Repair failures are failures of subsystem two — the system noticed the error but could not fix it.

- *Repair thrash.* Every retry produces a different wrong answer. The model is not converging. Mitigations: cap retry budget, vary prompt strategy across retries (different specialized tools, different VLM, different prompt template), recognize thrash patterns and escalate early.
- *Reflection drift.* The model's own critique drifts further from truth across iterations. Most often happens when the critique step is run by the same model that produced the original extraction, with no external grounding. Mitigations: external oracle in the critique step; rotate critic model.
- *Schema rigidity overcorrection.* The system tries to coerce an extraction into a schema that does not fit the document and produces a series of progressively worse answers. Mitigations: schema flexibility (accept fields the schema does not anticipate), and recognize when the document is structurally incompatible with the target schema.

Escalation failures are failures of subsystem three — the system escalated, but escalation did not solve the problem.

- *No HITL path.* The system raises an error or returns a “could not extract” signal, but there is no human reviewer queue downstream. The error becomes a customer complaint. Mitigations: build the queue; do not deploy without it.
- *HITL overload.* The system escalates too aggressively (often because confidence thresholds are too strict or because production traffic is OOD relative to development) and floods the reviewer queue. Mitigations: monitor escalation rate as a production metric; tune thresholds; investigate whether OOD documents have a different routing path than in-distribution documents.
- *HITL underload (silent acceptance).* The opposite failure: the system escalates almost nothing because confidence is structurally high on OOD data. Mitigations: track HITL acceptance rate (the rate at which reviewers correct vs. accept escalated extractions); a healthy system has escalation rate of 1–10% and HITL acceptance rate (i.e. reviewer agrees the system got it right) somewhere between 30% and 70%. Extremes in either direction suggest a broken loop.

The taxonomy is not exhaustive — the OpenReview paper and several follow-ups have catalogued additional failure modes in narrower contexts — but it covers the cases that most production teams will encounter. Run it as a checklist against a vendor's self-correction story. The vendors that can answer all nine specific mitigations have done the engineering. The vendors that hand-wave through the categories have not.

5.6 What self-correction does not do

The capability is real; the marketing claims around it are often larger than the capability. A few honest delineations:

Self-correction does not fix bad source documents. If the original page is illegible, smeared, partially redacted, or upside-down at a 30-degree angle the deskewer cannot handle, no amount of reflection will recover the underlying information. The model can detect that something is wrong (sometimes) but cannot conjure correct values from a document that does not contain them. Garbage in, garbage out — with more API calls.

Self-correction does not detect missing semantic context. An invoice with a missing required field is correctly extracted as “this field is missing.” Whether the field *should* have been present is a question about the document’s intended structure, which the model has no way to answer reliably. Production systems handle this with schema validation that flags missing required fields, plus HITL routing for the flagged cases. The model is not the right tool.

Self-correction does not provide compliance certification. It can support compliance work — grounded outputs, audit logs, escalation trails — but it does not produce compliance. A regulated workflow still needs a compliance review by humans who understand the regulatory regime. Vendors who sell self-correction as if it removes the need for compliance review are selling something they cannot deliver.

Self-correction does not eliminate HITL. It reduces the volume of HITL, often dramatically. Production deployments that pre-agentic ran 30–50% manual review can land at 5–10% escalation rates after a careful Pattern B or C deployment. That is transformative ROI. It is not zero. Plan for HITL forever. The vendors who promise HITL elimination produce the deployments that fail most spectacularly when they meet real production traffic.

5.7 What good self-correction looks like in production

A healthy 2026 agentic OCR deployment exhibits these properties simultaneously:

- **Escalation rate between 1% and 10%** of documents in steady state.
- **HITL reviewer agreement rate around 30–70%** — the reviewers correct the system often enough that escalation is doing useful work, but agree often enough that the system is not just dumping documents onto humans.
- **Confidence calibration within 5–10 percentage points** of accuracy across the production distribution, verified quarterly on a refreshed golden set.
- **Retry budgets observed (and respected) per document type**, with thrash detection that gives up faster on documents the model cannot handle.
- **External oracles wired in** — at minimum schema validation; preferably also entity lookups and (for high-stakes) secondary-model cross-checking.
- **Drift monitoring** that catches the inevitable production distribution drift before it becomes a customer-visible incident.

No deployment achieves all six on day one. Most achieve them iteratively over the first six to twelve months in production. Teams that treat self-correction as something the vendor sold them, rather than as a discipline they own, do not achieve them ever.

💡 Named takes

Self-correction is not “the model will figure it out.” It is three concrete subsystems — detection, repair, escalation. If your vendor talks about it as one thing, they have one of the three working.

A reflection loop with no out-of-loop ground truth is a model becoming more confident in its first answer. Call it what it is.

Confidence scores from agentic OCR systems are an opinion the model has about its own work. Treat them as you would any other unverified opinion.

6. The Shift-Left Thesis

i Executive summary

- The cleanest 2026 evidence for the shift-left thesis: Databricks reports a **16% average performance gain across every agent framework they tested**, simply by inserting their `ai_parse_document` upstream of the agent’s reasoning step — with no change to the reasoning layer.
- “Shift-left” means moving structural and semantic understanding earlier — into parsing — rather than after parsing. The result is *not* lower cost per page; the result is lower cost per **correct extraction with provenance**, which is the denominator that actually predicts your monthly bill.
- Complexity does not disappear under shift-left. It moves from downstream Python scripts into eval infrastructure, HITL workflows, vendor management, and observability. Anyone who tells you the engineering goes away has not deployed this in production.
- Shift-left is not a thesis about model size. PP-StructureV3 at under 100M parameters matches Gemini-2.5-Pro on OmniDocBench. The principle is *where* the understanding lives, not how big the model is.

6.1 The cleanest number in the 2026 literature

Databricks published an essay in 2026 titled “*Why Frontier Agents Can’t Read Documents — and How We’re Fixing It*” [1]. Most of the essay is a sales pitch for their Document Intelligence product, and most enterprise blog posts can be safely skimmed. This one is worth reading carefully for one number.

The number is **16%**. Across every agent framework Databricks tested — different reasoning models, different orchestration patterns, different downstream tasks — inserting their `ai_parse_document` step upstream of the agent’s reasoning produced an average 16% performance gain. *They did not change the reasoning model.* They did not change the agent’s tools, the prompt, or the workflow. They changed only what the document looked like by the time the reasoning model saw it.

This is the cleanest single piece of evidence in 2026 for the shift-left thesis, and it is worth unpacking what it actually shows. It does not show that agentic OCR is better than classic OCR; that was already known. It shows something stronger: that the *ceiling on agent quality* is set by document understanding, not by reasoning ability. A 16% gain across the board, with no change to the reasoner, means that 16% of the apparent reasoning failures were not reasoning failures at all. They were document-reading failures, masquerading as reasoning failures. The agent’s downstream behavior looked confused, hallucinatory, or inconsistent — but the cause was upstream, in what the agent was being asked to reason over.

The implication for procurement and architecture is direct: **if you have an agent that is struggling, your first investigation should be in the parsing layer, not the reasoning layer.** This sounds obvious in retrospect and is routinely ignored in practice. Teams that observe poor agent behavior reach for better reasoning models, more elaborate prompts, more sophisticated tool chains. Sometimes those are the right fixes. Often the underlying issue is that the agent is reasoning correctly over wrong inputs.

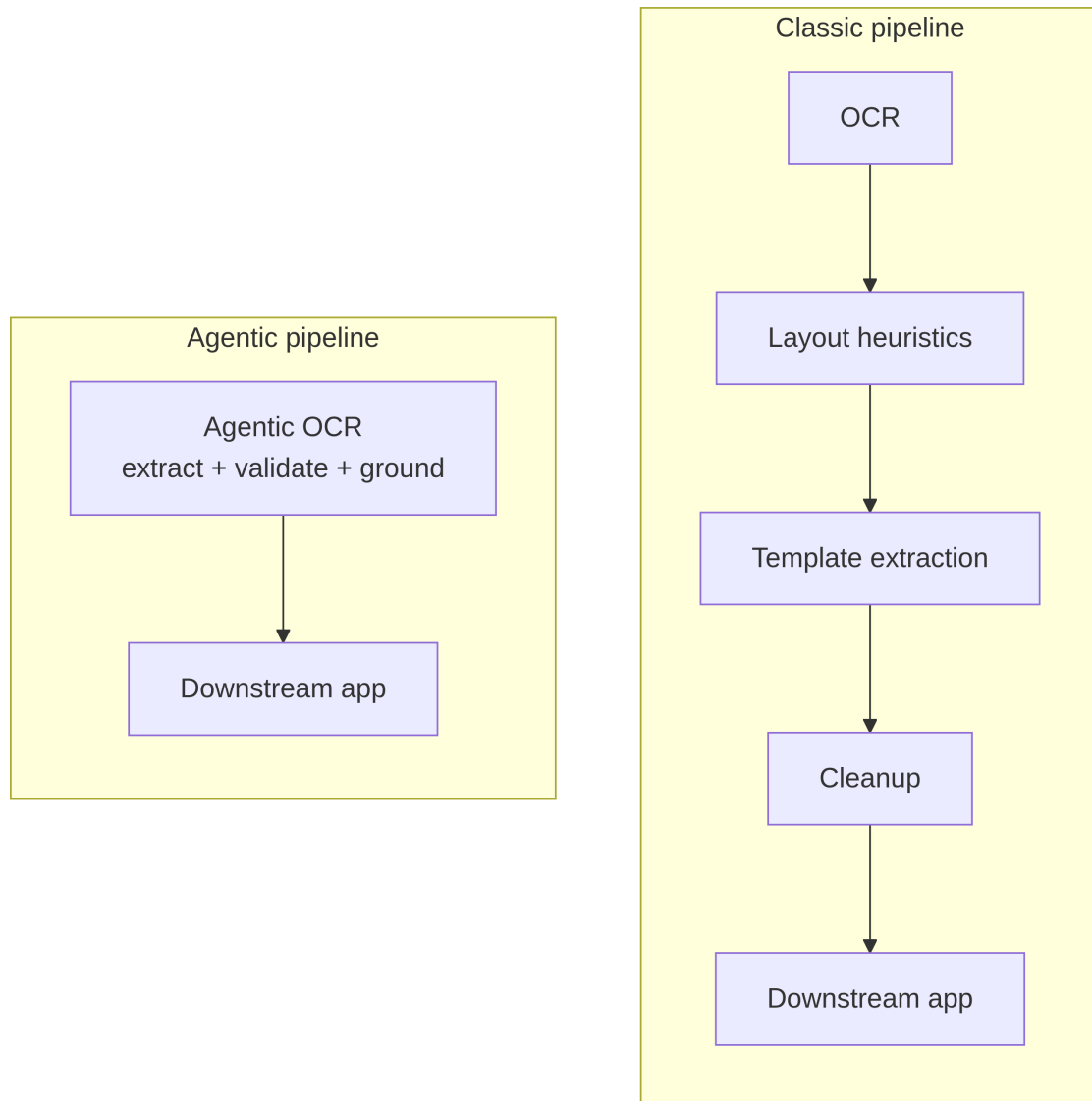
That, in one paragraph, is the shift-left thesis. The rest of this chapter unpacks what it means architecturally, what it means economically, and where it does and does not apply.

6.2 What “shift left” actually means

The metaphor comes from software engineering, where “shift left” historically referred to moving testing and validation earlier in the development cycle — from production back into CI, from CI back into the developer’s editor. The shared structure is moving expensive work to a point where mistakes are cheaper to catch.

In document processing, the work being shifted is **structural and semantic understanding**. The two ends of the pipeline that the shift moves work between:

- **Classic pipeline (pre-shift):** OCR converts the document to text. Layout heuristics try to recover structure post-hoc. Templates pull out specific fields. Cleanup logic, often hand-written, fixes the inevitable errors. Downstream applications receive structured-ish data and apply business logic.
- **Agentic pipeline (post-shift):** Agentic OCR emits structured, grounded, validated output ready for direct downstream use. Downstream applications receive structured data with provenance and apply business logic. No layout heuristics, no templates, no cleanup logic in between.



Where the work moves under the shift-left thesis

The diagram is deceptively neat — it makes the shift look like five boxes collapsed into one. The reality is more like: the five boxes’ worth of functionality has been moved into one *larger* box, and the surrounding scaffolding (eval, HITL, observability) has grown to handle the consequences.

6.3 The economic argument, with numbers

The economic argument is where the shift-left thesis lives or dies, and it is worth running the math carefully.

Per-page parsing cost goes **up** under shift-left. There is no honest version of the thesis that hides this. Classic OCR runs at fractions of a cent per page; agentic OCR runs at one to five cents per page, sometimes more. A team that switches from Textract Basic to LlamaParse Agentic is paying roughly 8× more per page on parsing. If that were the whole calculation, the thesis would be wrong.

The whole calculation includes downstream costs. In the classic-pipeline world, the per-page parsing cost is the smallest line item; the engineering cost of building, maintaining, and debugging the layout-

heuristics, template, and cleanup layers is much larger, and the cost of downstream errors (rework, customer complaints, missed extractions that lead to bad decisions) is larger still. Studies of pre-2024 IDP deployments routinely find that **parsing API spend is under 10% of the total cost of running the pipeline**. The other 90% is engineering, ops, error handling, and rework.

The shift-left thesis says: spend more on parsing, save more than the extra spend on the downstream stack. The Databricks number — “5–7× lower cost than comparable VLM-only pipelines” for equivalent accuracy [1] — is one data point in that direction. A customer cited in the same essay (Loopback Analytics, clinical-note extraction) reports about 90% lower cost for the same downstream entity-extraction outcome. These are vendor-reported figures, with the usual caveats, but the direction is consistent across vendors and customer case studies.

A worked example. Consider a healthcare claims-processing pipeline at 100,000 pages per month.

Component	Classic IDP	Pattern B agentic	Pattern C hybrid
Parsing (per page)	\$0.005	\$0.015	\$0.020
Monthly parsing	\$500	\$1,500	\$2,000
Error rate	12%	3%	1.5%
Errors / month	12,000	3,000	1,500
Downstream rework (\$5/error)	\$60,000	\$15,000	\$7,500
HITL escalation (% × \$3/page)	25% × \$3 = \$75,000	8% × \$3 = \$24,000	5% × \$3 = \$15,000
Total monthly	\$135,500	\$40,500	\$24,500
Cost per correct extraction	\$1.54	\$0.42	\$0.25

The numbers in this table are illustrative — they are derived from the public data in Chapter 7. and Chapter 9. but the specific values vary by vendor, vertical, and document mix. What is robust across realistic parameter choices is the *shape* of the answer: the parsing column changes by 4×, the total monthly cost changes by 5×, and the **cost per correct extraction** — the denominator that should actually be governing the decision — changes by 6×. Pattern C costs four times as much per page as classic IDP and produces output that is six times cheaper per correct extraction. That is the shift-left thesis in one table.

Three caveats are worth flagging before the reader takes the table as a universal recommendation.

The error costs matter enormously. The thesis depends on the cost of a downstream error being non-trivial. If your downstream consumer of the parsed output is a search index that nobody complains about when results are imperfect, the rework column is close to zero and the math flips. The shift-left thesis is strongest for high-error-cost workflows (healthcare, legal, finance, KYC) and weakest for low-error-cost workflows (bulk archiving, analytics ingestion of large unstructured corpora).

The HITL rate is a real engineering parameter. The table assumes 25% / 8% / 5% escalation rates for the three options, which are roughly consistent with vendor-reported numbers, but every team’s actual rates depend on document mix, schema strictness, and reviewer-acceptance thresholds. A team that runs Pattern C at a 20% escalation rate has effectively chosen the worst of both worlds.

Engineering and infrastructure are not in this table. Building and operating Pattern C requires more engineering effort than building and operating classic IDP. The table assumes both are deployed and operating; it does not include the up-front cost of getting there. For low-volume pipelines (under 10,000 pages / month), the up-front engineering cost can dominate, and the thesis can fail to apply on TCO grounds even though it is structurally correct.

6.4 Where complexity actually moves (not disappears)

The honest version of the shift-left thesis is that engineering complexity does not disappear — it *moves*. Specifically, it moves out of three places and into three other places.

Engineering complexity moves **out of**:

- **Hand-written layout heuristics.** The Python script that figures out which lines on the page constitute the line-items table, which row is the header, where the totals live. This logic was the bulk of pre-2024 IDP codebases and is essentially gone in the post-shift world.

- **Per-vendor templates.** The thousand templates the classic IDP era required, each broken individually by every layout change. These are gone.
- **Downstream cleanup logic.** The if-this-vendor-then-rewrite-that-field code that handled the long tail of OCR errors. Mostly gone, though not entirely.

Engineering complexity moves **into**:

- **Eval infrastructure.** Golden sets, regression testing, vendor version pinning, drift monitoring. This is now first-class engineering work that pre-shift teams did not need to do at all. Chapter 11 unpacks why.
- **HITL workflow design and operations.** The reviewer queue, the reviewer-facing UI, the feedback loop back into the system, the operational discipline around queue response times and reviewer agreement rates. These existed in classic IDP but were a small ops function; under shift-left they are a non-trivial engineering effort.
- **Observability for agentic systems.** Tracing extraction lineage, monitoring escalation rates as a primary production metric, detecting silent hallucinations before they become customer-visible incidents. This category is new in 2026 and is one of the loci of competitive differentiation among vendors (Chapter 15 returns to it).

The net is that the *amount* of engineering work has not gone down. The *shape* has changed. The new categories of work compound across document types and verticals — an investment in eval infrastructure pays off across every workload your team handles, where the equivalent classic-pipeline investment in templates paid off only on one vendor’s invoice format. This is the structural reason the thesis is correct on TCO even when per-page cost goes up.

6.5 Honest counter-arguments

The thesis is not universal. It is worth being explicit about where it does not apply.

Per-page cost can dominate at very high volume. A team processing 10 million pages per month of commodity invoices has an annual parsing bill of \$1.2M at \$0.01/page and \$5M+ at \$0.05/page. At that volume, the cost difference becomes large enough that the downstream savings may not cover it — particularly if the downstream consumer is tolerant of moderate error rates. Bulk archive indexing, large-scale analytics ingestion, and commodity invoice processing at the upper end of the volume curve often tip back toward classic IDP plus a thin cleanup layer.

Latency is real. Multi-pass reflection adds wall-clock time. A single-page document that completes in 200ms on Pattern A may take 2–5s on Pattern B and 3–7s on Pattern C. For batch workloads, latency is invisible. For interactive use cases — a chat interface over a document, a real-time form-fill assistant — the latency budget can rule out the shift-left architectures entirely.

Vendor lock-in is worse. A team that has built around Reducto, or LlamaParse, or Landing AI ADE, has more switching cost than a team that built around Tesseract plus a Python template engine. The dependency is sharper. The version drift is more consequential (a vendor’s behavior on edge cases changes between major versions in ways that broke prior agreements). Some teams legitimately decide the lock-in is unacceptable and accept the higher engineering burden of an open-source path.

The 16% has caveats. It is measured on Databricks’ own OfficeQA benchmark, which is new and Databricks-published. The number is directional, not authoritative. The shift-left thesis is robust in the general case, but no single number should be taken as the proof.

6.6 Shift-left is not a thesis about model size

The single sharpest counter to the popular reading of the shift-left thesis is the PP-StructureV3 result.

PP-StructureV3 is a layout-and-structure model in the PaddleOCR family. It runs at under 100 million parameters. On OmniDocBench, it achieves an edit distance of 0.145 on English documents and 0.206 on Chinese documents [17], which is roughly at parity with Gemini-2.5-Pro — a model with many orders of magnitude more parameters and many orders of magnitude more compute cost per inference.

The model is small. The model is specialized. The model is shifted *all the way left* — it does structural understanding at the parsing stage, with layout-aware architecture rather than scale.

The reading “shift-left means use a big VLM” misses the lesson. Shift-left is a thesis about where the structural understanding lives, not about how much compute you throw at it. A small layout-aware model that emits clean structural output is shifting left in the sense the thesis cares about. A 70B-parameter general VLM called twice in a reflection loop is doing something different — also useful, also defensible, but not what is happening when PP-StructureV3 beats Gemini-2.5-Pro on the relevant metric.

This is one of the reasons the open-source landscape in 2026 (Chapter 8) is so interesting. The open-source frontier is full of small, specialized, layout-aware models that are out-performing large general VLMs on extraction tasks while being orders of magnitude cheaper to run. That is the shift-left thesis playing out at the model layer rather than the architecture layer.

6.7 How to act on this thesis

If you are a buyer:

- Measure your **downstream error cost** before you measure parsing accuracy. The thesis is only meaningful if downstream errors are expensive.
- Compute **cost per correct extraction with provenance**, not cost per page. Cost per page makes Pattern A look great. Cost per correct extraction often does not.
- Insist on **grounded outputs**. Without grounding, the thesis's downstream savings (in audit and HITL efficiency) cannot be realized.

If you are an implementer:

- Treat **eval infrastructure as the primary investment**, not the parsing layer. The classic-pipeline engineering effort goes away; the eval-infrastructure engineering effort takes its place.
- Build **HITL workflows as a first-class subsystem**, not a side queue. Production agentic OCR's economic case depends on HITL being efficient.
- Plan for **observability** that is shaped around agentic workflows — extraction lineage, escalation rates, calibration tracking — rather than around web-app metrics.

If you are a skeptic:

- Look at the **shape** of the cost change, not the per-page numbers. Pattern A's cost-per-page is hard to beat. Pattern C's cost-per-correct-extraction is hard to beat. They are different things.
- Check whether your workload has the **error-cost asymmetry** the thesis requires. If a missed extraction is genuinely cheap, the thesis does not apply to you, and the right answer is to stick with classic IDP.

💡 Named takes

Shift-left is a thesis about where structural understanding lives. It is not a thesis about model size. PP-StructureV3 at 100M params is shifting left harder than a 70B model called twice.

If you can articulate your ROI only in cost-per-page, you have not understood the shift-left thesis. The right denominator is cost-per-correct-extraction-with-provenance.

The 16% across-the-board agent gain is one number. The point of the number is not the number; it is that document reading is the ceiling on agent quality, and reading is mostly solved.

III

Commercial Players

Part III – The 2026 Landscape

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7. Commercial Players

i Executive summary

- The commercial landscape divides into four tiers in 2026: **AI-native specialists** (LlamaParse, Reducto, Landing AI ADE, Mistral Document AI), **hyperscaler APIs** (Google Document AI, AWS Textract, Azure Document Intelligence), **classic IDP retrofitted** (ABBYY, Hyperscience, UiPath, Klippa, Rossum) and AI-native commercial IDP (Nanonets, Docsumo), and the newer **platform-integrated** tier (Databricks Document Intelligence).
- AI-native specialists lead on benchmark scores and pricing flexibility; hyperscaler APIs lead on procurement maturity and broad compliance certification; classic IDP leads on HITL and audit posture; the platform tier is a new strategic shape that makes the shift-left thesis institutionally credible.
- The right vendor is rarely the one with the highest benchmark score. The right vendor is the one that minimizes your cost-per-correct-extraction-with-provenance on **your** documents, evaluated against **your** golden set.
- Pricing across the tiers spans roughly two orders of magnitude per page. The cheapest credible parser (Mistral OCR 3 Batch at \$0.001/page) and the most expensive tier of premium agentic systems (LlamaParse Agentic Plus at \$0.056/page; LandingAI ADE Visionary tier) bracket the cost question.

7.1 How to read this chapter

This chapter is not a buyer's guide that recommends a vendor. It is a map. Buyers' guides date badly in this category — every quarter at least one vendor releases a major update that changes the ranking — and the goal here is to give you a stable mental model that survives the next twelve months of releases. The vendor names will move. The categories should hold.

Three angles for reading the chapter. If you are a procurement-led reader, focus on the pricing matrix and the per-vendor cost notes. If you are an architecture-led reader, focus on each vendor's pattern (A, B, or C from Chapter 4.) and what they have engineered well around the model layer. If you are a strategy-led reader, focus on the tier structure — the move from “buy a vendor” to “buy a platform” is the strategic shift this chapter is really about.

7.2 The four tiers

The 2026 commercial landscape resolves into four tiers, each with characteristic strengths, characteristic blind spots, and characteristic pricing.

AI-native specialists. Companies founded specifically to do agentic OCR. LlamaParse (part of LlamaIndex), Reducto, Landing AI's ADE product, Mistral Document AI. These vendors lead the public benchmarks, ship the most aggressive product cadences, and have the most flexible pricing — but their procurement processes are less mature, and their classic-enterprise compliance certifications often lag the hyperscalers and the classic IDP vendors. Most of them are series-A through series-C startups in 2026; some will not survive consolidation.

Hyperscaler APIs. Google Document AI, AWS Textract, Azure Document Intelligence. Mature, broadly compliance-certified, integrated with the rest of the buyer's cloud stack. Architecturally template-leaning and lagging the agentic frontier on adversarial workloads (especially tables) by a measurable margin. The default pick for buyers who care more about procurement risk than benchmark score, which is a larger fraction of the market than the AI-native specialists like to admit.

Classic IDP, retrofitted. ABBYY, Hyperscience, UiPath Document Understanding, Klippa, Rossum, plus the newer AI-native commercial IDP entrants Nanonets and Docsumo. Strongest HITL infrastructure of any tier; strongest compliance posture (SOC 2, HIPAA, GDPR built into the core product, not retrofitted

at sales time); weakest agentic story in the strict five-attribute sense, with notable exceptions. The right tier if HITL is the load-bearing requirement and the agentic loop is secondary.

Platform-integrated agentic OCR. Databricks Document Intelligence is the canonical 2026 example. Not a pure-play OCR vendor; an integrated set of AI functions inside a larger data platform. The strategic significance is bigger than its current market share: a hyperscaler-adjacent platform saying that document reading is the gating constraint on agent quality is air cover for the shift-left thesis (Chapter 6) at the institutional level. Snowflake, Microsoft Fabric, and the major LLM-platform vendors are all building or buying into this tier.

7.3 AI-native specialists

7.3.1 LlamaParse (LlamaIndex)

LlamaParse is the front-runner on the most-cited 2026 benchmark, ParseBench, with their Agentic tier scoring 84.9% overall at approximately \$0.012 per page [4]. The product offers four tiers — Fast (1 credit/page), Cost-Effective (3), Agentic (10), and Agentic Plus (45) — at a conversion rate of 1,000 credits to \$1.25 and 10,000 free credits per month for new accounts.

Architecturally LlamaParse runs Pattern B (agentic loop with reflection) at the Agentic and Agentic Plus tiers, and something closer to Pattern A with light validation at the Fast and Cost-Effective tiers. Strengths: messy enterprise documents, financial filings, scientific papers, and integration with the rest of the LlamaIndex ecosystem (which is itself winning meaningful market share in the agent-orchestration layer). Weaknesses: the system is at its best on text-heavy documents and lags Pattern C specialists on adversarial tables; pricing at the top tier is among the highest in the AI-native category.

The honest framing: LlamaParse is the right pick when you want a single vendor for parsing-plus-orchestration, when your downstream system is built on or compatible with LlamaIndex, and when your documents are text-heavy enough that the Agentic tier (not Agentic Plus) is sufficient. Caveat the ParseBench number with the usual vendor-published warning — the benchmark is LlamaIndex-maintained, and the rule selection inevitably shapes the leaderboard.

7.3.2 Reducto

Reducto is the Pattern C standard-bearer in the commercial landscape. Their hybrid CV-plus-VLM architecture is the most explicit implementation of “specialization beats generality” in the commercial category, and their result on RD-TableBench (~0.90 cell-alignment similarity, ~20 percentage points ahead of the hyperscaler APIs) is the cleanest data point [6].

Strengths: complex tables, scanned PDFs with mixed content, handwritten regions, and any workload where layout matters more than text fluency. Their published work on multi-pass agentic OCR [15] is the most technically detailed public account from any commercial vendor in 2026 — worth reading regardless of whether you intend to buy.

Pricing runs \$0.005–\$0.020 per page. SOC 2 and HIPAA available. The honest caveat: RD-TableBench is Reducto-published. Their lead on it is real, but you should cross-validate on a non-Reducto benchmark before signing — OmniDocBench or your own golden set are both useful.

7.3.3 Landing AI — Agentic Document Extraction (ADE)

ADE moved to credit-based monthly subscriptions in 2025–2026, ranging from \$39/month for 120,000 credits up to \$2,199/month for 900,000 pages [18]. The credit formula — $\text{ceil}((\text{input_chars}/5,000) + (\text{output_chars}/1,000), 0.1)$ — makes the actual per-page cost dependent on document size, which is honest if confusing.

ADE’s 2026 feature set is the most ontology-rich among the AI-native specialists. The product can identify attestations, ID cards, logos, barcodes, and QR codes as first-class entities — not just as “image regions” that downstream code has to interpret. Agentic table captioning, RBAC, HIPAA on Team and Enterprise plans, and one-click Zero Data Retention put ADE in a strong position for regulated workloads.

ADE is the right pick when ontology richness matters (insurance underwriting packets, healthcare records, KYC document review), when HIPAA or ZDR are non-negotiable, and when the documents have a complex enough structure that the broad entity classification pays off. It is not the cheapest tier, and the credit formula needs a few sample runs against your real documents to predict monthly cost confidently.

7.3.4 Mistral Document AI (OCR 3)

Mistral OCR 3 sits at the price floor of credible agentic-grade parsing: **\$2 per 1,000 pages** standard, **\$1 per 1,000 pages** via Batch API, with throughput of up to 2,000 pages/minute per node [20]. The model identifier `mistral-ocr-2512` is callable via API, and self-hosted deployment is available for organizations with strict data-sovereignty requirements.

The internal reported metric is a 74% win rate over Mistral OCR 2 across a customer-document workflow set, particularly strong on forms, handwritten content, and table-heavy documents. Output is Markdown, with tables reconstructed using HTML tags including `rowspan` and `colspan` — usable for downstream chunking and embedding without further parsing.

OCR 3 is the right pick when cost-per-page is the binding constraint, when throughput requirements push the hyperscalers' price-tier into the same range as a specialist, and when self-hosting is required for compliance reasons that exclude the SaaS-only specialists. Architecturally it is closer to Pattern B than to Pattern C — the agentic loop is real but the specialized-tool routing is less elaborate than what Reducto offers.

7.4 Hyperscaler APIs

The three hyperscaler ecosystems — Google Document AI, AWS Textract (plus Bedrock Data Automation), Azure Document Intelligence (plus Azure AI Vision Read) — are mature, integrated, and significantly more nuanced than a single “hyperscaler OCR API” framing suggests. Each is really a *family* of sub-products at different price points, with different output shapes, and with different positions on the spectrum from “cheap text OCR” to “agentic-flavored extraction.” Buyers who pick the wrong sub-product within the right hyperscaler are a common 2026 failure mode.

This section walks each ecosystem by sub-product rather than as a monolith, because the procurement-relevant choices live at the sub-product level.

7.4.1 Azure — Document Intelligence and AI Vision Read

Microsoft ships document AI through two distinct services that are commonly confused:

Azure AI Vision Read is a standalone OCR endpoint inside the broader Azure AI Vision service. It returns text with reading order; it does **not** return structured layout or fields. Cost is roughly \$1–\$1.50 per 1,000 pages. The right pick when you only need clean text — accessibility transcription, search indexing, bulk archive digitization — and do not want to pay for structural extraction you will not use.

Azure AI Document Intelligence (formerly Form Recognizer) is the full document AI product, and it splits into four sub-models worth understanding individually:

- **Read** — \$1.50/1k pages. OCR plus reading order plus basic key-value pair extraction. Cheapest tier of Document Intelligence; the right pick when Vision Read is not quite enough but you do not need tables or structural metadata.
- **Layout** — \$10/1k pages. Extracts text, tables, paragraphs, section headings, selection marks (checkboxes), and figure regions with bounding boxes. **This is the sub-model most relevant to agentic OCR pipelines.** It serves the same architectural role as PP-StructureV3 or Granite-Docling in a Pattern C hybrid — a strong layout-detector primitive that downstream stages can route off. Many in-house agentic OCR systems built on Azure use Layout as the parsing layer and add their own reflection loop on top.
- **Prebuilt models** — \$10/1k pages. Pretrained extractors for common document types: invoices, receipts, ID documents, business cards, W-2, 1098, contracts, marriage certificates, mortgage documents, health insurance cards. The right pick when your documents are one of the supported types and the extraction schema matches your downstream consumer's needs.
- **Custom** — \$30/1k pages. Train your own extractor on labeled samples. Template models (free training) for narrow stable layouts; neural models (free for the first 10 hours of training, \$3/hr thereafter) for variable layouts. The custom-model story is the strongest of any hyperscaler — Azure has invested deeply here and it shows.

High-volume commitments drop pricing meaningfully: at 8 million pages per month, the effective rate falls to around \$0.53/1k pages — roughly 50% off list. Free tier covers the first 500 pages per month.

Honest framing: Azure DI in 2026 is the most *complete* hyperscaler offering for buyers who want to compose their own agentic pipeline from primitives. Layout + Prebuilt + Custom + Azure AI Foundry

orchestration gives you most of a Pattern C architecture, with the compliance posture (HIPAA BAA, SOC 2 Type II, FedRAMP High) that classic IDP vendors fought to match.

7.4.2 Google — Document AI and the Layout Parser

Google Document AI is similarly a *family* of processors, each with its own price point:

- **Document OCR** — approximately \$0.65/1k pages for basic OCR. Cheapest tier; text + reading order, no structure.
- **Layout Parser** — \$10/1k pages, and the most strategically interesting Google offering in 2026. It uses **Gemini under the hood** to do layout extraction *plus initial chunking* in a single call. The output is structured for direct downstream use in RAG and agentic pipelines, which makes it the closest hyperscaler equivalent to the AI-native specialists' agentic mode. Worth shortlisting for buyers already committed to GCP who want to skip the "compose your own pipeline" engineering.
- **Form Parser** — \$30/1k pages below 1M/mo, \$20/1k above. Forms plus tables in one call. Older product, still useful.
- **Specialized processors** — pretrained extractors for invoices, US driver licenses, US passports, IRS 1040, W-2, and a dozen other specific document types. Pricing varies; broadly comparable to Form Parser.
- **Custom Document Extractor** — \$30/1k pages below 1M/mo, \$20/1k above. Train your own. Hosting a deployed custom processor adds \$0.05/hour (~\$438/year for always-on).
- **Document AI Workbench** — the orchestration layer that lets you chain processors. Increasingly the entry point Google recommends for new agentic workloads.

New customers get \$300 in free GCP credits, which goes a long way at these unit costs.

Honest framing: Google's Layout Parser is the **hyperscaler product most under-appreciated** by 2026 procurement teams. A team that needs Pattern B parsing and is on GCP can shortlist Layout Parser alongside LlamaParse Agentic and find it competitive on quality at a comparable price point — without the third-party vendor relationship.

7.4.3 AWS — Textract and Bedrock Data Automation

AWS has the messiest sub-product surface, partly because Textract grew incrementally and partly because the newer **Amazon Bedrock Data Automation** product is now the recommended starting point for new IDP workloads, but Textract remains the right answer for many existing pipelines.

Textract sub-features:

- **DetectDocumentText** — approximately \$1.50/1k pages for basic OCR. Text only, no structure.
- **AnalyzeDocument** — feature-stackable, \$15–\$65/1k pages depending on which features you turn on:
 - *Forms* (\$50/1k) — key-value pair extraction.
 - *Tables* (\$15/1k) — table structure extraction.
 - *Queries* (\$15–\$65/1k) — ask questions like "what is the invoice number" and get answers.
 - *Layout* (\$10–\$25/1k) — paragraph, header, footer, and reading order detection. AWS's equivalent to Azure DI Layout, added later and less mature. Stack them: a typical real pipeline uses Forms + Tables + Layout together.
- **AnalyzeID** — \$2.50/1k. Prebuilt extractor for driver licenses and passports.
- **AnalyzeExpense** — \$10/1k. Prebuilt for receipts and invoices.
- **StartLendingAnalysis** — multi-document workflow for mortgage and lending packets. Premium pricing.
- **Amazon Augmented AI (A2I)** integration — the HITL queue layer, well-engineered and the strongest HITL story among hyperscalers.

Amazon Bedrock Data Automation (BDA) — released late 2024, production-ready by 2026 — is AWS's agentic-flavored answer to LlamaParse Agentic and Azure DI Layout. Per-page pricing is **\$0.010/page for parsing plus \$0.040/page for Custom Output** (with an incremental charge if your blueprint defines more than 30 fields). The pricing is flat per document, which makes cost forecasting straightforward. BDA handles classification + extraction through a single multimodal API — and the API itself is multimodal beyond documents (audio, video, image), reflecting AWS's strategic positioning of Bedrock as the broader unstructured-data platform.

AWS's own guidance recommends BDA as the starting point for *new* IDP projects, and Textract for incremental work on existing pipelines. That is the right framing for buyers as well: if you are greenfield, start with BDA; if you have Textract in production already, the migration cost may not justify switching.

Honest framing: AWS in 2026 is the **most fragmented** of the three hyperscalers and also the **most actively evolving**. The cost-effective configurations for an agentic workload involve BDA for parsing + reasoning, with Textract feature-stacks as fallback or supplemental tools. Buyers who pick “AWS” without picking a specific configuration usually end up over-paying on Textract features they did not need.

7.4.4 The general framing across all three

Two patterns hold across hyperscaler ecosystems:

1. **The “Layout” sub-product is the load-bearing piece for agentic workloads.** Azure DI Layout, Google Document AI Layout Parser, AWS Textract AnalyzeDocument (Layout feature). These are the primitives that practitioners actually use as the parsing layer in custom agentic pipelines, including Pattern C hybrid architectures. The basic OCR tiers (Vision Read, Document OCR, DetectDocument-Text) are too thin; the prebuilt/extraction tiers are too narrow.
2. **The agentic story is the newest layer.** Google Layout Parser (Gemini-powered) is the most production-credible hyperscaler-native agentic offering. AWS Bedrock Data Automation is close behind and improving. Azure’s agentic story currently relies on chaining Layout + Custom + Foundry orchestration rather than a single packaged agentic product. Watch this race through 2027 — it is the one closing fastest.

The honest framing of the hyperscaler tier overall: these are the right pick for buyers whose procurement processes require an established cloud-vendor relationship, whose compliance requirements rule out smaller vendors, and whose documents fit within the hyperscalers’ design envelope. Choosing the wrong sub-product within the right hyperscaler is the failure mode this section is calibrated to prevent.

A particular failure mode worth naming: teams that pick a hyperscaler because “the procurement is easier” and then discover six months later that the documents they actually want to process are in the gap between what the hyperscaler does well and what the workload requires. The savings on procurement are real; the costs of the wrong architecture are larger. Run the cross-validation on your golden set before committing — and use the right sub-product when you do.

7.5 Classic IDP, retrofitted

The classic IDP vendors — ABBYY Vantage, Hyperscience, UiPath Document Understanding, Klippa, Rossum — have been incumbents in this space for fifteen to twenty years. Their strengths and weaknesses are inverted relative to the AI-native specialists.

Strengths: HITL is native, not retrofitted. The reviewer queue, the reviewer UI, the case-management workflow, the audit trail — these have been the core product for two decades. The agentic specialists are now building this functionality and discovering it is harder than it looks. Compliance posture is similarly mature: ABBYY, Hyperscience, and UiPath all carry the full enterprise compliance certification suite as a default rather than a sales-time question.

Weaknesses: the agentic story is largely retrofitted. The model layer at these vendors is being upgraded to include VLMs and reflection loops, but the surrounding architecture was designed for a different world. The “agentic” features in classic IDP vendor pitches in 2026 frequently fail the five-attribute test from Chapter 3 — typically missing tool-using and goal-driven attributes, or having only weak versions of self-correction.

Per-page pricing in this tier runs \$0.020–\$0.150, with Hyperscience at the high end of the range. The pricing reflects what classic IDP has always been about: HITL infrastructure and compliance, with parsing as one component rather than the whole product.

A separate sub-tier worth calling out is the AI-native commercial IDP entrants: **Nanonets** and **Docsumo**. These vendors are younger than the classic IDP incumbents and are explicitly marketing agentic capabilities. Nanonets prices in workflow blocks (\$0.30 per complex AI block, leading to under-\$2-per-invoice end-to-end pricing), supports 100+ languages, and reports 34% of Fortune 500 as customers (vendor claim; treat accordingly). Docsumo claims 150+ document types out of the box at 95%+ accuracy, with a free 14-day / 1,000-page trial and tiered subscriptions thereafter.

The honest framing of this tier: choose classic IDP or AI-native commercial IDP when HITL is the load-bearing requirement, when the documents are within the classic-IDP design envelope (invoices, claims forms, KYC, contract abstraction), and when your procurement process is more comfortable with established vendor relationships than with the AI-native startup risk. Do not choose this tier expecting the agentic capabilities to match the specialist vendors at the same price point.

7.6 The platform tier — Databricks, Snowflake, Microsoft Fabric

The most important structural shift in 2026 commercial agentic OCR is the rise of a **platform tier** distinct from pure-play vendors. Document parsing has been absorbed as a primitive into the major enterprise data platforms — Databricks Document Intelligence, Snowflake Cortex AI, and Microsoft Fabric AI Functions. For buyers already committed to one of these platforms, the procurement story has changed: agentic OCR is no longer a separate vendor relationship but a SQL function or AI Function call billed against existing platform capacity.

Three vendors, three slightly different shapes:

7.6.1 Databricks Document Intelligence

Databricks launched Document Intelligence in 2026 as a set of chainable SQL/AI functions: `ai_parse_document` (generally available), `ai_classify`, and `ai_extract` [1]. The integration with Lakeflow, Unity Catalog, and Agent Bricks is tight enough that the product is best understood not as an OCR API but as a primitive in the Databricks data platform.

The publicly reported numbers are aggressive. **5–7× lower cost than comparable VLM-only pipelines** for equivalent accuracy. A customer case study (Loopback Analytics, clinical-note extraction) reports approximately **90% cost reduction** for the same downstream entity-extraction outcome. The 16% across-the-board performance gain from inserting `ai_parse_document` upstream of agent frameworks (Chapter 6.) is the cleanest single piece of evidence for the shift-left thesis in any 2026 publication.

The strategic significance is larger than the product. A platform vendor at Databricks' scale saying — with public benchmarks and customer case studies — that document reading is the gating constraint on agent quality is a meaningful institutional endorsement of the shift-left thesis.

7.6.2 Snowflake Cortex — AI_PARSE_DOCUMENT

Snowflake matches Databricks' move with **AI_PARSE_DOCUMENT**, a Cortex AI function that extracts text, data, layout elements, and images from documents stored on internal or external stages [21]. The naming convention is not an accident — the two products converged on the same SQL-function shape, the same chainable-with-other-AI-functions architecture, and the same “agentic OCR as a primitive inside the data platform” positioning.

Supported formats: PDF, DOCX, PPTX, JPEG, JPG, PNG, TIFF, TIF, HTML, TXT. Image extraction (added January 2026, in preview) lets **AI_PARSE_DOCUMENT** emit images embedded in documents — extracted images can be written to stages or passed directly to other Cortex AI functions for further analysis, which is the chaining pattern Snowflake recommends for end-to-end agentic pipelines.

Billing is per page (with HTML/TXT chunked at 3,000 characters per billable page), charged against Snowflake credits. Pricing varies meaningfully by warehouse size and credit rate; the field-reported cost is roughly competitive with Databricks at typical configurations, with the caveat that **runaway costs are a known failure mode** — there are public 2026 incident writeups of single-query costs in the thousands of dollars when **AI_PARSE_DOCUMENT** is run against a large corpus without throttling. The cost-monitoring patterns in Chapter 11. apply doubly here.

7.6.3 Microsoft Fabric — AI Functions + Foundry Tools

Microsoft Fabric takes a slightly different shape. Rather than ship a single `parse_document` function, Fabric integrates Azure AI Document Intelligence (via Foundry Tools) as a callable AI Function inside the Fabric environment [22]. AI Functions in Fabric support multimodal input — images, PDFs, text — and use Fabric authentication, with all usage billed against the customer's Fabric capacity rather than separately metered.

The practical effect: a Fabric customer running document workflows is using **Azure DI Layout** (or DI Prebuilt/Custom) under the hood, but through the Fabric interface rather than as a separate Azure subscription. SharePoint Premium (formerly Syntex) is the Microsoft 365 side of the same integration, oriented at end-user document workflows rather than analytics pipelines.

The architectural distinction from Databricks and Snowflake is real: Databricks and Snowflake built their own document AI; Microsoft integrated Azure's. Both approaches work; both have customers; the choice is downstream of which platform you already use.

7.6.4 The strategic significance

A platform-tier offering is what every cloud-native data team encounters first now when they search “how do I parse documents in [platform].” This changes vendor selection at the procurement level: for buyers already on one of these three platforms, the platform-tier product is the default candidate, and pure-play specialists have to actively justify being added on top.

The honest caveats are familiar across all three: vendor-published numbers, benchmarks maintained by the platform vendor, and architecture that is platform-locked (each product is meaningfully useful only if you are already on the platform). For non-Databricks / non-Snowflake / non-Fabric customers, the platform tier is a data point about where the institutional landscape is heading, not an option you can buy directly. The pure-play specialists are not wrong to be nervous — but they are also not displaced from workloads outside these three platforms, which is most of the addressable market.

7.7 Pricing matrix

The data in `data/commercial-players.csv` and the per-vendor cost notes above resolve into the following compact summary:

Tier	Vendor / Sub-product	Per-page low	Per-page high
AI-native	LlamaParse	\$0.00125	\$0.05625
AI-native	Reducto	\$0.005	\$0.020
AI-native	Landing AI ADE (monthly)	\$39/mo	\$2,199/mo
AI-native	Mistral Document AI	\$0.001	\$0.002
AI-native commercial IDP	Nanonets	\$0.05/block	\$0.30/block
AI-native commercial IDP	Docsumo	(subscription)	(subscription)
Hyperscaler — Azure	AI Vision Read	\$0.001	\$0.0015
Hyperscaler — Azure	DI Read	\$0.0015	\$0.0015
Hyperscaler — Azure	DI Layout	\$0.010	\$0.010
Hyperscaler — Azure	DI Prebuilt	\$0.010	\$0.010
Hyperscaler — Azure	DI Custom	\$0.030	\$0.030
Hyperscaler — Google	Document AI OCR	\$0.00065	\$0.0015
Hyperscaler — Google	DocAI Layout Parser	\$0.010	\$0.010
Hyperscaler — Google	DocAI Form Parser	\$0.020	\$0.030
Hyperscaler — Google	DocAI Custom Extractor	\$0.020	\$0.030
Hyperscaler — AWS	Textract DetectDocumentText	\$0.0015	\$0.0015
Hyperscaler — AWS	Textract AnalyzeDocument	\$0.015	\$0.065
Hyperscaler — AWS	Textract AnalyzeID / AnalyzeExpense	\$0.0025	\$0.010
Hyperscaler — AWS	Bedrock Data Automation	\$0.010	\$0.040
Classic IDP	ABBYY Vantage	\$0.020	\$0.100

Tier	Vendor / Sub-product	Per-page low	Per-page high
Classic IDP	Hyperscience	\$0.030	\$0.150
Classic IDP	UiPath DU	\$0.020	\$0.080
Classic IDP	Klipa	\$0.010	\$0.040
Classic IDP	Rossum	\$0.020	\$0.080
Platform	Databricks Document Intelligence	(platform-tier)	(platform-tier)
Platform	Snowflake Cortex AI_PARSE_DOCUMENT	(platform-tier)	(platform-tier)
Platform	Microsoft Fabric + Foundry Tools	(Fabric capacity)	(Fabric capacity)

Two orders of magnitude separate the floor and the ceiling. The salient point is not which vendor is cheapest; the salient point is that **pricing differences across vendors look big until you remember that error rate is the multiplier**. A vendor that is four times cheaper per page and has four times the field-level error rate is, on cost-per-correct-extraction, identical to the more expensive vendor. The pricing matrix is only meaningful when paired with the error-rate column, and the error-rate column only becomes meaningful when you measure it on your own documents.

7.8 How to read this landscape

The chapter's organizing principle, restated as actionable advice:

- **Do not pick by benchmark rank.** Benchmark leaders frequently produce worse production results than benchmark followers on specific workloads. The benchmark is a filter, not a ranking.
- **Shortlist three vendors across three tiers** — one AI-native, one hyperscaler or classic IDP, one Pattern C specialist if your workload is high-stakes. Run all three on your golden set. Decide on data, not demos.
- **The right vendor is the one that minimizes cost-per-correct-extraction-with-provenance on your documents.** This is a measured number, not a marketing number. The only honest way to get it is to run the parsing on a representative sample of your real workload.
- **Procurement maturity matters more than buyers' guides admit.** A vendor with the right benchmark score but with a six-month security-review timeline can lose to a vendor with a lower benchmark score but a two-week procurement path, simply because the higher-benchmark vendor's project does not ship within the fiscal-year window. Factor this in honestly.

The chapters that follow build on the same picture. Chapter 8. covers the open-source side of the landscape, which has caught up on parsing quality more than the commercial vendors will acknowledge. Chapter 9. turns the landscape into a decision framework, including the explicit “should I not build this at all” rubric from Section 9.5. Chapter 11. gives you the eval methodology you need to make the vendor-selection numbers honest.

🔍 Named takes

The hyperscaler APIs are still where most enterprise procurement lands, and they are still the wrong choice for 2026 agentic workloads. They were designed for templates.

Pricing differences across vendors look big until you remember that error rate is the multiplier. A 4× cheaper parser with a 4× higher field-error rate is the same cost-per-correct-extraction.

The platform tier (Databricks today; Snowflake and Fabric next) is the most important strategic shift in 2026 commercial agentic OCR. The pure-play specialists are not wrong to be nervous.

8. Open-Source Players

i Executive summary

- **The open-source parsing-quality gap closed in 2026.** The top of the olmOCR-Bench leaderboard (Chandra at 83.1, OlmOCR-2 at 82.3) is open-weight and license-friendly to most enterprises. Open-source models match or beat proprietary systems on raw extraction quality across multiple benchmarks.
- The remaining differentiator that keeps commercial vendors viable is **agentic orchestration, HITL tooling, and grounding/audit infrastructure** — not raw extraction. The open-source side has the models; the commercial side has the surrounding platform.
- IBM Granite-Docling-258M is the most production-credible open-source agentic OCR pick for 2026 enterprise deployment: Apache-2.0, single 258M-parameter model, unified output representation, IBM-backed. PaddleOCR’s PP-StructureV3 matching Gemini-2.5-Pro at under 100M parameters is the strongest argument that shift-left is about architecture, not scale.
- Licensing is not a footnote. AGPL (MinerU) excludes meaningful commercial deployments. OpenRAIL (Chandra) is permissive but rare in enterprise legal review. Apache-2.0 (Granite-Docling, OlmOCR-2, PaddleOCR) is the safe choice. Read the license before you commit, not after.

8.1 The state of the open-source ecosystem

The Hugging Face “*Supercharge your OCR Pipelines with Open Models*” blog post, published October 21, 2025 [23], is the canonical 2026 reference for the open-source OCR landscape. The blog catalogs eight primary models in a single comparison table — a snapshot of an ecosystem that, two years prior, did not produce competitive document-extraction models at all.

The salient observation is that the gap closed quietly. While the commercial side was loudly announcing agentic capabilities at trade shows, the open-source side was shipping models that, on the public benchmarks where direct comparison is possible, are at or near parity with the closed-source frontier. Chandra (DataLab-to, 9B parameters, OpenRAIL license) sits at the top of olmOCR-Bench with a score of 83.1 ± 0.9 . OlmOCR-2 (AllenAI, 8B, Apache-2.0) is at 82.3 ± 1.1 . dots.ocr (Rednote-hilab, 3B) is at 79.1 ± 1.0 . These are open weights, free to inspect, fine-tune, and self-host. They were not in the conversation two years ago.

The implication is not that commercial vendors are obsolete — they are not, and this chapter and the next two will be honest about why. The implication is narrower and sharper: **if you are buying a commercial agentic-OCR product because you believe the proprietary models are meaningfully better, you are paying for a gap that has been closing fast and may already be closed for your workload.** The remaining things commercial vendors do well — orchestration, HITL, grounding infrastructure, compliance certification, support contracts — are real and worth paying for. The model quality itself is no longer the differentiator.

This chapter walks the open-source landscape with that framing. It is organized first by parameter budget (heavyweight, efficient, compact), then by the specific projects worth knowing in 2026, then by a license-and-capability matrix, then by the question of when to choose open-source over commercial.

8.2 Three groups by parameter budget

The 2026 open-source landscape splits cleanly along compute requirements.

Heavyweight VLMs ($\geq 8\text{B}$ parameters). Chandra (9B), OlmOCR-2 (8B), Qwen3-VL (9B). Strong all-rounders, capable of running on a single H100 or split across two A100s. These are the models you reach for when accuracy is paramount and compute is available.

Efficient VLMs (3–4B parameters). dots.ocr (3B), DeepSeek-OCR (3B), Nanonets-OCR2-3B (4B). Better cost-per-page on commodity hardware. These are the models for high-volume self-hosted deployments where the marginal compute savings compound.

Compact specialists (under 1B parameters). PaddleOCR-VL (0.9B), Granite-Docling-258M (258M), PP-StructureV3 (<100M). Stunning efficiency relative to capability. The defining 2026 result for this tier is PP-StructureV3 matching Gemini-2.5-Pro on OmniDocBench at under 100M parameters — a result that makes the “bigger is better” reading of agentic OCR look obviously wrong.

The compact tier is where the most interesting work is happening, both architecturally and commercially. A model that fits on a laptop GPU and produces production-quality structured output is a different kind of object than a frontier VLM — different deployment story, different cost profile, different security posture for sensitive workloads.

8.3 IBM Granite-Docling and Docling

IBM released **Granite-Docling-258M** in January 2026 under Apache-2.0 [16]. It is a single 258-million-parameter VLM that parses and processes documents in one shot, with output flowing through a unified representation called **DoclingDocument** that exports to Markdown, JSON, HTML, or DocTags (an XML-like layout-preserving format).

Architecturally, Granite-Docling replaces the SmolLM-2 language backbone used in the earlier experimental SmolDocling-256M-preview (March 2025) with a Granite-3 language model, and replaces the SigLIP visual encoder with the updated SigLIP2. Training data is 81,000 manually-labeled pages spanning patents, manuals, and 10-K filings; reported accuracy is within roughly 5 percentage points of human accuracy on page-element identification.

The Docling framework itself (MIT-licensed, separately maintained) is the orchestration layer — pipeline glue that wraps Granite-Docling and other models into customizable ensembles. The two together (model plus framework) form a complete open-source agentic OCR stack with the closest analogue to a commercial Pattern B or C deployment available without a vendor relationship.

Granite-Docling is, by some distance, the most **production-credible** open-source agentic OCR pick in 2026:

- **Apache-2.0** is the safest enterprise license. Legal review will not flag it.
- **258M parameters** is small enough to deploy on commodity hardware. Inference cost is meaningfully lower than the heavyweight tier.
- **IBM backs the project** institutionally. This matters more than it should for enterprise adoption, but it does matter. The cultural acceptability gap between “IBM” and “small startup on GitHub” is wide in regulated industries.
- **DoclingDocument as a unified format** simplifies downstream integration. Tooling that consumes DocTags or Markdown works across model versions and even across different models that emit the same format.

The honest caveats. Granite-Docling-258M is single-shot — it produces output in one pass and does not by itself implement the reflection loop that defines Pattern B in Chapter 4.. For a true agentic loop, you wrap Granite-Docling in the Docling framework with additional validation and retry logic, or you combine it with NuExtract-style downstream extraction. The result is a Pattern B or C architecture; the model is a strong primitive within it.

8.4 Marker and Surya (DataLab-to)

Two of the most-used open-source tools in 2026, often missing from formal vendor surveys because they sit slightly outside the “model” category — they are **end-to-end document conversion pipelines** built on top of specialized models, maintained by the same team that ships Chandra.

Marker (originally [VikParuchuri/marker](#), now [datalab-to/marker](#)) is the practitioner’s default for one-shot PDF → Markdown conversion. It accepts PDF, image, DOCX, PPTX, XLSX, HTML, and EPUB inputs, and emits Markdown, JSON, HTML, or document chunks. Under the hood it composes Surya for OCR, a layout model, and reading-order detection, producing structured output that is direct-consumable by downstream RAG and agentic systems.

Marker has become the de-facto answer in 2026 to “what’s the safest default open-source PDF-to-Markdown tool if I’m only going to install one.” It is widely cited in the field and benchmarked frequently against Docling, MinerU, and pdf-craft as a peer.

Surya is the OCR + layout + reading-order toolkit that powers Marker but is also useful standalone. Strong on 90+ languages, with separate detection and recognition modules that can be used as primitives in custom pipelines. Surya is what you reach for when you want to build your own document pipeline around an open-source OCR engine and need more than Tesseract.

License caveat worth flagging. Both Marker and Surya are GPL-3.0 licensed for open-source use. DataLab-to offers a separate commercial license for use in proprietary products — GPL excludes most SaaS productization scenarios, and getting the commercial license is a normal procurement conversation if Marker/Surya are central to your product. For internal use (HITL workflows, internal-only RAG systems, research) GPL is fine. For SaaS resold to customers, you need the commercial license or a permissively-licensed alternative.

This is the same licensing trade-off as MinerU (AGPL-3.0) — and the same team behind Marker also ships **Chandra** (covered separately above with the OpenRAIL license), so DataLab-to is increasingly the open-source equivalent of a focused commercial vendor: multiple complementary products, real engineering depth, with a real license-review conversation attached.

8.5 OpenDataLab MinerU and MonkeyOCR

OpenDataLab is the open-source AI research arm of Shanghai AI Laboratory; they are the most active institution in 2026 open-source document AI.

MinerU 2.5 (1.2B parameters, AGPL-3.0 license, available on Hugging Face as `MinerU2.5-2509-1.2B`) is OpenDataLab’s flagship parser. The model is excellent on scientific PDFs — papers, theses, technical reports — with strong handling of mathematical formulas, complex tables, and multi-column layouts. The AGPL-3.0 license is the elephant in the room: AGPL excludes meaningful classes of commercial deployment (you cannot easily use MinerU 2.5 in a SaaS product without open-sourcing the surrounding service), and many enterprise legal review processes block AGPL by default. For internal use, research, and academic deployments, MinerU 2.5 is excellent; for SaaS-resold downstream products, it is the wrong choice.

MonkeyOCR v1.5 (released November 2025, technical report on arXiv as 2511.10390 [24]) is OpenDataLab’s most recent contribution and the current state-of-the-art on OmniDocBench v1.5. The reported gains are concrete: MonkeyOCR v1.5 beats PP-OCR-VL by 0.15% and MinerU 2.5 by 2.34% on OmniDocBench, and beats PP-OCR-VL by 8.2% on the OCRFlux-complex dataset (a complex-layout-focused benchmark).

Architecturally MonkeyOCR is a specialized parser rather than a general VLM — closer in shape to PP-StructureV3 than to Qwen3-VL. It is a strong open-source option for complex-layout workloads, though production-readiness is closer to “research-quality with eval discipline required” than to “drop into production.” Treat MonkeyOCR as a Pattern C component (a strong specialized extractor that fits inside a hybrid architecture) rather than as a standalone Pattern B system.

8.6 AllenAI olmOCR and OlmOCR-2

AllenAI’s olmOCR project is one of the most rigorously engineered open-source document parsers, in the same intellectual tradition as their other open releases (OLMo language models, Tulu instruction-tuned variants, the Open Language Model series).

OlmOCR-2 (8B parameters, Apache-2.0) is the current release as of mid-2026 [25]. olmOCR-Bench score: 82.3 ± 1.1 . Optimized for batch processing with strong English-language performance; the model is somewhat English-centric and weaker on multilingual workloads. The cost anchor reported in the Hugging Face survey is **approximately \$178 per million pages** running on an H100 at \$2.69/hour, which is the cleanest open-source cost number in the 2026 ecosystem.

The associated **olmOCR-Bench** (also AllenAI-maintained) is the most useful unit-test-style benchmark in the open ecosystem, despite its limitations (English-only, annotations generated by closed-source VLMs). The combination of model and benchmark from the same group gives a self-consistent reference point for regression testing — you can fine-tune or modify the model and immediately see what changes on a reproducible benchmark.

The honest framing of olmOCR-2: it is the right pick for high-volume English-language batch workloads where self-hosting is the right deployment model and where the engineering team has the capacity to operate model serving and downstream orchestration. The model itself is one of the best in the open ecosystem; the surrounding tooling (HITL queues, audit logs, vendor-equivalent observability) you build yourself.

8.7 The new generation: Chandra, dots.ocr, DeepSeek-OCR

Three models released in the second half of 2025 form the current frontier of open-source agentic-OCR quality.

Chandra (DataLab-to, 9B parameters, OpenRAIL license) is the current top of olmOCR-Bench at 83.1 ± 0.9 . The model handles 40+ languages, supports grounding (region-pointer outputs), and emits Markdown, HTML, or JSON. Chandra is the strongest open-source quality leader for English-heavy work in 2026; the OpenRAIL license is permissive but worth a legal review pass before commercial commitment.

dots.ocr (Rednote-hilab, 3B parameters) sits at olmOCR-Bench 79.1 ± 1.0 . The model is strong on grounding and handwriting — both important attributes for compliance-oriented and form-heavy workloads. At 3B parameters it is significantly cheaper to operate than Chandra or OlmOCR-2 while remaining within a few points on benchmark quality.

DeepSeek-OCR (DeepSeek, 3B parameters) at olmOCR-Bench 75.4 ± 1.0 is the model to reach for when **multilingual coverage** and **throughput** dominate. It handles approximately 100 languages and reports ~200,000+ pages per day on a single A100. A particularly useful feature is **chart and table rendering to HTML**, which is a useful primitive for downstream RAG pipelines that need structured representations of visual content.

These three together — Chandra, OlmOCR-2 with the older olmOCR underneath, and dots.ocr — define the open-source quality ceiling for general-purpose document parsing in 2026. Any of them can serve as the parsing layer in an open-source Pattern B agentic OCR architecture.

8.8 PaddleOCR and PP-StructureV3 — the small-model story

PaddleOCR is the PaddlePaddle (Baidu) family of document-parsing models. Its 2026 entries deserve their own section because they make the strongest single argument for the “shift-left is about architecture, not scale” thesis from Chapter 6..

PP-StructureV3 is the most striking data point. The model achieves an OmniDocBench edit distance of **0.145 on English documents** and **0.206 on Chinese documents** — at **under 100 million parameters** [17]. The result matches Gemini-2.5-Pro’s billion-scale VLM accuracy on the same benchmark at a tiny fraction of the compute cost.

The lesson is structural. PP-StructureV3 is a layout-aware model — its architecture is designed around document structure rather than around general image-to-text generation. It is, in the Chapter 3 sense, **specialized**. The result is what you would expect: a specialized model trained for a specific task at modest scale outperforms a general-purpose model at enormous scale on that task. This generalizes beyond OCR, but the OCR result is the cleanest public demonstration in 2026.

PaddleOCR-VL (0.9B parameters, Apache-2.0) is the VLM-enriched variant covered in the Hugging Face survey. Supports 109 languages — broader multilingual coverage than any other model in the ecosystem — with Markdown, JSON, and HTML output. Prompt-based task switching makes it useful for varied workloads from a single model.

The honest caveat about the PaddleOCR ecosystem: the documentation is improving but historically lagged the Western open-source projects in English-language usability. The community is smaller in the West. Both gaps have narrowed sharply through 2025 — by mid-2026 PaddleOCR is a credible choice for serious deployments. The lesson should still be read carefully: the most efficient model in the open ecosystem, by a wide margin, comes from outside the Western open-source mainstream.

8.9 Qwen3-VL — the general-VLM-as-OCR option

Qwen3-VL (Alibaba, 9B parameters, open weights) is included in the Hugging Face survey not as an OCR-specialized model but as a general VLM that can be used for OCR. The model supports 32 languages and is particularly strong on ancient text recognition, handwriting, image extraction, and chart understanding.

The trade-off relative to OCR-specialized models is direct: Qwen3-VL is weaker on character-perfect transcription and stronger on **semantic understanding** of visual content. For documents where the intent is to extract meaning rather than to produce a faithful textual reproduction — charts that need to be summarized rather than transcribed, screenshots of dashboards, diagrams with annotations — Qwen3-VL is often the better tool than an OCR-first model.

This is a useful framing for the open-source landscape: not every “OCR” problem is an OCR problem. Some are visual-understanding problems wearing an OCR shape. A general VLM is sometimes the right tool, and Qwen3-VL is currently the strongest open-weight general VLM with serious document capabilities.

8.10 NuExtract (NuMind) — separating extraction from OCR

The NuExtract series from NuMind is a slightly different category and deserves its own treatment because it makes a clean architectural argument that the rest of the field still routinely confuses.

NuExtract is not OCR. It is structured extraction — document-or-text to JSON — that runs after a parser. The system takes already-parsed text as input and produces schema-validated JSON output. The model family has gone through three major releases:

- **NuExtract 1.5** (multilingual; reportedly beats GPT-4o on English extraction while being approximately 500× smaller).
- **NuExtract 2.0** (May 2025; adds vision, abstraction, in-context learning; open-source variants in the 2B–8B parameter range).
- **NuExtract 2.0 PRO** (API-only; reports +9 F-score over GPT-4.1 on extraction benchmarks).
- **NuExtract3** (released 2026; latest in the series).

The reason this matters for the open-source landscape is that NuExtract proves a sharp architectural point: **“agentic OCR” is two problems pretending to be one.** Parsing (converting an image to structured content) and extraction (pulling specific fields from structured content) are separable problems with different optimal solutions. A vendor that bundles them at a premium (LlamaParse Agentic + Extract, Landing AI ADE) is sometimes charging you twice for an engineering team that could be split. A pipeline that uses Granite-Docling or PaddleOCR-VL for parsing and NuExtract for downstream extraction is often cheaper and more flexible than a bundled commercial product.

The honest framing: NuExtract is the right downstream extraction layer for any open-source pipeline where the parsing layer is already settled. Use it whenever the bundled commercial offerings look expensive and you have engineering capacity to operate two model layers.

8.11 Throughput contenders

For self-hosted deployments at very high volume, the throughput numbers matter more than the benchmark scores. Three models are worth knowing.

LightOnOCR-1B (LightOn AI) claims **approximately 493,000 pages per day on a single H100**. The throughput record-holder among open models. Newer release, less production validation than the more established models, but the throughput number is striking enough that it warrants serious evaluation for high-volume workloads. License is open; details on the Hugging Face model card.

DeepSeek-OCR (already covered above) reports ~200,000+ pages per day on a single A100, with the additional benefit of broad multilingual coverage.

PaddleOCR-VL is the highest-throughput sub-billion-parameter model in the Hugging Face survey, though specific page-per-day numbers vary by hardware and document complexity.

For perspective: at 493k pages per day on a single H100 at \$2.69/hour, the per-page parsing cost is roughly **\$0.000135**. That number — about 1/700th of LlamaParse Agentic and roughly 1/15th of OlmOCR-2 — is what makes self-hosted open-source compelling at high volume. The trade-off is engineering capacity. You need a team that can operate model serving, autoscaling, eval pipelines, HITL queues, and observability. If you have the team, the economics are unanswerable in your favor.

8.12 A note on frontier proprietary VLMs as the model layer

This is an open-source chapter, but a fair fraction of “open-source agentic OCR pipelines” in 2026 actually use a **frontier proprietary VLM** as their model layer — Claude (Anthropic), GPT-4o or GPT-4.1 (OpenAI), or Gemini (Google) — wrapped in open-source orchestration. The pattern deserves naming because it is widespread and because the cost profile is meaningfully different from the pure-open-source path.

Three frontier VLMs are routinely used this way in 2026:

- **Claude (Anthropic)** supports native PDF input via the API and is widely used as the model layer when extraction quality and instruction-following matter more than per-token cost. Many internal agentic pipelines combine Claude for the extraction/reflection steps with open-source layout detection (Surya, PP-StructureV3) for the structural stage.
- **GPT-4o / GPT-4.1 (OpenAI)** is the most common frontier VLM in the wild for document workflows. Native vision input via API. Structured-output and function-calling support are mature. Many of the “I built a custom Pattern B” stories you hear at conferences use GPT-4o under the hood.

- **Gemini (Google)** is the model that powers Google Document AI Layout Parser under the hood. Available directly via the Gemini API for custom pipelines. Strong on multimodal and long-context document tasks.

The cost profile is different in two ways. **Per-page cost is dominated by output tokens**, not by a fixed parsing fee — a long document costs proportionally more, and the model’s verbosity matters. **Compliance posture is the API provider’s**, not yours — using Anthropic’s API for healthcare documents requires Anthropic’s BAA (now available), the same for OpenAI and Google. Self-hosting is structurally not an option, which is the line where buyers with hard data-sovereignty requirements peel off to the open-source side.

The honest framing: frontier VLMs sit in an *adjacent* category to open-source agentic OCR. The two complement each other — open-source primitives for layout and table extraction, frontier VLM for high-stakes extraction and reflection, orchestrated by open-source frameworks (LangChain, LlamaIndex OSS, custom Python). A team building a custom Pattern B or C pipeline in 2026 is more likely to use this hybrid than either pure-open-source or pure-commercial-vendor.

8.13 Tesseract and the still-useful primitives

Tesseract (Apache-2.0) is still in the ecosystem. It is **not** agentic OCR, by Chapter 3’s definition, and any vendor marketing a “Tesseract + LLM cleanup” pipeline as agentic OCR is wrapper-washing (Chapter 13).

Tesseract is still the right answer for narrow workloads: very high volume, clean English text, low stakes, minimal layout variation. Invoice line-item extraction at extreme scale, bulk indexing of scanned books, OCR for accessibility purposes — these are legitimate Tesseract use cases. The mistake is using Tesseract for workloads it cannot handle (tables, complex layouts, handwriting) and then either accepting the resulting accuracy or, worse, bolting an LLM cleanup step on top and calling the result agentic.

The legitimate use of Tesseract in 2026 is as a **primitive within a Pattern C hybrid architecture**: the layout detector sends pure-text regions to Tesseract because Tesseract is cheap and good at pure text, and routes everything else (tables, figures, handwriting, signatures) to specialized models. That is a reasonable architectural choice and does not violate any of the Chapter 3 attributes — the surrounding architecture provides the agentic loop; Tesseract is one tool in the chain.

8.14 Capability and license matrix

The data in [data/opensource-players.csv](#) consolidates the per-model details. The compact summary:

Model	Params	License	Best for	Caveat
Chandra	9B	OpenRAIL	Highest open-source quality on English	License review needed
OlmOCR-2	8B	Apache-2.0	Batch-optimized English; ~\$178/M pages on H100	English-centric
Qwen3-VL	9B	Open	General VLM; semantic over textual fidelity	Not OCR-specialized
dots.ocr	3B	Open	Grounding + handwriting at modest scale	Smaller community
DeepSeek-OCR	3B	Open	Multilingual (~100 langs); high throughput	Slightly behind frontier on accuracy
Nanonets-OCR2-3B	4B	Unclear	Signatures, checkboxes, flowcharts	License clarity issue

Model	Params	License	Best for	Caveat
MonkeyOCR v1.5	N/A	Apache-2.0	SOTA on OmniDocBench v1.5 complex layouts	Research-quality
MinerU 2.5	1.2B	AGPL-3.0	Scientific PDFs	AGPL excludes commercial SaaS
PaddleOCR-VL	0.9B	Apache-2.0	109 languages; high throughput	Documentation maturing
PP-StructureV3	<100M	Apache-2.0	Matches Gemini-2.5-Pro on OmniDocBench at tiny scale	Less agentic by itself
Granite-Docling-258M	258M	Apache-2.0	Most production-credible open-source pick	Single-shot; wrap for Pattern B
Docling (framework)	N/A	MIT	Pipeline glue for ensembles	Not agentic by itself
Marker	N/A (orchestrator)	GPL-3.0 (commercial option)	Most popular open-source PDF→Markdown tool	GPL excludes SaaS resale without commercial license
Surya	N/A	GPL-3.0 (commercial option)	OCR + layout + reading order primitive	Same GPL caveat as Marker
LightOnOCR-1B	1B	Open	Throughput leader (~493k pages/day H100)	Newer; less validated
NuExtract3	2B–8B	Open	Best-in-class structured extraction post-parse	Extraction-only
AgenticOCR	N/A	Apache-2.0	Reference implementation for agentic loops	Less mature tooling
Tesseract	N/A	Apache-2.0	Cheap text-only OCR primitive	Not agentic; not for tables
<i>Claude (frontier VLM)</i>	N/A	Closed API	Strong model layer in custom pipelines	Closed-source; pay-per-token
<i>GPT-4o / GPT-4.1 (frontier VLM)</i>	N/A	Closed API	Most common model layer in 2026 custom pipelines	Closed-source; pay-per-token
<i>Gemini (frontier VLM)</i>	N/A	Closed API	Powers Google DocAI Layout Parser; native PDF	Closed-source; pay-per-token

The matrix above is the most actionable artifact of this chapter. Reading it: pick the model whose “best for” column matches your workload, whose license fits your deployment context, and whose caveat does not disqualify it for you specifically. Then evaluate on your own golden set.

8.15 When to choose open-source over commercial

Four conditions push the right answer toward open-source. The presence of any one of them is suggestive; the presence of two or more is usually decisive.

Data sovereignty or regulatory hard-requirement. Documents that must never leave your network — defense, intelligence, certain healthcare and legal workloads, EU GDPR scenarios with strict transfer restrictions — make self-hosted open-source the default. The commercial-vendor self-host options exist (Mistral Document AI has one) but the open-source story is usually cleaner.

Per-page cost dominated by API spend at scale. Above roughly 1 million pages per month, self-hosted open-source pays for itself fast. OlmOCR-2 at ~\$178 per million pages is one to three orders of magnitude cheaper than commercial API pricing, even before the LightOnOCR-1B throughput numbers are factored in. The crossover point depends on your engineering loaded cost and your tolerance for operational complexity, but for volumes north of a few million pages per month, open-source is rarely the wrong economic choice.

Engineering capacity to operate the stack. Open-source assumes you can run model serving, eval pipelines, HITL workflows, autoscaling, and monitoring. If you cannot, the commercial vendors are earning their margin honestly. The mistake is choosing open-source for cost reasons and then not staffing the team to operate it.

Auditability requirements that commercial vendors cannot satisfy. Open-source models can be inspected, fine-tuned, and modified. Commercial models are black boxes. For compliance-heavy industries where understanding model behavior at a low level is part of the audit posture, open-source has structural advantages that no commercial vendor can match.

The corresponding conditions that push the answer toward commercial: low-volume workloads where API cost is dominated by engineering overhead, organizations without ML/serving operational capacity, teams that need vendor support contracts as a procurement requirement, and regulated industries where commercial-vendor certifications (SOC 2 Type II, HIPAA BAA, GDPR DPA) are easier to procure than to build internally.

No team picks one side and stays there forever. The right pattern, often, is to start commercial — to learn the workload, build the eval, get the HITL workflow right — and migrate parsing to open-source after twelve to eighteen months when the engineering team has the capacity and the operational maturity. The reverse migration (open-source to commercial) is rarer in 2026 but happens when growing compliance demands outpace the team's ability to certify their own stack.

Named takes

By the end of 2026, the open-source parsing-quality gap is closed. The reason you'll still pay for commercial agentic OCR is that nobody wants to operate the orchestration, HITL queues, and audit infrastructure themselves.

PaddleOCR-VL at 0.9B params matching Gemini-2.5-Pro on OmniDocBench is the most important data point in 2026 open-source OCR. Read it again.

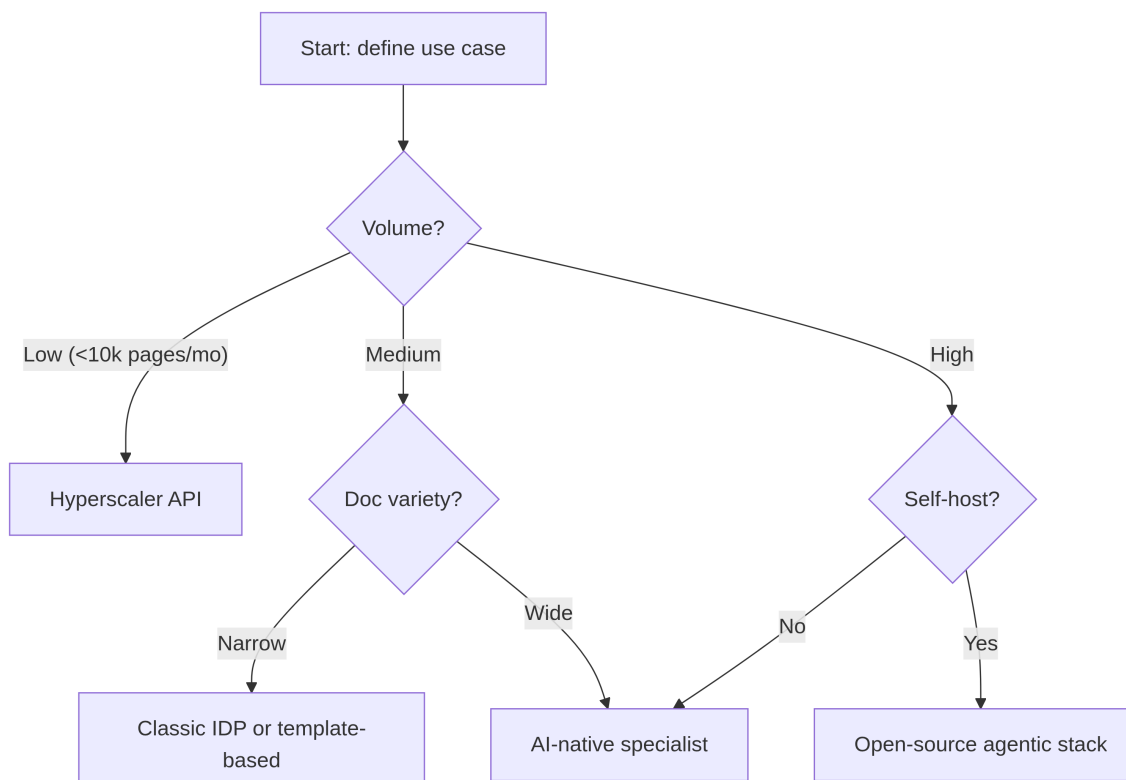
NuExtract is the proof that "agentic OCR" is two problems pretending to be one. Parsing and extraction are separable, and the vendor that bundles them at a premium is often charging you twice for one engineering team.

9. Vendor Selection

i Executive summary

- Vendor selection is a stack of four orthogonal questions — **volume, document variability, compliance posture, self-host vs. SaaS** — that collapse the vendor space by roughly 80% before benchmark scores enter the conversation. Start there, not with a vendor’s marketing site.
- Healthcare prior-authorization (post-CMS rule), legal contract review, KYC document review, and financial filings are the clearest 2026 ROI cases for Pattern C agentic OCR. Generic invoice processing, commodity document indexing, and any sub-cent-per-page workload typically are not. Section 9.5 details the “should I build this at all” question with explicit cost math.
- **Buy the parser, build the eval.** The single most important principle in vendor selection. Vendors can do parsing well; only you can define correctness for your documents.
- A shortlist of three vendors across three tiers, evaluated on a 200–500 document golden set drawn from your production traffic, is the methodology that consistently produces good decisions. Anything cheaper than that is gambling.

9.1 A decision framework



Vendor-selection decision tree

The decision tree above is the compressed version of this chapter, and you should be able to use it on its own without reading further. The expansion that follows is for the cases where the compressed answer is not quite right for your situation — which is most cases, because vendor selection is genuinely workload-specific.

The four orthogonal questions, in the order they should be asked:

Volume. Below 10,000 pages per month, a hyperscaler API in Pattern A mode is almost always the right answer regardless of the rest of the analysis. The math does not support investing in Pattern B or C — and certainly not in a vendor relationship — until volume justifies it. Between 10,000 and 1,000,000 pages per month, the AI-native specialists and Pattern B architectures dominate; this is the 2026 sweet spot for the “buy from a vendor” answer. Above 1,000,000 pages per month, self-hosted open-source becomes economically compelling, and engineering capacity becomes the binding constraint rather than vendor pricing.

Document variability. Narrow and stable (one vendor template, three field types, layouts changing once per year) → template-driven classic IDP can still win on TCO. Wide and unstable (mixed vendors, evolving formats, occasional novel document types) → AI-native specialists with Pattern B or C architectures. The mistake is paying for variability handling you do not have; the mirror mistake is choosing template-driven IDP when your variability is wider than the templates can cover, and discovering this at scale.

Compliance posture. HIPAA, SOC 2 Type II, GDPR DPA, FedRAMP, ZDR (Zero Data Retention), data residency requirements. These collapse the vendor space sharply. If you need HIPAA with a BAA, you immediately rule out vendors that do not offer one. If you need on-premises deployment for data sovereignty reasons, you rule out SaaS-only vendors and most of the hyperscaler offerings. Compliance is rarely the most discussed criterion in technical decks; it is often the most consequential in actual procurement.

Two additional 2026 compliance filters worth naming explicitly because they have become procurement-decisive faster than the older certifications:

- **EU AI Act.** In force since August 2024, with phased compliance obligations rolling through 2027. Document AI systems used in the EU for “high-risk” purposes (employment decisions, access to essential services, law enforcement, migration, justice administration) face the heaviest obligations — risk management, data governance, transparency, human oversight, and post-market monitoring. For most enterprise document workflows the obligations are lighter, but the act’s transparency requirements still apply (output must be identifiable as AI-generated where users would otherwise reasonably assume it is human-generated). Vendor selection in 2026 increasingly involves an “AI Act conformity” question alongside the older GDPR DPA question; vendors with substantial EU customer bases (Mistral, ABBYY, Klippa, Rossum) are typically ahead on this.
- **FedRAMP authorization.** Required for US federal government workloads. The three hyperscalers (AWS GovCloud, Azure Government, GCP Assured Workloads) all hold FedRAMP High. The AI-native specialists are mostly **not** FedRAMP-authorized in 2026 — a meaningful gap for buyers in federal civilian, defense, or intelligence space. Some specialists offer GovCloud-deployed instances via the hyperscaler relationship, which can be a workable bridge.

Self-host vs. SaaS. Closely related to compliance but distinct. Some teams have data-sovereignty requirements that mandate self-hosting; others have engineering capacity that makes self-hosting attractive on economic grounds; others have neither and should not self-host. The honest framing: self-hosting open-source agentic OCR is not “cheaper” by default. It is cheaper *if* your team can operate it, and most teams discover the operational cost is higher than they expected.

These four questions, answered before any vendor demo is scheduled, eliminate most of the false-economy decisions that teams make. The remaining decision — which specific vendor, on which specific tier — is then a tractable shortlist exercise rather than an open-ended landscape sweep.

9.2 Use-case → architecture mapping

The right vendor depends on the right architecture, which depends on the workload. The mapping from workload shape to architecture (and therefore to vendor tier) is reasonably stable.

High variability, high stakes. Healthcare prior-authorization, legal discovery, claims adjudication, KYC review on documents from heterogeneous sources. The 2026 default is **Pattern C (hybrid CV+VLM)** with explicit HITL gates. Specialization beats generality where the stakes justify the engineering, and the public benchmarks make the gap concrete: Reducto-style hybrid architectures sit roughly 20 percentage points ahead of single-pass systems on adversarial tables, and the gap is wider on documents with mixed modalities (handwriting + print + tables on the same page).

High variability, lower stakes. Customer onboarding with mixed-vendor identity documents, invoice processing across many small vendors, contract abstraction for moderate-stakes commercial agreements. **Pattern B (agentic loop)** is the right choice here — LlamaParse Agentic, LandingAI ADE

on the Team tier, Mistral OCR 3 in its agentic configuration. The reflection loop catches the errors that matter; the absence of specialized routing keeps cost and engineering modest.

Low variability, any stakes. Single-vendor invoices at scale, standardized government forms, regulated forms with strict layouts that change once a year. **Pattern A (single-pass VLM) with strong schema validation**, or **classic template-driven IDP**, often wins here on TCO. Adding agentic complexity to a workload that does not need it is paying for variability you do not have.

Long-form reasoning. Financial filing analysis, multi-document loan packets, scientific paper synthesis, e-discovery on document collections. This is where the DABstep result matters: extraction alone is not enough, and even perfect extraction does not yield correct downstream answers without a separate reasoning layer. The right architecture is **Pattern B for the parsing layer + a dedicated reasoning agent** that operates on the parsed output. Selecting only a parsing vendor for a long-form reasoning use case is the most common shape of “we picked the wrong tool” complaint in 2026.

9.3 The wrong question to start with

Most procurement processes begin with the wrong question: *which vendor has the highest benchmark score?* This question feels rigorous and is mostly counterproductive.

The reason it is counterproductive: benchmark leaders frequently produce worse results than benchmark followers on specific workloads. ParseBench measures certain things; OmniDocBench measures certain things; RD-TableBench measures certain things. None of them measures what *your* documents demand. A vendor that scores 84.9% on ParseBench may score 71% on your golden set; a vendor that scores 79% on ParseBench may score 88% on yours. The benchmark rank is one signal; it is not the signal that predicts your production behavior.

The right starting question is sharper: “**What is my error budget per output, and what does each error cost me downstream?**”

This question is answerable, even before vendor evaluation begins. A healthcare prior-auth shop knows roughly what a wrong decision costs (denial → appeal → potential reversal → reputation cost, often hundreds to thousands of dollars amortized). A KYC team knows what a wrongly approved high-risk account costs (regulatory exposure, sometimes seven figures in fines). A scanning-and-archive workflow knows that an error in a single page costs near zero except for the occasional user complaint when something turns out to be unsearchable.

Once you have the error-cost number, vendor selection collapses to a single objective: **minimize cost-per-correct-extraction-with-provenance**. The provenance requirement is non-negotiable for any use case that will face an audit; the “correct” definition is workload-specific and is captured by your golden set; the cost is straightforward arithmetic given the per-page parsing cost, error rate, and HITL escalation rate.

This reframing turns vendor selection from a comparison-shopping exercise into an optimization problem. The optimization can be done on a golden set in a week. The comparison-shopping exercise can absorb months without producing a defensible answer.

9.4 ROI per vertical

Specific verticals have specific shapes that worth treating concretely. The four below are the ones where the 2026 economics are clearest.

9.4.1 Healthcare prior-auth

This is the strongest 2026 ROI case in the agentic OCR landscape. The CMS Interoperability and Prior Authorization Final Rule [3], with public-reporting requirements effective March 31, 2026, made manual prior-authorization workflows untenable for health plans operating at any meaningful scale. Plans must now publicly report turnaround times, denial rates, appeal rates, and overturn rates — and the political and reputational pressure of those numbers being public has accelerated agentic OCR adoption faster than any other vertical force.

The public data point is striking. Anterior AI’s workflow on Reducto’s parsing achieved **99.24% accuracy** on a prior-authorization golden set [15]. The deployment was profitable at any commercial-tier parsing price point because the alternative — manual review at roughly 30 minutes per case — was so expensive on a per-correct-decision basis that even premium parsing fees were a rounding error in the total cost.

The architectural default for healthcare prior-auth is Pattern C with explicit HITL gates. Vendor candidates: Reducto, LandingAI ADE (Team or Enterprise plan with HIPAA), LlamaParse Agentic Plus, Mistral

Document AI self-hosted for data-sovereignty cases. The decision among these vendors comes down to integration story, HITL workflow quality, and how well the vendor's golden set predicts behavior on your specific clinical-document types. All four are credible; none is uniformly best.

9.4.2 Legal — contract review and discovery

The legal vertical splits into two sub-verticals with quite different shapes.

Contract review is a moderate-volume, high-precision workload. Agentic OCR helps with clause extraction, redline detection, schema-bound field extraction (parties, effective dates, governing law, indemnification terms). Pattern B agentic systems are usually sufficient — LlamaParse Agentic and ADE both handle the workload well. The HITL queue exists but is small (a few percent of contracts go to human review for ambiguous clauses).

Discovery is the inverse: high-volume, low per-document stakes — until a smoking-gun document appears, at which point the stakes for *that document* become enormous. The dominant pattern is classic IDP (or basic OCR + indexing) for bulk processing, with agentic OCR applied selectively to documents flagged by retrieval or relevance scoring. Trying to apply Pattern C across the entire discovery corpus is economically infeasible and is not the right architectural choice.

ROI in legal varies wildly by sub-domain. Contract abstraction at scale (typical for M&A due diligence, banking documentation reviews, real-estate portfolio analysis) is comfortably above the 25¢-per-page threshold from Section 9.5. General document indexing for litigation is often below the 5¢ threshold, and agentic OCR rarely makes sense as the primary tool.

9.4.3 Financial services — KYC, loan packets, filings

Three sub-shapes within financial services.

KYC document review (identity documents, proof-of-address, source-of-funds documentation) is well-suited to Pattern B with strong schema validation. Vendors that perform consistently here include Reducto (for the document-quality requirements of regulated KYC) and LlamaParse Agentic (for the schema flexibility across document types). The economic case is comfortably above the 25¢ threshold — a single wrongly-approved high-risk customer can generate enforcement risk that dwarfs years of parsing fees.

Loan packets — multi-document submissions for mortgages, commercial loans, asset financing — are the DABstep-style challenge in operational form. The packets contain heterogeneous documents (tax returns, bank statements, employment letters, asset valuations) with cross-document dependencies (the mortgage application references the appraisal; the appraisal references the property tax records; the tax records reference the income statement). Extraction alone is necessary but not sufficient; the system also needs to reason across documents. The right architecture is Pattern B for parsing each document plus a dedicated reasoning layer for cross-document consistency checks. Trying to do this in a single agentic OCR system over-loads the loop and underperforms.

10-K and other financial filings are dense, table-heavy, footnote-laden documents where LlamaParse Agentic Plus is purpose-built for the workload. The premium tier's pricing (~\$0.056 per page) is justified by the structural complexity; in normal Agentic-tier mode, accuracy on filings is meaningfully worse. This is one of the few cases where "pay the premium tier" is the right answer rather than a vendor up-sell.

9.4.4 Customer onboarding (generic)

Generic customer onboarding — account creation, ID verification for moderate-risk products, address verification, basic eligibility — is high-volume, low-to-moderate-stakes, latency-sensitive. Cost-per-page matters because volumes are large; latency matters because users are waiting; accuracy matters but with looser tolerance than KYC.

Mistral OCR 3 at \$1–\$2 per 1,000 pages is hard to beat for the parsing layer. NuExtract (or NuExtract3) for downstream structured extraction gives you a cheaper, more flexible pipeline than the bundled commercial offerings. The architectural pattern is usually A or B; Pattern C is rarely justified at this end of the stakes/volume curve.

A useful generalization: **the higher-volume / lower-stakes end of the workload spectrum is where the AI-native commercial IDP entrants (Nanonets, Docsumo) and the cheap end of the AI-native specialists (Mistral, LlamaParse Cost-Effective) tend to win.** The vertical-specific premium vendors (Reducto, LandingAI Enterprise) earn their margin on the high-stakes end.

9.5 When the right answer is “don’t build this”

Most chapters in this book assume you have a use case that justifies agentic OCR and the question is *which* system to deploy. This section is for the reader who has not yet answered the prior question: *should I deploy agentic OCR at all?*

The honest answer, in 2026, is “frequently no.” The category is real, the technology works, and there are plenty of workloads where it is the right call — but there are at least as many workloads where building or buying it is the most expensive mistake the team can make. The point of this section is to give you a concrete economic and architectural threshold so you can rule yourself out cheaply, before you have spent a quarter on a pilot that was never going to break even.

9.5.1 Start with the per-page math

Realistic all-in cost-per-correct-extraction for **true agentic OCR** in mid-2026 breaks down as follows.

Cost component	Range per page	Notes
Premium agentic parsing (LlamaParse Agentic Plus, ADE Visionary tier)	\$0.050 – \$0.060	The expensive end
Standard agentic parsing (LlamaParse Agentic, ParseBench top entry)	\$0.010 – \$0.020	The 2026 sweet spot
Hybrid CV+VLM (Reducto, LandingAI ADE Team)	\$0.005 – \$0.020	
Cheapest “agentic-flavored” parser (Mistral OCR 3 Batch)	\$0.001	The floor, but closer to single-pass-with-validation than true agentic
Self-hosted open-source (OlmOCR-2 on H100)	~\$0.00018	Requires the team to operate it
HITL amortized at typical 5% escalation rate	\$0.05 – \$0.25	Often the largest line item
Eval infrastructure, vendor management, observability	\$0.01 – \$0.10	Hidden but real

Total all-in for a true agentic OCR deployment runs \$0.05 – \$0.30 per correctly-extracted page. That number is the one you should hold against your downstream value per page.

9.5.2 The one-cent threshold

If your downstream business value is around **one cent per page** — for instance, a workflow where the entire economic gain from extracting a page correctly is on the order of a penny — agentic OCR is structurally infeasible.

The math:

- A standard agentic parser at \$0.012/page consumes 1.2× your entire per-page ROI **on parsing alone**.
- A premium tier at \$0.056/page consumes 5.6× the ROI.
- Even the cheapest credible parser (Mistral OCR 3 Batch at \$0.001/page) eats roughly 10% of the per-page ROI before you have paid for engineering, HITL, or anything else.
- The only parsing-cost configuration that fits is self-hosted open-source. But the open-source path requires an engineering team to operate model serving, evaluation, HITL queues, observability, and vendor-equivalent infrastructure. That team’s loaded cost has to be amortized across enough pages to pay for itself — typically meaning you need *millions* of pages per month before self-hosted open-source pays for itself against a \$0.01/page ceiling.

The honest framing: **at \$0.01/page downstream ROI, agentic OCR has already eaten your budget before you have written a single line of integration code.** The category is wrong for the use case, regardless of how impressive the demos are.

A useful rule of thumb: **agentic OCR is economically defensible when your downstream per-page value is roughly 5 cents or higher, and clearly defensible at 25 cents and above.** Healthcare prior-authorization decisions, contract clause extraction, KYC document review, and clinical-note coding all sit comfortably above the 25-cent threshold. Commodity invoice line-item extraction, receipt digitization for analytics, and bulk archive indexing typically do not.

9.5.3 When you should not build agentic OCR

Beyond the per-page math, several other workload shapes argue against agentic OCR even when the budget exists:

- **Your real bottleneck is downstream, not parsing.** If the documents are native digital PDFs or HTML, classical PDF text extraction plus light cleanup often produces output that is functionally identical to agentic-OCR output. The DABstep result is the reminder that **extraction is not reasoning** — a system that retrieves answers poorly will not be saved by better parsing, and “we need better OCR” is often a \$100k–\$500k diagnostic mistake. Chapter 12 covers this in depth; the short version is *measure where your chain is failing before you upgrade the parsing layer*.
- **Your workload is narrow and stable.** One vendor template, three field types, layouts that change once a year. This is the workload that template-driven IDP was designed for, and it is still the cheapest answer on a TCO basis. Adding agentic OCR to a stable narrow workload is paying for variability you do not have.
- **Your stakes are low and your volume is enormous.** Hundreds of millions of commodity pages per year, where a 3% error rate is tolerable because the downstream cost of an error is pennies. Hyperscaler basic OCR plus a thin cleanup layer is roughly 50× cheaper for the same downstream outcome. The shift-left thesis (Chapter 6) does not apply at this end of the volume curve.
- **Your latency budget is sub-second.** Multi-pass reflection adds wall-clock time. If users expect interactive responses on documents — a chat interface over a PDF, a real-time form-fill assistant — Pattern A (single-pass) is the only architecture that fits, and a single-pass system is by definition not agentic OCR.
- **Your compliance posture is satisfied by a simpler system.** Some regulated workflows specifically require deterministic, auditable extraction — not “model with reflection” but “this template, this field, every time.” Agentic OCR’s flexibility can be a liability where determinism is the requirement.
- **You have not built a golden set.** Without a golden set against which to measure, you cannot tell whether agentic OCR is improving your accuracy or hurting it. The first investment is the golden set (Chapter 11), not the parser. Teams that skip this step usually deploy the wrong vendor and discover the mistake six months later.

9.5.4 “Don’t build for the sake of it”

The dominant 2026 anti-pattern at the *procurement* level — distinct from the per-vendor wrapper-washing anti-pattern in Chapter 13 — is building agentic OCR because the field is hot, not because the use case demands it. The pattern shows up as a pilot whose business case was assembled after the vendor selection, with ROI projections that conveniently round up. The pilot ships, the team learns the technology, the ROI does not materialize, the pilot quietly winds down, and the budget is consumed.

The way to avoid this trap is to invert the order. Establish the per-page downstream value **first**, with a real number that survives stakeholder scrutiny. Establish the all-in cost-per-correct-extraction **second**, using the table at the top of this section as the floor. Only if the value clears the cost — by at least 3× to give room for the inevitable scope creep — does the question “which vendor?” become live. If the gap is closer than 3×, the right answer is usually to wait. The pricing of agentic OCR is dropping faster than any other component of the modern data stack; today’s marginal workload becomes next year’s comfortable margin.

9.5.5 What to do instead

If the math says agentic OCR is not for you, the alternatives in roughly increasing order of capability are:

- **Native digital text extraction** (PDF text layer, HTML parser) for clean documents.
- **Classic OCR** (Tesseract, PaddleOCR base) plus regex/template extraction for narrow stable workloads.
- **Template-driven IDP** (ABBYY, Hyperscience) when the workload is narrow but the volume justifies a vendor.
- **Hyperscaler OCR APIs** (Textextract, Document AI, Azure DI) for moderate-volume, moderate-variability workloads where the cost-per-page is genuinely below 0.5 cents.

- **Single-pass VLM** (any frontier model with a structured-output prompt) when you need some flexibility but not multi-pass orchestration.
- **Agentic OCR** when the use case clears the cost threshold and the documents have the structural complexity to justify the loop.

The right answer is the cheapest one on this list that meets your accuracy and compliance requirements. There is no prize for using the most sophisticated technology; there is a meaningful penalty for using technology your use case cannot pay for.

9.6 The buy-vs-build question

For teams whose workload clears the “should we build this at all” filter, the next question is buy versus build. The framing that has held up in 2026: **buy the parser, build the eval**.

The case for buying the parser is straightforward. Parsing models are commodities now in the sense that any of a dozen vendors (or a dozen open-source models) can do credible work. The engineering effort to build a competitive parser from scratch — collecting training data, training a vision-language model, building the inference infrastructure — is multiples of the cost of using a vendor for the next three to five years. Almost no team’s competitive advantage lies in being slightly better at parsing than the vendor frontier.

The case for buying surrounding infrastructure (HITL queues, eval tooling, observability) is weaker but still usually positive. Vendors that have been building HITL workflows for fifteen years have engineered things that you would otherwise rediscover. The classic IDP vendors (Hyperscience especially) are often underrated specifically because their HITL infrastructure is genuinely best-in-class, even when their model layer is behind the AI-native specialists.

The case for **building your own eval** is overwhelming. Only you know what “correct” means for your documents. No vendor’s golden set matches your production distribution. No vendor’s evaluation rubric captures the specific failure modes that matter for your downstream consumers. The vendors that sell pre-built evaluation packages are selling you something that looks like an answer to the eval question but is not — it answers the *general* eval question, not the *yours* eval question. Always build your own.

The cases where the build option for the parser becomes correct rather than just defensible: data-sovereignty hard-requirements that no vendor can satisfy, scale at which API spend exceeds engineering loaded cost (typically several million pages per month), and audit requirements that demand a degree of model transparency that no commercial vendor provides. These are real but rare; most teams that think they have these requirements turn out, on close examination, to have softer versions that vendors can satisfy.

A common middle path: **use a vendor for the first 12–18 months while the team learns the workload and builds the eval discipline; migrate parsing to open-source self-hosted once the team has the operational maturity to operate it**. This pattern produces fewer disasters than either of the extremes (build everything from day one, or buy a vendor and never invest in operational depth).

9.7 A shortlist methodology

The methodology that consistently produces good vendor decisions in 2026 has four steps and takes one to two weeks per shortlist round.

Step 1: Build (or refresh) the golden set. 200–500 hand-labeled documents from your actual production traffic. Include the edge cases your team complains about most. Label them with the schema your downstream consumers actually use. This is the most expensive step and the highest-leverage one; teams that skip it produce vendor decisions that look defensible on paper and fail in production.

Step 2: Pick three vendors across three tiers. One AI-native specialist (LlamaParse, Reducto, ADE, Mistral). One hyperscaler (Document AI, Textract, Azure DI) — even if you do not think you will pick one, the hyperscaler baseline tells you what’s free or near-free in your existing cloud stack. One classic IDP or open-source option (Hyperscience or Granite-Docling, depending on whether HITL or self-hosting is the priority). Three vendors is the right number — fewer leaves you over-anchored on the first; more dilutes the comparison.

Step 3: Run all three on the golden set. Score on the metric that matches your downstream cost (cost-per-correct-extraction-with-provenance is the default; field-level F1 broken down by field is the diagnostic). Track the three numbers that matter: accuracy, cost, escalation rate. Latency too, if it is a binding constraint. Do not score on benchmark scores; you are running your own benchmark now, and yours is the only one that matters for this decision.

Step 4: Decide on data, not demos. The vendor that wins is the one that produces the lowest cost-per-correct-extraction-with-provenance on *your* golden set, subject to compliance and integration constraints. Resist the pull of the vendor whose demo was most impressive; demos are calibrated to make vendors look good, and the calibration is not your workload.

A separate methodological note: **the shortlist should be re-run annually.** The vendor landscape changes faster than procurement cycles. A vendor that was the right choice last year may not be this year; a vendor that was not credible last year may have caught up. Annual re-evaluation is good hygiene and is cheaper than the alternative (sticking with a vendor whose relative position has deteriorated).

9.8 A walked example: 80k pages/month insurance underwriting

To make the framework concrete: an insurance team running underwriting at 80,000 pages per month, mixed digital and scanned PDFs, strict compliance requirements (SOC 2 Type II, HIPAA BAA), three-month deployment window.

Applying the framework:

Volume: medium (80k pages/month). Rules out the very-low-volume “just use a hyperscaler” answer; rules out the very-high-volume “self-host open-source” answer (the team is not yet operating at the scale where self-hosted economics are decisive). Sits squarely in the AI-native-specialist sweet spot.

Variability: wide. Mixed insurance carriers, mixed document types within each carrier’s submission. Rules out template-driven IDP as the primary solution.

Compliance: SOC 2 + HIPAA. All four AI-native specialists clear SOC 2; only LandingAI ADE and Reducto have published HIPAA BAA options at the time of this writing. (LlamaParse and Mistral can typically negotiate HIPAA on enterprise plans but it is not a published default.) The compliance filter narrows the shortlist to ADE and Reducto, plus one hyperscaler or classic IDP as a sanity-check baseline.

Self-host vs. SaaS: SaaS is acceptable here given the BAA and ZDR options. Self-hosting is not required.

Stakes: moderate-to-high. Wrong underwriting decisions cost five to six figures per incident; volume-times-error-rate produces meaningful expected loss. Justifies Pattern C investment if the parsing layer cannot meet accuracy targets in Pattern B.

The shortlist that emerges: **Reducto, LandingAI ADE Team plan, and AWS Textract** (as the hyperscaler baseline). Run all three on a 300-document golden set of representative underwriting submissions. Score by cost-per-correct-extraction-with-provenance plus HITL escalation rate. Whichever wins on those metrics — and the answer is genuinely not obvious in advance for this workload — gets deployed.

This shape of decision, walked through with explicit numbers from a real-ish use case, is what good vendor selection looks like. The math is straightforward; the discipline is the rare part.

💡 Named takes

Most teams pick a vendor before they pick a metric. This is backwards. Choose the metric that determines your downstream cost, then choose the vendor that minimizes that metric.

Build-vs-buy is the wrong frame. Buy the parser, build the eval. Reuse vendor parsers; never reuse vendor evals.

The shortlist methodology — three vendors, one golden set, one cost-per-correct-extraction number — is unglamorous and consistently produces better decisions than any amount of vendor-deck reading.

IV

Benchmarks

Part IV — 75

Evaluation & Integration

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10. Benchmarks

i Executive summary

- **OmniDocBench has saturated.** Top scores cluster within a percentage point. LlamaIndex's own blog declared as much in mid-2026. If you are still picking vendors by OmniDocBench rank in late 2026, you are not paying attention.
- Use the public benchmarks as **filters** (eliminate the obviously worst), not as **rankings** (pick the best). The benchmark that actually predicts your production behavior is your golden set, evaluated on metrics that match your downstream cost (Chapter 11).
- **DABstep is the benchmark to watch.** The top reasoning model scores about 14.55% on its hard tasks. The number is low enough to function as a permanent reality check on anyone selling magic, and it punctures the most common 2026 fallacy: that better parsing solves the long-form-reasoning problem.
- Vendor-published benchmarks (ParseBench, RD-TableBench, OfficeQA) are useful but biased toward the vendor's strengths. Cross-validate; never let a single vendor's leaderboard decide procurement.

10.1 Why benchmarks matter — and where they mislead

Benchmarks are the field's shared coordinate system. Without them, every claim about agentic OCR is a vendor anecdote. With them, the conversation has a substrate of public, reproducible numbers that buyers, builders, and researchers can argue over. That much is real progress, and the 2026 benchmark landscape is dramatically more useful than the 2024 landscape was.

What benchmarks do not do — and what most procurement processes implicitly assume they do — is predict production behavior. A benchmark is a measurement of how a system performs on a specific distribution of documents under a specific evaluation rubric. Your production distribution is not the benchmark distribution. Your evaluation rubric is not the benchmark rubric. The benchmark score is a signal about the system's general competence; it is not a forecast of how the system will handle your documents.

Three failure modes are worth naming up front.

Benchmark saturation. When top scores cluster within a few percentage points, the benchmark stops discriminating between vendors. This has happened to OmniDocBench in 2026. It happens to every benchmark eventually. The right response is to treat saturated benchmarks as filters (eliminate the bottom of the field) rather than rankings (pick the top), and to seek out newer, harder benchmarks for discrimination among leaders.

Vendor publication bias. Benchmarks published by vendors are useful but selectively shaped. ParseBench is run by LlamaIndex; RD-TableBench by Reducto; OfficeQA by Databricks; SCORE-Bench by Unstructured.io. None of these are dishonest, and all of them are valuable signals. But each one is calibrated to the strengths of its publisher, and each one's top vendor on its own leaderboard is, more often than not, the publisher. Take vendor-published rankings as one signal among many, not as the deciding answer.

Distribution mismatch. The most common production failure mode is a team that picks the highest-scoring vendor on a public benchmark and then watches their accuracy fall by five to ten percentage points on Day 1 in production. The vendor's score was real; the production gap is real; the difference is that the benchmark and the production distribution were not the same. Chapter 11 deals with this in depth; the relevant point here is that no benchmark, however well-designed, predicts production behavior on a distribution it does not match.

10.2 The 2026 benchmark catalog

The benchmarks worth knowing about in 2026, ranked by current usefulness for vendor selection:

10.2.1 ParseBench (LlamaIndex + Kaggle)

Approximately 2,000 human-verified pages and 167,000 test rules across five evaluation axes: tables, charts, content fidelity, formatting, and visual grounding [4]. Targets high-stakes enterprise use cases — insurance claims, financial filings, healthcare records — where small parsing errors materially affect downstream decisions.

The live leaderboard at parsebench.ai is the most-cited 2026 reference for end-to-end agentic-OCR vendor comparison. As of mid-2026, the top result is **LlamaParse Agentic at 84.9% overall**, at approximately \$0.012 per page. The leaderboard includes 14 methods spanning vision-language models, specialized document parsers, and the various LlamaParse tiers.

Use it for: getting a broad sense of where vendors sit in the commercial-agentic landscape, comparing pricing-vs-accuracy tradeoffs across tiers, and tracking the field's overall accuracy frontier.

Caveat: published by LlamaIndex. The rule selection inevitably shapes which behaviors the benchmark rewards. LlamaParse's lead on its own benchmark is a useful data point, not a definitive ranking. Cross-check on OmniDocBench, olmOCR-Bench, or your own golden set before letting ParseBench drive a procurement decision.

10.2.2 OCRBench v2

Approximately 10,000+ QA pairs, bilingual English plus Chinese. Tests text recognition, layout understanding, and visual question-answering. Useful when your workload has heavy Chinese-language content; less useful when your production contract is structured extraction rather than visual text reasoning.

OCRBench v2 is the benchmark to track if you are choosing between models for a bilingual or Chinese-heavy deployment. For English-only or English-dominant workloads, olmOCR-Bench is a sharper signal.

10.2.3 olmOCR-Bench (AllenAI)

The most rigorously engineered benchmark in the open-source ecosystem. Approximately 7,000+ test cases across 1,400 documents, with a unit-test-style evaluation approach: each test verifies a specific property (table cell relationships, reading order, field-level extraction correctness) rather than producing a single aggregate score [5].

English-only. Annotations generated by closed-source VLMs (a known but transparent limitation; the AllenAI team has been candid about it). The current top scores as of mid-2026:

- **Chandra (9B, OpenRAIL):** 83.1 ± 0.9
- **OlmOCR-2 (8B, Apache-2.0):** 82.3 ± 1.1
- **dots.ocr (3B):** 79.1 ± 1.0
- **DeepSeek-OCR (3B):** 75.4 ± 1.0

Use it for: open-source model comparison, regression testing as you iterate on prompts or fine-tune models, and as a reproducible benchmark you can re-run on your own infrastructure. The unit-test structure makes it the easiest of the major benchmarks to incorporate into a CI pipeline.

Caveat: English-only. Annotations from closed-source VLMs introduce a slight bias toward those VLMs' interpretations.

10.2.4 OmniDocBench v1.7

1,651 pages plus a new 296-page hard subset added in April 2026. Covers 10 document types, 5 layout types, and 5 languages. Evaluates layout, formulas, tables, HTML/LaTeX rendering, and reading order [26].

OmniDocBench has saturated. This is the most important fact about it. The top of the leaderboard clusters within a percentage point or two; the discriminating signal that the benchmark provided in 2024–2025 has compressed. LlamaIndex published an explicit post titled “OmniDocBench is saturated, what's next for OCR benchmarks?” in mid-2026, which is a useful read for understanding the field's view of where benchmark evolution needs to go next.

The 296-page hard subset added in April 2026 partially addresses saturation by introducing more difficult formulas, tables, and layouts. Top scores on the hard subset still discriminate meaningfully. If you are using OmniDocBench in 2026, use the hard subset rather than the full leaderboard.

PP-StructureV3’s result on OmniDocBench — edit distance 0.145 on English documents and 0.206 on Chinese documents, at under 100M parameters — remains the cleanest data point in the benchmark’s history. The story is not “PP-StructureV3 is the best model on OmniDocBench”; the story is “a sub-100M-parameter specialized model matches Gemini-2.5-Pro’s accuracy, which means scale is not the variable that matters here.”

Use it for: cross-validation against vendor-published benchmarks, multilingual evaluation (limited but present), and the historical record of how the field has progressed.

Caveat: saturated at the top; use the hard subset for discrimination.

10.2.5 RD-TableBench (Reducto)

1,000 adversarial tables, PhD-annotated, evaluated using Needleman-Wunsch alignment for table-cell matching [6]. The most demanding public benchmark specifically for table extraction.

Top result: **Reducto at approximately 0.90 cell-alignment similarity**, approximately 20 percentage points ahead of the hyperscaler APIs. The gap is real and reproducible.

Use it for: evaluating any system whose workload involves complex tables (financial filings, scientific papers, regulatory documents, invoices with nested line items). RD-TableBench’s adversarial nature — the tables are explicitly chosen to be hard — makes it a sharper signal of real-world table handling than general benchmarks.

Caveat: published by Reducto, with Reducto at the top of the leaderboard. The pattern repeats across vendor-published benchmarks; treat the leadership as one signal, cross-validate before deciding.

10.2.6 DABstep (Adyen + Hugging Face)

The most important 2026 benchmark, and the one most likely to be ignored by procurement processes that should be paying attention.

450+ real-world data-analysis tasks pulled from Adyen’s operational workloads [7]. Mixes structured (CSV, JSON) and unstructured (text, manuals, internal documentation) data. Requires agents to demonstrate multi-step reasoning — extraction is necessary but not sufficient.

Top score: approximately 14.55% on hard tasks. The best-performing model (o4-mini with reasoning prompting) scores 14.55%; the overall best-in-class on the full benchmark is around 16%.

That ceiling is low enough to be the field’s most useful reality check. The story DABstep tells: **frontier reasoning models, given parsed data and tools, fail roughly 85% of multi-step analysis tasks**. The framing that the field has unlocked complex document understanding is contradicted by this number more thoroughly than by any other 2026 result.

DABstep is the benchmark to cite when arguing that: - Extraction and reasoning are separate problems, and solving one does not solve the other. - “Better OCR will fix our agent” is sometimes a \$500k mistake (Chapter 12 returns to this). - The field has more headroom for improvement than the saturated parsing benchmarks suggest.

Use it for: forecasting where agentic-OCR-plus-reasoning systems will struggle, evaluating any system that claims to handle multi-document or multi-step workloads, and as a permanent humility check.

Caveat: small N (450 tasks); narrow distribution (Adyen’s operational workloads). The ceiling is real; the specific number may move as models improve.

10.2.7 SCORE-Bench (Unstructured.io) — new in 2026

1,000+ page enterprise dataset spanning scanned invoices, nested tables, handwritten notes, schematics, patents, marketing collateral [27]. Built specifically for generative parsing systems with an evaluation framework that separates legitimate representational diversity (two valid ways to represent the same content) from actual extraction errors.

SCORE = Structural and Content Robust Evaluation. The interpretation-agnostic design is the most novel aspect: traditional benchmarks penalize models that emit a structurally valid output that differs from the reference (a Markdown table vs. an HTML table containing the same content, for instance). SCORE accepts equivalent representations and only penalizes actual extraction errors.

The benchmark is new and worth watching. As of mid-2026 it has been used to compare Reducto, LlamaParse, Docling, Snowflake, Databricks, and NVIDIA — a useful set of cross-vendor comparisons that did not exist before. The methodological contribution may end up being more important than the leaderboard.

Use it for: cross-vendor parsing comparisons, especially for downstream use cases where the parsed output feeds a generative model (RAG, agentic workflows) and representational flexibility is acceptable.

Caveat: published by Unstructured.io; new enough that the field has not converged on how seriously to take it. Worth tracking.

10.2.8 OfficeQA (Databricks) — new in 2026

Enterprise document-workflow reasoning tasks. The benchmark on which Databricks reported their **16% across-the-board agent-performance gain** when `ai_parse_document` was inserted upstream of the reasoning step (Chapter 6).

The headline number on OfficeQA itself: **frontier agents score below 50% accuracy** on document-reasoning tasks without good upstream parsing. Add good parsing and the average improves by 16 percentage points across every agent framework tested.

The benchmark’s methodology and dataset are described in arXiv 2603.08655. The numbers are vendor-published (Databricks), with the appropriate caveats, but the directional finding — that reading is the ceiling on agent quality — is robust across the wider 2026 literature.

Use it for: framing the shift-left thesis in vendor pitches and internal architecture discussions, citing the 16% number as the cleanest piece of evidence for upstream parsing as an agent-quality lever.

Caveat: new benchmark, Databricks-maintained, used primarily to demonstrate the value of Databricks Document Intelligence. The thesis the benchmark supports is broader than the vendor’s product.

10.3 What each benchmark actually measures

The data in `data/benchmarks.csv` consolidates the per-benchmark details. The compressed cross-comparison:

Benchmark	Year	Scope	Top score	Top vendor / model	Best use
ParseBench	2026	End-to-end agentic parsing	84.9%	LlamaParse Agentic	Commercial vendor comparison
OCRBench v2	2025	Text + layout + QA (EN+ZH)	—	Varies	Bilingual workloads
olmOCR-Bench	2025	Open-source unit tests (EN)	83.1	Chandra (9B)	Open-source regression
OmniDocBench v1.7	2026	Diverse docs, multilingual	Saturated	Cluster within 1pp	Cross-validation; use hard subset
RD-TableBench	2025	Adversarial tables	~0.90	Reducto	Table extraction
DABstep	2025	Multi-step reasoning	~14.55%	o4-mini	Reasoning ceiling check
SCORE-Bench	2026	Generative-parser eval	—	Varies	Cross-vendor representational fairness
OfficeQA	2026	Doc-reasoning workflows	<50% (without parsing)	—	Shift-left thesis evidence

The compressed reading: ParseBench and olmOCR-Bench for vendor and model selection respectively; RD-TableBench for table-heavy workloads; OmniDocBench’s hard subset for cross-validation; DABstep for humility; SCORE-Bench and OfficeQA for new-methodology signals worth tracking.

10.4 What benchmarks miss

Benchmarks are useful precisely because they constrain what they measure. The constraint is also their limitation. Five things are systematically under-measured in the 2026 public benchmark landscape:

Latency. Most benchmarks score accuracy and ignore wall-clock time. Production deployments care a great deal about latency, especially for interactive use cases. A vendor with 2% higher accuracy and 10× higher latency is not better than the alternative for many real workloads. Track latency in your own evaluation; do not let benchmark scores conceal it.

Cost-per-correct-output. Benchmarks report accuracy *or* cost separately, rarely jointly. The number that predicts your monthly bill is cost-per-correct-extraction-with-provenance, not accuracy. The combination is workload-specific and is not in any public benchmark.

Long-form reasoning. DABstep is the only public benchmark that meaningfully tests this, and its ceiling is ~14.55%. Most agentic-OCR systems sold for “intelligent document understanding” have never been benchmarked on long-form reasoning at all. The gap between extraction accuracy and downstream-task accuracy is real and large; benchmarks that elide it produce false confidence.

HITL acceptance rate. No public benchmark measures the metric that matters most in production — the rate at which human reviewers accept the system’s extractions versus the rate at which they correct them. This is a production-only metric, and the absence from public benchmarks is one of the deeper structural gaps in the 2026 evaluation landscape.

Domain shift. Every benchmark has a distribution. Your production distribution is different. The gap between benchmark performance and your production performance is workload-specific, and no benchmark, however well-designed, predicts it. The only honest answer is to evaluate on your own data.

10.5 When to ignore the leaderboard

A few specific situations where the right answer is “stop reading the leaderboard”:

- **When the top 5 cluster within 1–2 percentage points** on a metric your team does not directly care about. This is OmniDocBench in mid-2026 — the discrimination has compressed to inside the benchmark’s own measurement noise.
- **When the benchmark is published by the vendor whose model leads it**, and you have not yet cross-validated on a non-vendor benchmark. Vendor leaderships on vendor benchmarks are real but biased. One vendor benchmark is a signal; three independent benchmarks plus your golden set is a decision.
- **When the benchmark has not been updated in 12+ months.** OCR benchmark leaderboards have a short half-life. A 2023 benchmark in 2026 is documenting a different problem than the one you have.
- **When your golden set says something different.** Your golden set wins. Always. The benchmark may say Vendor A scores 84%; your golden set may say Vendor A scores 71%. The 71% is the number that matters. The benchmark documented a behavior on a distribution that is not yours.

The last point is the strongest. Public benchmarks are valuable as a shared coordinate system for the field. They are not a substitute for measuring on your own data. Teams that internalize this run vendor selection well; teams that do not get burned with predictable regularity.

10.6 A note on benchmark evolution in 2027

The 2026 benchmark landscape will look different in 2027. Three predictions worth flagging:

- **OmniDocBench will be deprecated or replaced** as its saturation deepens. SCORE-Bench is the leading candidate to replace it; a new benchmark from a research group not yet active in 2026 is also plausible.
- **A long-form reasoning benchmark will overtake DABstep in attention** as the parsing-quality benchmarks saturate. DABstep is the leading edge of this transition; the next benchmark will likely combine multi-document parsing with multi-step reasoning at greater scale.
- **Cost-aware benchmarks will appear.** No 2026 benchmark integrates accuracy and cost into a single metric. Some research group will publish one in 2027, and it will be more useful for procurement than any of the accuracy-only leaderboards currently in circulation.

These are predictions, not promises. Track the space; do not assume the 2026 benchmark set is the permanent set.

 Named takes

OmniDocBench told us a year ago what works. It can't tell us anymore. If you're still picking vendors by their OmniDocBench number in late 2026, you're not paying attention.

DABstep is the only benchmark whose top score embarrasses the field. It is therefore the benchmark to watch.

Your golden set wins. Every public benchmark documents a distribution that is not yours. Treat them as filters, not rankings.

11. Evaluating Your Own Pipeline

i Executive summary

- The single most important investment in agentic OCR is a **golden set** of 200–500 hand-labeled documents drawn from your actual production traffic. Build it before you choose a vendor. Teams that skip this step pick the wrong vendor with predictable regularity.
- **Metric selection is domain-dependent.** Character Error Rate is wrong for structured extraction. Field-level F1 is the default. Table-cell accuracy needs alignment (Needleman-Wunsch), not Jaccard. Schema-validity rate is cheap and surprisingly diagnostic. Pick the metric that predicts your downstream cost.
- The HITL queue is **part of your eval**, not a fallback bolted on the side. Reviewer accept/correct rates are the most honest production-accuracy signal you will get, and most teams underutilize them.
- The number that predicts your monthly bill is **cost-per-correct-extraction-with-provenance**. It is computable from three measured numbers (API cost, accuracy, HITL escalation rate $\times$ HITL cost). It is the answer; the rest is mostly noise.

11.1 Why the public benchmarks won't save you

A common 2026 procurement story: a team picks the highest-scoring vendor on ParseBench, deploys to production, and watches accuracy fall by 5–10 percentage points on Day 1. The benchmark score was real. The production gap is real. The difference is that ParseBench's distribution and the team's production distribution are not the same.

This is not a problem with ParseBench. The benchmark is well-designed for what it does, and what it does — measure end-to-end agentic-parsing quality on a curated set of high-stakes enterprise documents — is genuinely useful. The mistake is treating any public benchmark as a forecast of your production behavior. No public benchmark, however well-designed, predicts your performance on documents the benchmark has never seen.

The discipline that closes the gap is straightforward to describe and operationally rare to execute. Build a golden set. Measure the right metrics. Run regression tests. Treat HITL as part of the eval. Compute the cost-per-correct-extraction number. Repeat quarterly.

This chapter walks each of those steps. Most of it is not new — every team operating agentic OCR in production has run into these requirements eventually. The hope of this chapter is that you hit them deliberately rather than after the third production incident that the public benchmarks did not predict.

11.2 Golden sets — the only thing that matters

A golden set is a collection of hand-labeled documents from your production traffic, against which every parsing decision is evaluated. Its purpose is to give you a measurement that *does* predict your production behavior, because it *is* your production behavior — sampled, labeled, and made into a stable evaluation target.

The discipline has four parts:

Selection. Pull 200–500 documents from your actual production traffic. Stratify by document type, source vendor, and difficulty. Include the edge cases your team complains about most loudly — the ones that the system gets wrong consistently — not the easy cases your current pipeline already handles. The temptation to over-represent clean documents because they are faster to label is the most common golden-set construction error.

Labeling. Hand-label each document with the schema your downstream consumers actually use. Field by field. Not the schema in a vendor's documentation — your schema, the one your downstream system

expects. Label the cases the system finds ambiguous: missing fields, illegible regions, contested values. The labels embed the team’s judgment about correctness, and that judgment is the load-bearing artifact.

Provenance metadata. For each labeled field, capture the source region (page number, bounding box) where the value originated. This makes the golden set usable not just for accuracy measurement but for **grounding accuracy** measurement — does the vendor’s reported source region match the actual region. Grounding measurement is a separate axis, and the golden set is the only place you can measure it reliably.

Maintenance. Add 50 new documents per quarter, drawn from recent production. Retire documents that have become unrepresentative (vendors that no longer exist, document types that no longer appear). The golden set is a living artifact, not a one-time project. Teams that label once and never update produce evaluations that predict behavior on the 2024 distribution while their production distribution moved on.

The labor cost is real. A 200-document golden set with thorough labeling and provenance metadata typically takes one to three weeks of focused effort by a labeler who understands the domain. The labor cost is also small relative to the cost of picking the wrong vendor. Teams that resist building a golden set on cost grounds routinely spend ten times the labor cost on a wrong-vendor rollback six months later.

A specific recommendation worth highlighting: **label the golden set with the same people who will be operating the HITL queue.** Their judgment about what constitutes a correct extraction is the judgment that will govern the production system. Labels from a separate “evaluation team” drift away from operational reality and produce eval scores that do not match what the HITL reviewers find in practice.

11.3 Metric selection per domain

The wrong metric quietly destroys vendor selection. The choice of metric needs to match the downstream cost of an error, and the right choice varies by workload type.

Character Error Rate (CER). The metric that classic OCR has used for thirty years. Correct for plain-text OCR — does the system get the characters right. Wrong for structured extraction — penalizes formatting differences that do not matter (extra whitespace, equivalent date formats, capitalization variants on enum values) and under-penalizes structural errors (a wrong field value in a correctly-extracted character stream).

CER is the right metric for: scanning books, indexing email attachments, accessibility-focused OCR. It is the wrong metric for most agentic-OCR workloads.

Field-level F1. The default metric for structured extraction. Per-field precision (of the fields the system extracted, what fraction match the golden label) and recall (of the fields in the golden label, what fraction did the system extract). F1 is the harmonic mean.

Field-level F1 is the right metric for: invoices, forms, contracts, claims, KYC documents, any workload where the output is a fixed-schema JSON object. The right granularity is per-field, not aggregate: per-field F1 tells you which specific fields the system handles well and which it struggles with, which is operationally diagnostic. Aggregate F1 averages across fields and hides the variance.

Table-cell accuracy with alignment. Tables need their own metric. Jaccard similarity on the set of cell values is wrong because it does not respect cell position — a system that gets every cell value right but in the wrong row and column scores perfectly on Jaccard and is, in fact, useless.

The right alignment metrics are **Needleman-Wunsch** (used in RD-TableBench [6]) or **Hungarian assignment**, both of which match predicted cells to ground-truth cells with positional constraints and penalize misalignments. The exact choice between Needleman-Wunsch and Hungarian depends on whether row/column order matters more (Hungarian) or whether sequence alignment is the right model (Needleman-Wunsch). For most enterprise tables, Hungarian is the safer default.

Schema-validity rate. Cheap, deterministic, and surprisingly diagnostic. Does the system’s output parse as your downstream schema? Are required fields present? Are dates in valid ranges? Are enums populated with allowed values?

Many extraction errors that confidence checks miss show up as schema failures. A system with 95% field-level F1 and 80% schema-validity rate is producing per-field correct outputs that aggregate into structurally invalid documents — a pattern you want to catch in eval, not in production.

Grounding accuracy. For compliance workloads, does the vendor’s reported source region for a field actually contain the extracted value? Measured as the IoU (intersection over union) between predicted and ground-truth bounding boxes, or as a binary “does the predicted region contain the value” check.

Without grounding accuracy, you cannot audit. Without auditability, regulated workloads cannot deploy. Make this metric part of the eval from day one, not a feature added at compliance-review time.

Reasoning accuracy. For DABstep-style downstream tasks where the agentic system’s output feeds into a reasoning step. This is a separate eval from extraction; conflating them produces misleading results.

If your workflow has a reasoning layer, evaluate both extraction (with field-level F1) and downstream task success (with task-specific metrics) separately.

A general principle: **pick one to three metrics that match your downstream cost, track them per-field where applicable, and resist the urge to add more.** A team that evaluates on twelve metrics is implicitly saying it has no model of which errors matter most. Pick the metrics. Defend the choice. Track them religiously.

11.4 Regression testing

Vendor versions change. Even when the vendor name on your contract does not change, the model behind the API gets retrained, retuned, or rebuilt — and the new version sometimes behaves worse than the old version on your specific documents. Regression testing is the discipline that catches this before it ships.

The cadence that works in production:

- **Weekly: 50-document subset.** A fast sanity-check on the highest-traffic document types. Run automatically after any vendor version-bump notification.
- **Monthly: full golden set.** The full 200–500 document evaluation. Track per-field F1 over time. Look for drift in specific fields, not just aggregate.
- **Quarterly: golden-set refresh.** Add new documents, retire stale ones. Re-baseline the per-field expectations on the refreshed set.
- **Vendor major-version bumps: full re-eval before promotion.** Vendors that release major versions (LlamaParse v2 → v3, Reducto major-version updates, etc.) should not be promoted to production until the full golden set has been re-evaluated. This is the case where the rollback risk is highest.

Pin vendor versions in production. Most commercial vendors support version pinning; use it. The cost is missing out on improvements between pinned-version and current; the benefit is that production behavior changes only when you decide to upgrade, not when the vendor decides to roll out a new model overnight.

Track regression as a separate operational metric. A healthy regression cadence catches roughly one to three regressions per year on a stable golden set. Zero regressions per year is a sign your golden set is not sensitive enough; more than five per year suggests vendor instability that may itself be a procurement signal.

11.5 HITL feedback loops

The HITL queue is part of the eval. This is the point that most teams underweight, and it costs them.

Three production metrics live in the HITL queue:

Escalation rate. The percentage of documents the system routes to a human reviewer. Healthy 2026 deployments sit in the 1–10% range in steady state. Below 1% suggests the system is silently accepting errors that should have escalated; above 10% suggests extraction is too unreliable for the workload (or eval rules are too strict).

Acceptance rate. Of escalated documents, the percentage where the reviewer accepts the system's extraction as-is (versus correcting it). Healthy range: 30–70%. Below 30% means the system is escalating documents it should have caught itself but the model still produced acceptable output; above 70% means the system is escalating documents it actually got right and wasting reviewer time.

Correction patterns. When reviewers correct, *what* do they correct? Track the per-field correction rate. A field that reviewers correct 40% of the time is a field the system handles poorly, regardless of what the per-field F1 on your golden set says. Production reviewer corrections are the most honest signal about model performance on novel documents — by definition, those documents were not in your golden set.

Closed-loop is the goal. Reviewer corrections should feed back into:

- The golden set (high-disagreement documents become labeled examples).
- Prompt tuning (systematic errors trigger prompt revisions).
- Vendor selection (a vendor whose corrections cluster in your high-value fields is a worse fit than one whose corrections cluster in low-value fields, even at equal aggregate accuracy).
- Schema design (corrections that flag fields as ambiguous suggest the schema may need a “needs review” sentinel rather than an aggressive default).

The anti-pattern is HITL as a dead-letter queue — reviewers correct documents one at a time, and the system never learns. This is the pattern that produces flat or declining accuracy over time while leadership wonders why the agentic-OCR investment is not paying off.

11.6 Grounding and auditability for compliance

For regulated workloads, grounding is not a feature, it is a requirement.

Practically, this means every extracted field carries metadata identifying the source region — page number, bounding box, optionally specific text span. The downstream consumer can render the output alongside the document with the cited region highlighted. The compliance auditor can verify the extraction. The HITL reviewer can correct it efficiently.

The eval implication: **grounding accuracy is a first-class metric, not a checkbox**. Measure it on the golden set. Track it over time. A vendor whose field-level F1 is 95% but whose grounding-accuracy is 70% is a worse fit for a compliance workload than a vendor at 92%/95% — the second vendor's outputs are auditable; the first vendor's are not.

The retention discipline matters too. Audit logs for grounded extractions need to persist for the regulatory retention window — six years for HIPAA, varying by jurisdiction for GDPR and SOX-equivalent regimes. Verify that the vendor's audit-log retention matches your regulatory window before signing, not after the first audit.

For ZDR (Zero Data Retention) commitments — LandingAI ADE Team plan, Mistral self-hosted, certain Reducto configurations — verify in eval that ZDR is actually honored in production. The verification is operationally non-trivial: you cannot easily prove a negative (“the vendor did not retain my document”), but you can verify that the vendor's audit logs and infrastructure are consistent with the ZDR claim. Some compliance regimes require a SOC 2 Type II report that specifically attests to ZDR; insist on this rather than the bare contractual claim.

11.7 Cost-aware evaluation

The metric most teams report — accuracy — is the wrong target. The metric that predicts the monthly bill is **cost-per-correct-extraction-with-provenance**, which combines accuracy with cost and HITL load:

$$\text{cost-per-correct} = \frac{\text{API cost per page}}{\text{accuracy}} + \text{HITL cost per escalation} \times \text{escalation rate} \quad (11.1)$$

Worked example. A vendor at \$0.012 per page with 90% accuracy and a 5% escalation rate at \$3 per escalation:

- API contribution: $\$0.012 \div 0.90 = \0.0133 per correct extraction.
- HITL contribution: $\$3 \times 0.05 = \0.15 per page; $\$0.15 / 0.90 = \0.167 per correct extraction.
- Total: \$0.180 per correct extraction.

A competing vendor at \$0.008 per page with 82% accuracy and an 8% escalation rate at the same \$3 per escalation:

- API contribution: $\$0.008 \div 0.82 = \0.00976 per correct extraction.
- HITL contribution: $\$3 \times 0.08 = \0.24 per page; $\$0.24 / 0.82 = \0.293 per correct extraction.
- Total: \$0.303 per correct extraction.

The second vendor is 33% cheaper per page and 68% more expensive per correct extraction. The cost-per-page number is misleading; the cost-per-correct number is the actual answer.

Run this calculation on every vendor in your shortlist. Most teams find the ranking changes — sometimes substantially — relative to ranking by cost per page or accuracy alone. The combined metric is the one that should govern the decision.

A practical note on the HITL cost estimate. A reviewer who is a domain expert (a paralegal for legal documents, a nurse for clinical records) loads at roughly \$50–\$150 per hour. At three to five minutes per review, that translates to \$2.50–\$12.50 per escalation. Plug your team's actual numbers in; the order of magnitude is robust, but the specific cost-per-correct number is sensitive to this input.

11.8 Production cost observability

Most teams treat cost as a finance problem rather than an evaluation problem. In agentic OCR specifically this is the wrong framing. Cost is volatile in this category in ways that traditional infrastructure cost is not: a vendor changes its pricing tiers, a model regression bumps the average retry budget, a misclassified batch flows through the most expensive premium tier instead of the standard one, a runaway query

against the platform-tier products hits five figures in a single run. The failure modes are operational, not financial, and they belong in the observability stack.

Three things every production agentic OCR deployment should be tracking from week one.

11.8.1 Per-document cost attribution

Every processed document should carry an attached cost record. Not “we spent \$X this month on parsing” — that’s a finance report. What’s needed is **per-document cost broken down by step**: classification cost, parsing cost (and which tier was invoked), retry cost (how many retries triggered, and at what unit price each), validation cost, HITL escalation cost where applicable.

The cost record is what lets you answer the question that matters in incidents: *why did this document cost forty cents when the median is three?* Without per-document attribution, the answer is unfindable. With it, the answer is usually “the reflection loop retried six times, each at the agentic-plus tier” — which is then a fixable failure mode.

The implementation: emit a structured cost event to your observability backend alongside every extraction. Most teams under-engineer this and discover they cannot trace cost back to documents when a budget alarm fires. Engineer it once, properly, before you need it.

11.8.2 Budget alerts and circuit breakers

Two thresholds worth defining explicitly:

- **Daily/monthly budget envelope.** A soft alert at 60% of expected spend, a hard alert at 90%, an action threshold at 110%. The action threshold should trigger a circuit breaker — automatic throttling, rejection of low-priority work, or in the worst case a temporary pipeline pause — rather than continuing to burn through budget until someone notices on the bill.
- **Per-document cost ceiling.** A single document should not cost more than 3–5× the median for its document class. When it does, something has gone wrong (retry-thrash, mismatched tier, classification error routing to a more expensive pipeline). The right response is to stop processing that document and escalate to HITL or a dead-letter queue rather than let the reflection loop burn an unlimited retry budget.

Both of these need to be implemented at the application layer, not relied on at the vendor layer. Vendors will happily process the runaway document and bill you for it. Your team’s spend discipline is your problem.

A 2026 incident worth knowing about: a Snowflake Cortex customer reported a single `AI_PARSE_DOCUMENT` query that ran up roughly \$5,000 in credits when run against a large corpus without throttling [28]. The pattern is common across all platform-tier products — the convenience of “just call this SQL function” hides how aggressively the function will scale when given an aggressive query. Circuit breakers are the cure.

11.8.3 Vendor-cost drift detection

Vendors change prices. Models change behavior. A model retrain at the vendor can shift the median number of input tokens per document — at frontier-VLM API pricing that translates directly to a higher cost per page even though the per-token rate did not change. The vendor’s pricing page still says the old number; your monthly bill says the new number.

Track three metrics weekly on a representative subset of production traffic:

- **Median cost per page**, by document class and vendor tier.
- **Median cost per correct extraction** (i.e., cost / accuracy on the golden set).
- **P95 cost per document** — the long tail is where retry-thrash and runaway extractions hide.

A 10%+ shift week-over-week on any of these is a signal to investigate. Sometimes it is real (vendor pricing change, model retrain) and you adjust budgets. Sometimes it is a regression on your side (classification accuracy dropped, more documents now route to the expensive tier) and you fix it.

11.8.4 The cost / accuracy frontier

The denominator that should govern operational decisions — **cost-per-correct-extraction-with-provenance** — is itself a moving target. As vendors evolve and as your document distribution shifts, the optimal

point on the cost / accuracy frontier moves. Teams that rebalance vendor mix or tier choice on this signal every quarter outperform teams that pick a configuration once and ride it until budgets force a panic.

Practical pattern: maintain a small “challenger” eval — 5–10% of production traffic routed to a different vendor or tier — and compare cost-per-correct against the current champion. Promote when the challenger wins on the joint metric; iterate. This is the same idea as champion/challenger experimentation in ML deployment, applied to vendor selection.

11.8.5 Runaway-cost failure modes

A short catalog of cost incidents the field has accumulated by mid-2026:

- **Retry-thrash on a difficult document class.** The reflection loop burns its full retry budget per document because the model cannot converge on a particular layout. Per-document cost goes up 5–10×, escalation rate creeps up, monthly bill spikes. Fix: detect thrash early (per-document retry count > N triggers escalation to HITL rather than further retries).
- **Tier misrouting.** A classification mistake sends documents to the premium tier when the standard tier was appropriate. Fix: cross-check classification confidence against tier cost — if classification is below 70%, default to the cheaper tier with HITL on the output rather than burning premium-tier cost on uncertain documents.
- **Vendor pricing changes mid-month.** A vendor doubles a tier’s price; your budget assumptions are now wrong; the monthly bill is a surprise. Fix: weekly cost-per-page tracking, alerting on shifts.
- **Platform-tier “convenience runaway.”** Single `AI_PARSE_DOCUMENT`-style calls run against unscoped corpora, billing thousands in credits. Fix: enforce row-limit or document-count caps at the orchestration layer, not at the SQL function level.
- **Frontier-VLM context-window inflation.** A prompt change or document trend pushes input tokens up; cost rises silently because token-based pricing is invisible per-document until you aggregate. Fix: track input-token median per document class as a separate metric.

Each of these is preventable. None of them is automatically prevented by any vendor. Operational cost observability is the discipline that catches them before they catch you.

11.9 When your eval is the bottleneck

A few honest observations about eval that most vendor pitches will not surface:

If your eval is fast, your eval is suspicious. Real evals on 200–500 documents take minutes to tens of minutes to run end-to-end, depending on the vendor’s API and your eval rubric. An eval that completes in seconds is probably scoring against a too-small subset, a too-permissive rubric, or both.

If your eval is small (under 100 documents), it can’t reliably distinguish vendors within 5 percentage points. The statistical noise on a 50-document eval is large enough that a vendor scoring 87% and one scoring 82% are not reliably different. 200+ documents is the minimum for reliable cross-vendor comparison.

If your eval has no OOD slice, you don’t know how the system fails on novel documents. Hold out a slice of explicitly out-of-distribution documents — document types your production traffic has not yet seen, or vendor formats new to your pipeline — and measure separately. Most production incidents start as OOD failures; an eval that never tests OOD performance is silent about the most consequential failure mode.

Investment in eval is almost always under-funded relative to investment in model selection. The standard ratio is something like 10:1 in favor of evaluating vendors and against building the eval that makes vendor evaluation meaningful. Reverse this. Spend more time on the golden set, more time on metric selection, more time on regression cadence. The vendor decision becomes obvious once the eval is good.

11.10 What good eval looks like in production

A healthy 2026 agentic OCR deployment exhibits these eval properties simultaneously:

- **Golden set of 200–500 documents** from production traffic, refreshed quarterly.
- **One to three metrics** matched to the workload (typically: field-level F1, schema-validity, grounding accuracy).
- **Per-field tracking**, not just aggregate.

- **Regression cadence** — weekly subset, monthly full, quarterly refresh, vendor-version full re-eval.
- **HITL closed-loop**: corrections flow back into the golden set, prompt tuning, vendor selection.
- **Cost-per-correct-extraction** computed and reported as the primary economic metric.
- **OOD slice** held out and measured separately to surface novel-document failures.

No deployment achieves all seven on day one. Teams that work toward them iteratively, over twelve to eighteen months, produce robust, defensible eval infrastructure. Teams that treat eval as a setup task to be completed and forgotten produce evaluations that gradually lose connection to production reality and start lying without warning.

💡 Named takes

Golden sets are unfashionable, slow, and the single best ROI any agentic OCR team can deliver. Build one before you buy anything.

If your eval is faster than your inference, your eval is wrong. Real evals are not fast.

The HITL queue is part of the eval, not a fallback bolted on the side. The metric that tells you the most about your system in production lives in the reviewer's accept/correct decisions.

12. Agentic OCR and RAG

i Executive summary

- **Bad parsing kills RAG silently.** Retrieval returns plausible-looking chunks; the generator answers fluently from them; the answer is wrong; the system has no way to know. The user-facing failure rate diverges from internal retrieval metrics, and most teams discover this only after a high-visibility incident.
- Agentic OCR helps RAG in three concrete ways: **structural metadata** for semantic chunking, **table awareness** that respects cell relationships, and **grounding back to source** that makes citations real and audits possible.
- **OCR is not always your RAG bottleneck.** If your documents are clean native PDFs or HTML, the parsing layer is probably not the limit on quality; retrieval, embedding, or generation usually is. Diagnose the chain before you upgrade — “we need better OCR” is sometimes a \$100k–\$500k diagnostic mistake.
- RAG evaluation that does not include questions whose answers live in tables is RAG evaluation that lies. Tables are 30%+ of enterprise document content and 80%+ of the parsing errors. Build the table evals first.

12.1 A representative production failure

A common 2026 incident report, summarized to its operational shape:

A bank deploys an internal RAG system that lets compliance officers ask questions over its document corpus — policies, procedures, regulatory filings, vendor agreements. The team builds the system on a standard stack: PDF parsing (a hyperscaler API in default mode), fixed-window chunking, OpenAI embeddings, a vector store, and a frontier LLM for generation. They run an internal evaluation against 200 hand-curated QA pairs and the system scores 92%. Confident in the result, they ship to a pilot group of compliance officers.

Three months later, the pilot’s user-satisfaction survey reports an answer-quality score equivalent to about 41% correctness. The team is initially puzzled — the internal eval still shows 92% — and then they sit down with five of the pilot users and run real queries.

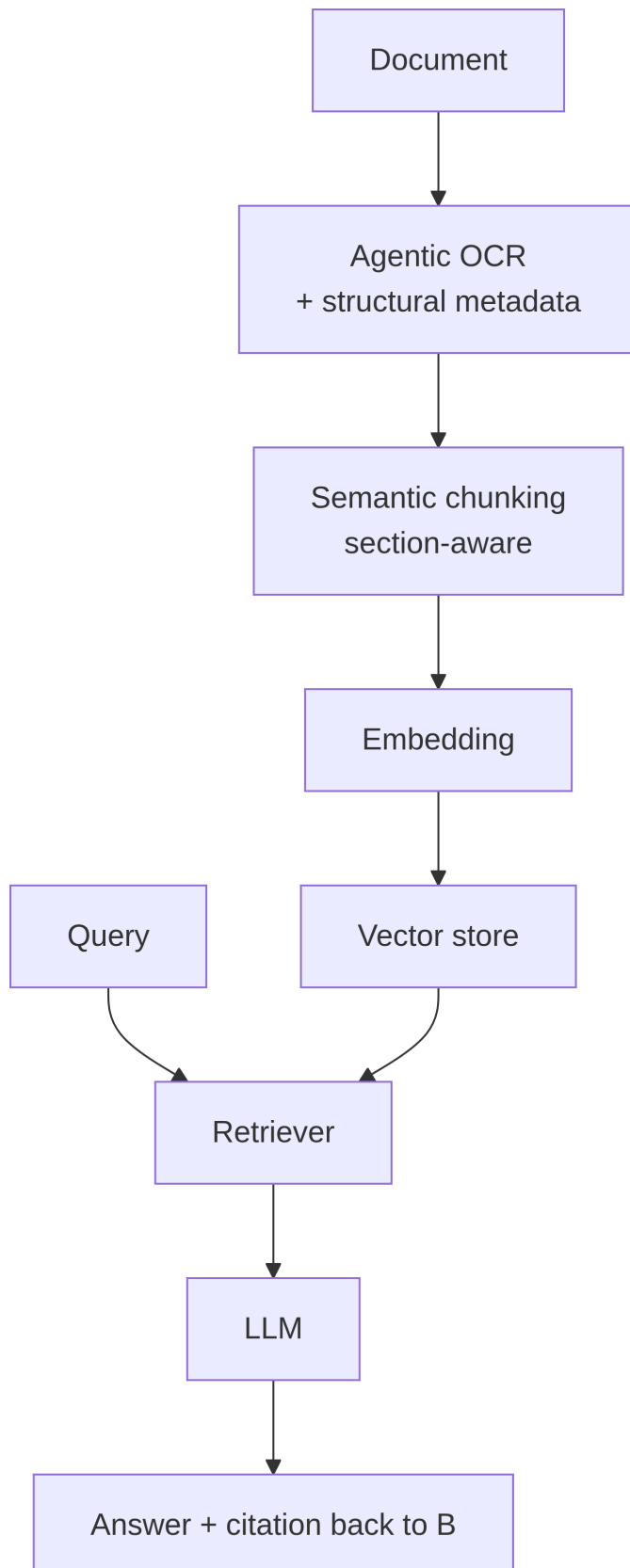
The problem becomes immediately visible. The system handles prose-based questions (“What is our policy on third-party data sharing?”) well. It handles structural-fact questions (“Who is the named contact in our Acme vendor agreement?”) well. It fails — silently, fluently, confidently — on questions whose answers live in tables. “What were our average prior-auth turnaround times for cardiology procedures in Q3?” The answer lives in a table on page 47 of a quarterly report. The parser, in default mode, dropped the table structure during extraction; the table is in the corpus as a sequence of text fragments with no row/column relationships preserved. The retriever found the fragments; the generator stitched them into a fluent-sounding answer; the answer is wrong.

The internal QA eval did not catch this because the eval, like most early RAG evaluations, had no table-based questions. The team built questions about the text in the documents because text is what they thought of when they thought of “what’s in the documents.” Tables are 30%+ of the content. They were invisible to the eval.

This is a representative shape for how agentic OCR and RAG intersect in 2026, and it is the shape this chapter walks through. The diagnosis is bad parsing; the consequence is invisible RAG failures; the fix is a combination of better parsing, table-aware chunking, and an evaluation regime that does not lie about what the system is doing.

12.2 Why bad parsing kills RAG silently

The failure mode in the example above is not specific to that team. It is structural in how RAG pipelines compose. Let's walk the chain carefully.



How agentic OCR feeds the RAG pipeline

Each step has its own failure mode, and the failures compound in a way that is invisible from any single step's metrics:

- **Parsing** drops tables, misreads numbers, loses structure. The output is text; the text is plausible; nothing screams that something is wrong.
- **Chunking** — if it does not know about structure — splits across semantic boundaries. A chunk may contain the last sentence of one section and the first sentence of the next, with no signal that they are unrelated. Fixed-window chunking is the dominant pre-2024 approach and is the leading source of this failure.
- **Embedding** turns broken chunks into vectors. Embedding is robust to small input errors but cannot fix structural breaks; a chunk that is two unrelated half-paragraphs embeds to a meaningless vector. Cosine similarity searches will still match it on surface features.
- **Retrieval** returns top-k chunks with high similarity scores. The scores are real; the chunks are bad. The retriever has no way to know.
- **Generation** answers fluently from bad chunks. LLMs are excellent at making plausible-sounding answers out of plausible-sounding inputs. This is the step that makes the failure invisible: a wrong answer with confidence indicators that say everything went fine.

The compounding effect is the central point. Each step has a small failure rate. Their product is large. A pipeline where each step succeeds 95% of the time has end-to-end success of about 77% — and that is assuming the failures are independent, which they are not. Failures correlated through bad parsing produce much sharper user-visible drops.

What makes this silent in metrics: every step's internal metric still looks fine. Retrieval precision (does the retrieved chunk contain text similar to the query?) is high. Generation perplexity is normal. The chunks have valid embeddings. The pipeline has no internal observability that connects the user-facing wrong answer to its upstream cause.

12.3 The three concrete contributions agentic OCR makes to RAG

Agentic OCR (Pattern B or Pattern C from Chapter 4.) helps RAG in three specific ways that classic OCR and single-pass VLMs cannot match. Each one addresses a specific failure in the chain.

12.3.1 Structural metadata that drives semantic chunking

A Pattern C parser emits a document representation that includes heading hierarchy, section labels, paragraph boundaries, and explicit structural relationships. A downstream chunker can use this to chunk *semantically* — a chunk is a section (or sub-section, or coherent block), not a fixed-window slice of the text stream.

The difference is dramatic. Compare two chunking strategies on a regulatory filing:

- **Fixed-window chunking (no structural metadata).** Chunks of 512 tokens, no respect for section boundaries. A chunk may start in the middle of a definitions section and end in the middle of a procedures section. Retrieval will match on surface features but the chunks themselves are semantically incoherent. Generation will produce answers that mix the two contexts as if they were one document.
- **Section-aware chunking (with structural metadata).** Chunks are sections or sub-sections, each preserving its heading context. Retrieval matches semantically meaningful units. Generation produces answers grounded in a coherent context. The same document, indexed twice, produces measurably different RAG quality.

Vendors that emit good structural metadata in 2026: Granite-Docling (via the DoclingDocument format), Reducto (their hybrid architecture preserves layout through to the output), LlamaParse Agentic and Agentic Plus (the parser explicitly produces heading hierarchies), Landing AI ADE (with the expanded chunk ontology covering attestations, IDs, logos). Vendors whose default output strips structure: hyper-scaler APIs in their default modes, single-pass VLM calls without explicit structure-output prompting.

The integration cost of switching from fixed-window to section-aware chunking is usually low — a few days of engineering once the parser emits the metadata. The quality gain is consistently large for any RAG workload that includes structured documents.

12.3.2 Table awareness that respects cell relationships

Tables are 30%+ of enterprise document content and produce 80%+ of the silent RAG failures. Treating them well is the single highest-leverage RAG-quality improvement available.

The wrong approach: treat the table as a stream of text. Cell values get concatenated with whitespace into a paragraph. Rows lose their relationship to row headers. The semantic structure is gone. Retrieval may match on cell values, but generation cannot produce a correct answer because the cell-row-column relationships are no longer recoverable from the chunk.

The right approach: parse tables as structured objects (HTML with rowspan/colspan preserved, or a custom JSON representation of the table grid), chunk tables **as atomic units** (do not split a table across chunks), and embed the table with a textual representation that preserves the structure (“Table: Q3 turnaround times by procedure. Cardiology: 4.2 days. Orthopedic: 5.1 days. Oncology: 3.8 days.”).

The downstream generation step can read structured table chunks and produce correct answers. The retrieval step can match queries to tables by content. The chain works.

Vendors with good table handling in 2026 (and the public benchmark evidence): Reducto (RD-TableBench leader at ~0.90), LlamaParse Agentic Plus (purpose-built for table-heavy documents like financial filings), Mistral OCR 3 (HTML tables with rowspan/colspan preserved). Open-source: Monkey-OCR v1.5 (SOTA on OmniDocBench v1.5 complex layouts), MinerU 2.5 (strong on scientific tables), PP-StructureV3 (small-model leader on layout-aware extraction).

Test your RAG on table-answer questions specifically. If you have not, the team in the bank example above is your team. The fix is not always switching parsers; sometimes it is using your existing parser’s table-aware mode that you did not turn on, or adding a table-specific extractor in front of the existing pipeline. Diagnose first.

12.3.3 Grounding that makes citations real

Pattern B and Pattern C parsers emit grounded outputs — every extracted field carries pointers to the source region (page number, bounding box). The RAG pipeline can preserve these pointers through chunking, embedding, retrieval, and generation, so that the final answer can cite the specific source region where each fact came from.

The user-facing impact is large. A RAG system that produces an answer with a “click to see source” link to the highlighted region in the original PDF is qualitatively different from one that produces an answer with a citation that says “from the company handbook.” Users trust the first; they correctly mistrust the second.

The compliance impact is larger. For regulated workloads (healthcare, legal, finance), the auditor’s question is “where did this answer come from?” A grounded RAG system can answer the question deterministically. An ungrounded one cannot.

The technical implementation: each chunk inherits the source-region pointers from its parsed source. The vector store stores chunk + pointer. The retriever returns chunk + pointer. The generator’s prompt includes the source pointers and is instructed to cite them. The UI renders the answer with clickable citations.

This is standard 2026 RAG engineering, but it requires the parser to emit groundings in the first place. Most public vendors do; some open-source models do (Chandra, dots.ocr, OlmOCR-2 all support grounding outputs); the hyperscaler defaults often do not without explicit configuration. Verify in eval that grounding metadata flows through your specific pipeline; do not assume.

12.4 Contextual drift, and how good parsing fixes it

A subtler failure mode that bad parsing introduces is **contextual drift**. Information that is correct in context becomes wrong when retrieved out of context.

A canonical example: a financial table includes a footnote that says “all amounts in thousands.” The table itself shows “Revenue: 124,500.” In context, this means \$124.5 million. When the table row is retrieved without the footnote, the chunk shows “Revenue: 124,500” — which a generator will, plausibly and confidently, interpret as \$124,500. The answer is off by three orders of magnitude. The retrieval was successful; the chunk was technically correct; the user got the wrong number.

Other examples: a “patient” mentioned in a clinical note may be ambiguous if the document has multiple patients and the chunk does not include the patient identifier. A “Q3” reference is ambiguous without the fiscal year context. A regulatory citation may be qualified by jurisdiction in a heading that the chunk does not include.

The general pattern: **chunks need to carry their parent context**. Structural metadata makes this possible — the chunker can attach heading context, footnote scope, units of measurement, and other implicit qualifiers as chunk-level metadata. Generation prompts can be instructed to use this metadata. Retrieval can index on it.

Vendors and open-source projects that handle this well in 2026 expose explicit “context” fields alongside chunks (Docling, LlamaParse with chunking enabled, Granite-Docling via DoclingDocument). Teams that do not get this right reproduce the contextual drift problem at scale and discover it through user complaints about wrong-by-orders-of-magnitude answers.

12.5 Reference patterns

The 2026 reference patterns for agentic OCR + RAG, in increasing order of sophistication:

Section-aware chunking. Chunks are sections, not fixed windows. Each chunk preserves its heading hierarchy as metadata. Generation includes the heading context in the prompt.

Table-preserving chunking. Tables are atomic chunks. Each table chunk includes a textual description of its structure (“Table titled X with columns A, B, C; N rows of data follow.”). Retrieval matches table chunks on content; generation can read the structured representation.

Hierarchical retrieval. Retrieve at the section level first, then at the chunk level within retrieved sections. This produces more coherent generation context than flat top-k retrieval, especially for multi-step queries.

Grounded answer with citations. Every generated answer includes source-region pointers that the UI can render as clickable citations. The compliance audit trail is automatic.

Context-augmented chunks. Chunks carry their parent context — heading hierarchy, footnote scope, units, named entities — as metadata available to both retrieval and generation. Contextual drift is addressed structurally rather than left to chance.

Most production RAG systems in 2026 implement two or three of these patterns. The systems that implement all five produce qualitatively better answers and qualitatively better audit trails. The cost is modest — most of this is chunking-and-retrieval engineering on top of a parser that emits the right metadata; the parser itself is the main investment.

12.6 When OCR is NOT your RAG bottleneck

A specific warning, because the failure mode is expensive: **agentic OCR is not always the right fix for a struggling RAG system**.

The most common shape of the wrong diagnosis: a RAG system is producing poor answers; the team assumes the parsing layer is the problem; they upgrade from a hyperscaler API to a premium agentic-OCR vendor; the per-page cost goes up by 8–10×; the answer quality does not noticeably improve. In retrospect, the bottleneck was retrieval, or embedding, or the generation prompt — and the parsing investment did not address it.

The diagnostic is straightforward and worth running before any parser upgrade:

Step 1: Sample 50 user queries that produced wrong answers. Real queries from real users, not synthetic test queries.

Step 2: For each, hand-trace the chain. Where did the failure originate? - Did the parser produce a faithful representation of the source document? (*If no → parsing failure.*) - Did the chunker produce semantically coherent chunks? (*If no → chunking failure.*) - Did the retriever return chunks that contained the answer? (*If no → retrieval failure.*) - Did the generator produce a correct answer from the retrieved chunks? (*If no → generation failure.*)

Step 3: Categorize the failures. If parsing failures are under 30% of the total, parsing is not your bottleneck. Upgrading parsing will not meaningfully improve the system. If parsing failures are over 60%, parsing is the bottleneck and the agentic-OCR upgrade is likely to help. The 30–60% range is mixed — fix both parsing and the dominant other failure mode.

The cheap diagnostic — a few hours of hand-tracing — saves significant amounts of money on premature parser upgrades. The DABstep result is the canonical reminder: extraction is not reasoning, and a 14.55% ceiling on multi-step reasoning over correctly-extracted data tells you that *reasoning failures look like parsing failures from the outside*. A team that upgrades parsing in response to a downstream-reasoning bottleneck will spend \$100k–\$500k and not move the needle.

12.7 RAG-specific evaluation

Building on Chapter 11’s general evaluation discipline, a few RAG-specific additions:

Question coverage. Your RAG eval needs questions whose answers live in tables, in footnotes, in section-level context (where a heading qualifies the answer), and in cross-document references. Most RAG evals built in 2024–2025 omit the first three categories entirely.

Citation accuracy. Not just “did the answer cite a source,” but “did the cited source contain the answer.” Score citation accuracy separately from answer accuracy. A system with 90% answer accuracy and 60% citation accuracy is dangerous for compliance work — the citations are wrong even when the answers are right.

Hallucination rate on missing-answer queries. Include queries whose answers are not in the corpus. A well-behaved RAG system answers “I don’t have that information” when the corpus does not contain the answer. A poorly-behaved system hallucinates plausibly. Measure the hallucination rate on missing-answer queries explicitly.

Latency under load. RAG latency is the sum of parsing (typically offline), retrieval, and generation. Production load can change the latency profile significantly. Test under realistic concurrent-query loads, not just single-query latency.

12.8 What good RAG-with-agentic-OCR looks like

A healthy 2026 RAG-with-agentic-OCR deployment exhibits these properties:

- Section-aware chunking driven by structural metadata from the parser.
- Tables chunked as atomic units with structural representations.
- Grounded outputs flowing through to citations in the user-facing UI.
- Hierarchical retrieval that respects section structure.
- Context-augmented chunks that address contextual drift.
- RAG eval that includes table-, footnote-, and context-dependent questions.
- Citation accuracy and hallucination rate measured as separate metrics from answer accuracy.

No system achieves all seven on day one. Teams that work toward them iteratively, with eval that catches each category of failure as the pipeline matures, produce RAG systems that meaningfully outperform the 2024-era fixed-window-chunking baselines. Teams that treat “upgrade the parser” as the entire answer reproduce the bank’s 41% production-failure scenario from the opening of this chapter, just with a more expensive parser.

💡 Named takes

RAG eval that doesn’t include questions whose answer requires a table is RAG eval that lies. Tables are 30%+ of enterprise document content and 80%+ of the parsing errors.

“Better OCR will fix our RAG” is sometimes true and sometimes a \$500k mistake. Run the ablation first.

Grounding turns RAG citations from theater into audit infrastructure. For compliance workloads, this is the difference between a deployable system and a non-deployable one.



Part V — Reality Check

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13. Anti-Patterns and Misconceptions

i Executive summary

- Five dominant anti-patterns in 2026 agentic OCR: **wrapper-washing**, **schema rigidity**, **ignoring HITL**, **over-relying on confidence scores**, and **conflating extraction with reasoning**. Each has a one-line smell test you can apply to a vendor demo, an internal architecture review, or your own pilot.
- The most common failure shape is not technical; it is **procurement-level**. Buying the wrong category of tool, for the wrong workload, on the basis of impressive demos. The “don’t build for the sake of it” framing in Section 9.5 is the procurement-level companion to this chapter’s per-system anti-patterns.
- These anti-patterns are not always vendor villainy. They are also things diligent teams do to themselves when they have not yet absorbed the five-attribute test, the architecture vocabulary, or the eval discipline that the rest of the book has built up.
- Print the checklist at the end of this chapter. Use it before signing.

13.1 How to use this chapter

Each anti-pattern in this chapter follows the same structure:

- A short description of the pattern as it appears in the field.
- Why it works as a sales or design pattern despite being a bad idea.
- A specific *smell test* — one line you can apply during a demo or design review to detect the pattern.
- What the corresponding *correct* implementation looks like.

The five anti-patterns are not exhaustive; the field has more pathologies than five. They are the five that, in the author’s field experience, account for the largest fraction of disappointed agentic-OCR deployments. A system that fails any one of these checks is a system to interrogate further, not necessarily a system to reject — but the burden of proof should sit with the vendor or the architect to explain why the pattern is being avoided.

13.2 Anti-pattern 1: Wrapper-washing

The pattern. A classic-OCR pipeline (typically Tesseract or a hyperscaler API) with an LLM cleanup or post-processing step, rebranded as “agentic OCR.” The marketing collateral describes the system using all the right vocabulary — multi-pass, self-correcting, schema-aware, tool-using — but the substantive engineering is the same shape as a 2020-era pipeline with a 2024-era model bolted on.

Why it works as a sales pattern. In 2026, “agentic” is the word that gets the conversation past procurement. The five-attribute test from Chapter 3 is not yet common knowledge in enterprise procurement processes. A vendor who can describe their system using the vocabulary, without necessarily implementing the architecture, can charge a premium for what is, structurally, a previous-generation product.

Why it is dangerous to deploy. A wrapper system inherits the failure modes of its base layer. The Tesseract underneath cannot read complex tables; the LLM on top cannot fix what was lost in extraction. The system looks agentic in demos that use clean documents and produces classic-OCR-quality output on the messy ones — i.e., the ones you actually deployed agentic OCR to handle.

Smell test. *Does the system satisfy all five attributes from Chapter 3’s scorecard — goal-driven, multi-pass, self-correcting, tool-using, and grounded?* Score them yes / no / partial. A system that satisfies three of five is not agentic OCR. A system that satisfies four of five is borderline. Five of five is the bar.

The most common attribute wrapper-washed systems miss is **tool-using**. A model with a `parse_pdf()` function call is *tool-capable*; tool-using means routine, orchestrated invocation of specialized tools during the workflow. Wrapper systems generally fail this test cleanly.

What correct looks like. A system with explicit tool-routing logic (layout detector → region classifier → specialized extractor per region type), an external oracle in the reflection loop (schema validator, secondary model, KB lookup), and grounded outputs that trace every field back to a source region. Reducto, LandingAI ADE, LlamaParse Agentic, and Granite-Docling-in-Docling-framework all satisfy this; wrappers do not.

13.3 Anti-pattern 2: Schema rigidity

The pattern. An agentic OCR system that is forced to produce a fixed schema with no flexibility for unexpected fields, document variations, or fields the document contains but the schema does not anticipate. Often paired with strict downstream validation that rejects any output that does not exactly match the schema.

Why it works as a design choice. Schema rigidity feels safe. Downstream systems that consume the output do not have to handle variation; the API contract is stable; the integration is simpler. For narrow, stable workloads this can even be correct.

Why it is dangerous for agentic OCR specifically. Agentic OCR’s central advantage over template-driven IDP is its ability to handle **schema-flexible extraction** — to capture whatever the document contains, not just what your schema anticipates. Forcing a strict schema discards this advantage. The system becomes a more expensive template extractor.

The specific failure mode: a document has an extra field that the schema does not include. Three things the system might do: 1. **Drop the field silently.** The document had useful information; the system threw it away. The schema rigidity is invisible until a compliance audit asks where the missing data went. 2. **Fail the whole document.** Stricter, but throws away the rest of the extraction along with the unexpected field. 3. **Capture the field in a flexible “extras” container, with a flag.** The right answer.

Most schema-rigid systems do option 1 or option 2. Option 1 is more common because it preserves the per-document success rate at the cost of silent data loss.

Smell test. *When a document has an extra field your schema does not include, what does the system do?* If the answer is “the system drops it” or “the document fails validation,” you have a schema-rigidity problem. If the answer is “the system captures the extra in a flexible container and flags it for review,” you do not.

What correct looks like. Schema-aware *but not schema-bound* extraction. The schema is a *priority signal* — the system prioritizes extracting the requested fields — not a *boundary*. Fields outside the schema are captured into a structured “additional fields” container with their own provenance metadata. The HITL queue can decide whether to add them to the schema (extending coverage) or to leave them in the extras bucket. Both options are operationally manageable; silent data loss is not.

13.4 Anti-pattern 3: Ignoring HITL

The pattern. Deploying agentic OCR with no human-review gate, on the theory that the reflection loop is the safety net. The system extracts; the output ships; if anything goes wrong, the customer notices.

Why it is tempting. The marketing pitch for agentic OCR routinely includes claims of “eliminates manual review” or “reduces HITL by 100%.” Teams that buy this story design their workflows accordingly. The cost savings on reviewer staffing show up in the business case for the pilot.

Why it fails. The reflection loop is not an external oracle (Chapter 5). It cannot catch silent hallucinations — the failure mode where the model is confidently wrong and the loop’s self-critique approves the wrong answer. The Databricks “\$10,000 → \$3,000” example is a textbook silent hallucination; the loop did not catch it; the error shipped because there was no human in the loop to catch it either.

The deeper structural point: some errors *require human eyes*. The model cannot tell, by inspection, that “\$10,000” on the page is the right number and “\$3,000” is the wrong one. Only a human with context (the bond’s face value should be in a specific range; the field is critical to the downstream decision) can. A system without a HITL gate has no recourse to that human, and the error rate ships at whatever the model’s silent-hallucination rate happens to be.

Smell test. *When the system outputs a high-confidence extraction that is, in fact, wrong, what happens next?* If the answer is “the wrong extraction ships,” you have an ignoring-HITL problem. If the answer is “the extraction is gated by HITL on high-stakes fields, sampled for QA on low-stakes fields, or escalated by an out-of-distribution detector before output,” you do not.

What correct looks like. Every high-stakes field gets a HITL gate. Escalation rates run 1–10% in steady state. Low-stakes fields can ship without HITL but are sampled at some rate (often 1–5%) for ongoing quality monitoring. The HITL queue is engineered as a first-class subsystem with reviewer UI, feedback loop, and operational discipline around response times.

Vendors that have engineered HITL well: the classic IDP incumbents (Hyperscience especially), Landing AI ADE, Reducto. Vendors whose HITL story is weaker: many of the newer AI-native entrants whose feature roadmap has not yet caught up with their model layer. Verify in vendor evaluation; do not assume.

13.5 Anti-pattern 4: Over-relying on confidence scores

The pattern. Production gating logic that treats VLM-emitted confidence scores as calibrated probabilities and uses them as the only signal for whether an extraction is good enough to ship.

Why it is appealing. Confidence scores look like probabilities. They live on the 0-to-1 scale. They go up when the model is more sure of itself. Gating on them is operationally easy: `if confidence > 0.85: ship`. The pattern is everywhere in 2026 production systems.

Why it fails. VLM-emitted confidence scores are not calibrated probabilities. They are at best ordinal signals — the model is more confident on extraction A than on extraction B — but the absolute values do not correspond to actual accuracy probabilities in any reliable way, especially on out-of-distribution data. A confidence of 0.95 on a document the model has structurally never seen before can be accompanied by an actual accuracy as low as 60%.

This is not a “the models will get better” problem. Calibration on OOD data is a fundamental research problem in machine learning, and the production reality is that any confidence-based gating system will have non-trivial silent-hallucination rates as long as production traffic includes OOD documents. Which it does. Always.

Smell test. *Plot confidence vs. accuracy on your golden set.* If the curve is smooth and monotone (high confidence → high accuracy, with a clean relationship), confidence is genuinely calibrated for your distribution and you can use it as a gate. If the curve is flat or noisy (high confidence corresponds to highly variable accuracy), confidence is at best an ordinal signal, and you should not use it as the only gate.

What correct looks like. Confidence is one of several inputs to the gating decision, not the only one. Other inputs: - **Schema validation** — does the output parse, with valid field values? - **External KB lookups** — does the extracted entity resolve against the relevant knowledge base? - **Secondary-model agreement** — does a second VLM with a different prompt produce the same answer? - **OOD detection** — is this document sufficiently unlike the model’s training distribution that we should escalate by default?

The combined gate uses confidence as one signal among multiple. The system escalates when the combined evidence suggests an error, not just when one model’s opinion of itself happens to be low. The engineering cost is modest; the production reliability gain is substantial.

13.6 Anti-pattern 5: Conflating extraction with reasoning

The pattern. Treating an agentic-OCR system’s correct extraction as a *guarantee* that downstream tasks will produce correct answers. The procurement decision is made on parsing accuracy; the deployment fails on multi-step or reasoning-heavy tasks; the team is surprised.

Why it is tempting. The vocabulary collapses the two. “Intelligent document understanding,” “agentic document AI,” “document intelligence” — these terms are sold as if extraction and reasoning are one capability. They are not.

Why it fails. The DABstep result is the cleanest evidence. The top reasoning model — *given correctly extracted data and access to tools* — scores 14.55% on hard tasks and around 16% overall. Frontier reasoning models, with perfect parsing, fail roughly 85% of real multi-step analysis tasks.

This means: even if your agentic-OCR system extracts every field correctly on every document, a downstream agent reasoning over those extractions can still fail catastrophically. Parsing accuracy is necessary; it is not sufficient. The two failure modes are separate. They have separate evals. They require separate fixes.

The most common production manifestation: a team deploys an agentic-OCR system, builds a downstream LLM-powered analysis layer, and observes poor downstream performance. The team upgrades the parser, expecting the downstream behavior to improve. Parsing accuracy does improve (slightly); downstream performance does not. The team is puzzled. The DABstep result is the explanation: the bottleneck was always in the reasoning layer, and a better parser was the wrong investment.

Smell test. *When a downstream answer is wrong, can you tell whether parsing or reasoning was at fault?* If the system has no instrumentation to attribute failures to one or the other, you cannot diagnose, and

you will spend money in the wrong place. The fix is to ablate: replace the parsing layer with a hand-curated perfect parse for a sample of downstream-failure cases. If downstream still fails, the failure was reasoning, not parsing.

What correct looks like. Keep parsing and reasoning evals separate. They are separate problems with separate failure modes. A system that bundles them in a single “answer accuracy” metric hides the diagnostic information you need to allocate engineering effort correctly. The eval discipline is roughly:

- *Extraction eval* — field-level F1, table-cell accuracy, grounding accuracy on a golden set.
- *Reasoning eval* — task-success rate on hand-curated downstream tasks, with parsing held constant (perfect parses or fixed parser output).
- *End-to-end eval* — combined system behavior, but with the ability to attribute failures to one layer or the other.

Vendors that bundle parsing-and-reasoning at a premium (the larger “document intelligence platform” pitches) are often genuinely useful for end-to-end workflows but make the layer attribution harder, not easier. Verify that the vendor’s tooling lets you instrument both layers independently before committing.

13.7 A short checklist for your own architecture

Print this. Use it before signing a vendor contract, before approving an internal architecture, before promoting a pilot to production.

1. The wrapper test. Does the system satisfy all five attributes from Chapter 3 — goal-driven, multi-pass, self-correcting, tool-using, grounded? Score them. Five of five is the bar.

2. The schema-flexibility test. What does the system do with an unexpected field in a document? “Captures it in a flexible extras container with a flag” is the right answer; “drops it” or “fails the document” is not.

3. The HITL test. When the system outputs a high-confidence extraction that is, in fact, wrong, what happens next? “Gated by HITL on high-stakes fields or sampled for QA on low-stakes fields” is the right answer; “ships without review” is not.

4. The calibration test. Plot confidence vs. accuracy on your golden set. Is the curve smooth and monotone? If not, confidence is an ordinal signal at best — do not let it be the only gate.

5. The decomposition test. When a downstream answer is wrong, can you tell whether parsing or reasoning was at fault? If not, your instrumentation is missing, and your future engineering will land in the wrong place.

If you fail any of these five tests, you have one of the anti-patterns. The system is not necessarily unredeemable — most of these are fixable with deliberate engineering — but the deployment is not yet ready for production at the stakes you are likely contemplating. Fix the system, not the marketing language.

13.8 Beyond the five — patterns to watch for

A few additional anti-patterns that are less common but worth naming briefly:

Eval theater. A vendor or internal team builds an evaluation that scores their preferred system favorably and uses it as the justification. The eval may technically measure something, but the metrics chosen, the documents included, and the scoring rubric are calibrated to produce the result the team wanted. The defense is the golden set — built independently by the team that will operate the system, evaluated on metrics defended on first principles, not on convenience.

Vendor-version drift. A vendor releases a model update that subtly changes behavior on edge cases your team did not include in regression tests. Production accuracy declines. The decline is gradual, not obvious from any single document, but visible over months in HITL queue volume or downstream error rates. The defense is version pinning plus regression testing on vendor major-version bumps (Chapter 11).

Compliance retrofitting. A team deploys an agentic-OCR system without grounding or audit logs, then discovers six months in that a regulator wants to see the source region for each extraction. The system cannot produce them. The defense is treating grounding as a Day 1 requirement, not a Day 180 requirement, for any workload that might face regulatory scrutiny.

Vendor lock-in by output format. A team’s downstream systems are built around the specific output format of one vendor. Switching vendors requires rebuilding the integration. The defense is treating the vendor’s output format as a contract you control — normalize to a vendor-neutral intermediate representation, even when one vendor’s native format would be easier. Six months of pain in vendor switching is more expensive than the marginal engineering effort to maintain a normalized representation.

None of these are immediate red flags in the way the five primary anti-patterns are. They are slower failure modes that emerge in production over six to eighteen months. Worth watching for; worth defending against deliberately.

💡 Named takes

Wrapper-washing is the dominant marketing pattern of 2026. The five-attribute scorecard is your defense.

Confidence scores from agentic OCR systems are an opinion the model has about its own work. Treat them as you would any other unverified opinion.

The most expensive 2026 mistake is conflating extraction with reasoning. DABstep is the reminder. The bottleneck is rarely where the team first looks.

14. What Agentic OCR Doesn't Solve

i Executive summary

- Agentic OCR's capabilities are real, but the marketing claims overstate them. Five categories of problem remain meaningfully unsolved in 2026: **hallucinations**, **domain semantics**, **cost**, **latency**, and the **HITL reality**.
- Hallucinations have not been eliminated. They are rarer than in 2024, harder to detect, and more expensive when they do happen. The Databricks "\$10k → \$3k" example is the canonical 2026 hallucination shape, not a 2024 one.
- "Eliminates manual review" is the marketing line that sells the most pilots and produces the most failed production deployments. Plan for HITL forever. Volume drops by 80%+ in healthy deployments. It does not drop to zero. It will not drop to zero in 2027.
- "Good enough for production" in 2026 looks like: 95%+ accuracy on a domain-specific golden set, <5s latency per page for batch, <\$0.10 cost-per-correct extraction for high-stakes work, 1–10% HITL escalation rate, <1pp accuracy drift per quarter. No deployment hits all of these on day one. Healthy deployments work toward them over twelve to eighteen months.

14.1 Hallucinations

Hallucinations have not been eliminated. They have been reduced — meaningfully, measurably, and in some workloads dramatically — but the residual rate is non-zero and the residual cases are systematically harder to catch than the failures of earlier generations.

The vendor-reported numbers tell part of the story. Classic OCR pipelines have reported failure rates of 15–20% on enterprise documents (varies wildly by document type, condition, and what "failure" means). Agentic systems report rates under 2% on the same workloads. That is a 10× improvement and it is real. The qualification is that the 2% errors are not a uniform random sample of the 20% — they are a different distribution.

What the residual 2% looks like in 2026:

Silent hallucinations. The model produces a wrong answer with high confidence. The reflection loop's self-critique does not flag it. The schema validator does not catch it (because the value is type-correct, just wrong). The system ships the error. The Databricks "\$10,000 → \$3,000" example [1] is the canonical shape. The OpenReview paper on "Tiny Silent Hallucinations in Agentic AI" [19] documents this category extensively.

Plausible-but-wrong extractions. A vendor name is captured correctly, except the system reads "Acme Corp" as "Acme Co." (because the document showed "Acme Corp." with a period and the model dropped the abbreviation). A date is captured as a valid date, just the wrong one. An amount is read as a valid number, just one digit shifted. The values are *plausible* — type-correct, range-valid, format-compliant — and *wrong*. These are the most expensive errors because they pass every automated check and only surface in downstream consequences.

Hallucinations from absence. A field is not present in the document, but the schema expects it. The model invents a plausible value. This is the failure mode that schema-rigid systems (anti-pattern 2 in Chapter 13) are particularly prone to. The defense is allowing the model to return "missing" explicitly, with schema validation that recognizes the sentinel.

Hallucinations from ambiguity. The document has two candidate values for a field — say, two dates on a contract, one of which is the effective date and one of which is the signature date. The model picks one; sometimes it picks the wrong one. The error is not a hallucination in the strict sense (the value exists in the document) but is functionally indistinguishable from one.

The honest 2026 framing: hallucinations are now rare enough that they will not show up in your pilot evaluation, frequent enough that they will show up in your production at meaningful rates, and consequential enough that the workloads where they matter most need explicit defenses against them. The

defenses are: external oracles in the reflection loop, secondary-model cross-checks on high-stakes fields, HITL gates on critical outputs, schema-validity flags that include “value is plausible but unconfirmed” states, and post-hoc audit sampling that catches errors weeks after they shipped.

None of these defenses is free. All of them are cheaper than the cost of a wrong-value-shipped incident in regulated industries.

14.2 Domain semantics

Agentic OCR ends at extraction. It does not begin at understanding.

A clinical record says “DX: hypertension.” A well-tuned agentic OCR system extracts `diagnosis: "hypertension"` correctly, ground-truths the source region, returns the structured output. This is parsing.

Whether hypertension is the correct diagnosis given the rest of the record — the patient’s age, the lab results, the prior diagnoses, the differential considerations — is medicine. The model that extracted the diagnosis correctly knows nothing about whether the diagnosis itself is right, and would not be qualified to opine if it did.

This delineation matters because the marketing pitches frequently elide it. “Intelligent document understanding,” “clinical document intelligence,” “legal AI for contracts” all blur the boundary between extraction and domain reasoning. The extraction is solved-ish; the domain reasoning is a separate problem, often handled by different models in different parts of the workflow, evaluated separately, regulated separately.

The failure mode for teams that miss this delineation: building an agentic-OCR system that extracts beautifully, deploying it into a domain workflow, and watching the domain experts (doctors, lawyers, financial analysts, compliance officers) push back against the system’s *domain judgments* — which the system never had any business making in the first place. The extraction layer was overburdened with responsibilities it could not carry.

The right architecture: extraction layer (agentic OCR) emits structured, grounded output. Domain layer (a separate model, often a different vendor, often fine-tuned for the domain) operates on that output. The two are evaluated separately. The domain layer’s errors are *domain* errors, attributable to the model that handles them; the extraction layer’s errors are *extraction* errors. The decoupling is operationally cheaper and architecturally more honest.

In practice: do not buy or build an agentic-OCR system expecting it to do clinical decision support, legal advice, financial analysis, or regulatory interpretation. Those are downstream domain problems that domain models handle. Agentic OCR’s job is to deliver good inputs to those models.

14.3 Cost

The shift-left thesis (Chapter 6) argues — correctly — that cost-per-correct-extraction is often lower under agentic OCR than under classic IDP plus downstream cleanup. The thesis depends on downstream error costs being non-trivial. For workloads where they are not, the thesis does not apply, and the cost picture inverts.

The categories of cost that remain real even in workloads where the thesis applies:

Per-page API cost. \$0.001 to \$0.056 per page depending on tier (Chapter 7), versus fractions of a cent for classic OCR. The increase is real and shows up directly in monthly billing.

HITL operational cost. Even at a healthy 5% escalation rate, HITL operations are usually the largest line item in an agentic-OCR deployment’s monthly cost. A reviewer earning \$50–\$150 per hour at three to five minutes per case at 5% of a 100k-pages-per-month workflow is \$12,500 to \$62,500 per month — multiples of the parsing API spend.

Engineering opportunity cost. Building eval infrastructure, HITL workflows, observability, and integration logic is real engineering. The shift-left thesis says this engineering is more leveraged than the classic-pipeline engineering it replaces, which is true; the engineering is not free.

Vendor lock-in cost. Switching vendors is expensive in agentic OCR. Output formats are non-trivially different across vendors. Eval rubrics calibrated to one vendor’s behavior may not transfer to another’s. Integration code carries vendor-specific assumptions. Realistic switching cost is months of engineering and a non-zero customer-visible disruption window.

Observability and audit cost. Drift monitoring, audit-log retention, eval-set refresh, compliance reporting. New categories of cost that did not exist (or existed at much smaller scale) in classic IDP deployments.

For high-error-cost workloads (healthcare, legal, finance, KYC, anything with regulatory exposure), the totals net out in favor of agentic OCR. For low-error-cost workloads (bulk indexing, analytics ingestion,

commodity invoice processing at extreme scale), the cost picture often does not net out, and the right answer is one of the simpler architectures in Section 9.5's "what to do instead" list.

14.4 Latency

Multi-pass reflection adds wall-clock time. The latency penalty is structural — it cannot be optimized away — and constrains the use cases agentic OCR can serve.

Rough 2026 latency profiles by pattern:

- **Pattern A (single-pass VLM):** 200ms to 2s per page.
- **Pattern B (agentic loop):** 2s to 5s per page.
- **Pattern C (hybrid CV+VLM):** 3s to 7s per page.

These are per-page; multi-page documents scale roughly linearly (sometimes super-linearly if the loop revisits earlier pages during reflection).

For batch workloads, latency is invisible. A 100k-page-per-month pipeline running overnight has time for all three patterns regardless of per-page latency.

For interactive workloads, the picture is different. A chat interface over a single PDF, a real-time form-fill assistant, an interactive document-review tool — these need sub-second responses to feel usable. Pattern A is often the only architecture that fits the latency budget, and Pattern A is by Chapter 3's definition not really agentic OCR.

The honest framing: agentic OCR (Pattern B or Pattern C) is designed for batch or near-batch workloads. Interactive workloads can use the parsing output of agentic OCR — chunked, indexed, retrieved — but they cannot wait for an agentic loop to complete on each query. Use cases that mix the two (parse documents in batch, serve queries against the parsed corpus interactively) work well. Use cases that demand per-query agentic parsing (parse this newly-uploaded document and answer my question about it, both in under two seconds) do not.

14.5 The human-in-the-loop reality

The single most consequential overstatement in 2026 agentic-OCR marketing is "eliminates manual review."

The reality: agentic OCR **reduces** HITL volume, often by 80% or more compared to classic pipelines. A workflow that previously ran 30–50% manual review can land at 5–10% escalation rates after a careful Pattern B or C deployment. That reduction is transformative ROI. It is also not zero. It will not be zero in 2027.

The structural reasons HITL remains:

Some errors require human eyes. The model cannot tell, by inspection, that a confidently-extracted value is wrong. Some errors are only catchable by a human who has context the model does not have (the patient's other records, the prior version of the contract, the typical pattern for this vendor's invoices).

Out-of-distribution documents. Production traffic includes documents that are unlike anything the model has been trained on. The model's behavior on OOD data is unreliable. The right response is to route OOD documents to HITL by default, not to trust the model's extractions on them.

Edge cases compound. Any production system handles edge cases that are not in its training data — uncommon document types, novel vendor formats, layout variants the model has not yet seen. Each one alone is rare; collectively they account for a meaningful slice of production traffic, and HITL is the right home for them.

Compliance requirements. Regulated workloads often *mandate* human review for certain decisions, regardless of model performance. The HITL gate is not optional; it is a regulatory feature.

The healthy pattern is to engineer HITL as a **first-class subsystem**, not a queue bolted on the side:

- Reviewer-facing UI that shows the document, the candidate extraction, the source region highlighted, and a one-click accept / correct / reject interface.
- Escalation routing based on document type, confidence, OOD detection, and field-level stakes — not just on aggregate confidence.
- Feedback loop from reviewer corrections back into the system: the corrections feed the golden set, drive prompt tuning, surface vendor-selection signals, and over time reduce the escalation rate for the document types the system learns to handle.
- Operational discipline: queue response times, reviewer SLAs, audit trails of correction history.

Vendors that engineer HITL well in 2026: Hyperscience (best-in-class; their HITL infrastructure is one of the genuinely under-appreciated strengths in the classic IDP tier), Landing AI ADE, Reducto. Vendors whose HITL story is weaker: many of the newer AI-native entrants whose model layer outpaced their workflow tooling.

The 2026 production reality: plan for HITL forever. Plan for a 1–10% escalation rate in steady state. Plan for HITL operational cost as a major line item. Plan for the HITL queue to be an organizational function with its own SLAs, staffing, and reporting. Teams that plan otherwise reproduce the dominant 2026 production-failure shape — agentic OCR deployed without HITL infrastructure, accuracy meets vendor benchmarks but downstream consequences accumulate, pilot quietly winds down, budget consumed.

14.6 What “good enough” actually looks like in production

A healthy 2026 agentic-OCR deployment exhibits the following metrics simultaneously:

- **Accuracy: 95%+ on a domain-specific golden set** for HITL-supplemented deployment; 99%+ for HITL-light deployment. Aggregate accuracy can be misleading; track per-field accuracy too.
- **Latency: under 5 seconds per page** for batch workloads; under 500ms for interactive workloads (which often means using Pattern A for the interactive path).
- **Cost: under \$0.10 per correct extraction** for high-stakes workloads; under \$0.01 for high-volume low-stakes workloads. The denominator matters — cost-per-page can hide error-rate-multiplied real cost (Chapter 11).
- **HITL escalation rate: 1% to 10%** in steady state. Healthy values cluster around 3–5%. Outside this range, investigate.
- **HITL acceptance rate: 30% to 70%**. Reviewers correcting more than 70% means the system is escalating documents it should have caught itself; reviewers accepting more than 70% means the system is escalating documents it actually got right.
- **Drift: under 1 percentage point of accuracy degradation per quarter** on a refreshed golden set. Anything more, investigate. Vendor model updates are the most common cause.
- **Grounding accuracy: 90%+** for any compliance-sensitive workload. Without it, audit is impossible.

No deployment hits all seven on day one. The first six months of a deployment are typically spent building the eval discipline that lets you measure these metrics; the next six to twelve months are spent tuning the system to hit them. Teams that treat these as starting requirements rather than maturity goals set themselves up for premature pilot termination. Teams that treat them as maturity goals — and invest the engineering effort to reach them iteratively — produce the deployments that survive contact with production reality.

14.7 Things people expect agentic OCR to fix that it does not

A short list, deliberately blunt:

- **Bad source documents.** Illegible scans, smeared faxes, redacted-then-photocopied artifacts — agentic OCR cannot conjure information that is not in the document. The model can detect that the document is bad; it cannot extract values that are not there.
- **Missing semantic context.** A document that omits a required field is correctly extracted as “field missing.” Whether the field should have been present is a question about the document’s intended structure, which the model cannot answer reliably.
- **Downstream reasoning quality.** The DABstep result. Even perfect extraction does not produce correct downstream answers on multi-step tasks. Reasoning is a separate problem.
- **Vendor lock-in.** Switching costs are worse, not better, than classic IDP. The marketing line “vendor-neutral output” is almost always less true than it sounds.
- **Compliance certification.** Agentic OCR can support compliance work — grounded outputs, audit trails — but does not produce compliance. That requires human review, regulatory expertise, and process discipline.

The list is not exhaustive. The pattern across all of them is the same: agentic OCR addresses one specific problem (structured extraction from documents) very well, and is routinely sold as if it addressed adjacent problems too. The adjacent problems are not the same problem. They require different tools, different models, different evaluations, and different organizational commitments.

The clear-eyed reader of this chapter will conclude that agentic OCR is a real, useful, expensive category of technology with specific applicability and specific limits. That conclusion is correct. The marketing materials want you to conclude something larger. Resist the marketing; deploy on the merits.

 Named takes

“Eliminates manual review” is the marketing line that sells the most pilots and produces the most failed production deployments. Plan for HITL. Forever.

Hallucinations in 2026 are rarer, harder to detect, and more expensive when they happen. The technology is better; the failure mode shifted, not disappeared.

The list of problems agentic OCR does not solve is longer than the list it does. The skill is matching the technology to the problem; the failure mode is the inverse.

15. Conclusion: Where 2027 Goes

i Executive summary

- **Parsing quality is commoditizing.** The open-source frontier (Chandra, OlmOCR-2, Granite-Docling, PP-StructureV3) is at or near parity with proprietary systems. By the end of 2027 there will not be a meaningful parsing-quality moat for English-language structured documents.
- The 2027 moats are forming around **agentic orchestration, observability, HITL workflows, compliance posture, and vertical specialization** — not around raw extraction. Vendors building those layers will outlast vendors competing on benchmark scores.
- Three open research problems will dominate the next eighteen months: **empirical failure-mode taxonomies** from production deployments, **long-form benchmarks** that go beyond DABstep, and **cost-aware evaluation** that integrates accuracy with TCO into a single metric procurement can use.
- The thesis of this book: **reading is the ceiling on agent quality, and reading is mostly solved.** The next ceiling is **trust** — observable, calibrated, auditable agentic behavior. That is the book to write next.

15.1 What's about to commoditize

The clearest 2026 signal in the public benchmarks is that parsing quality is approaching saturation.

OmniDocBench top scores cluster within a percentage point of one another. The olmOCR-Bench leaderboard's top three (Chandra, OlmOCR-2, dots.ocr) are within four points of each other, all open-source, all license-friendly to most enterprises. The PP-StructureV3 result on OmniDocBench — matching Gemini-2.5-Pro at under 100M parameters — argues that the parsing problem can be solved with specialized small models, which is the leading indicator for commoditization in any technology category.

The implication for 2027: **the parsing-quality moat is going to disappear** for English-language structured documents. By mid-2027, no vendor will be able to credibly claim “we have meaningfully better parsing accuracy than the open-source frontier” on the majority of enterprise document types. The vendors that survive will not be the ones with the best models; they will be the ones with the best surrounding platform.

What will commoditize fastest:

- **Single-pass VLM extraction.** Already commoditized by mid-2026; frontier models from any major lab can do credible structured extraction with a structured-output prompt.
- **Basic table extraction.** Reducto's RD-TableBench lead is real but the open-source side (MonkeyOCR v1.5, PP-StructureV3) is catching up fast. By 2027, “we handle tables well” stops being a differentiator and starts being table stakes.
- **Multilingual coverage.** PaddleOCR-VL at 109 languages, DeepSeek-OCR at ~100 languages, Chandra at 40+ languages — the open-source side is solving this faster than the commercial side. Vendors selling “we handle 50 languages” as a feature will sound dated by 2027.

What will commoditize more slowly:

- **Grounding accuracy.** The technical capability exists, but the surrounding infrastructure (audit trails, region-tracking through chunking, citation-rendering UIs) is engineering-heavy. Open-source models have grounding outputs; few open-source *systems* do grounding well end-to-end.
- **Scientific and technical documents.** MinerU 2.5 and the open-source scientific-document tier are competitive on this workload, but commercial vendors continue to lead on the long tail (rare equation notations, niche scientific layouts, languages with limited training data).
- **High-stakes vertical compliance.** Healthcare BAA workflows, regulated financial document extraction, legal discovery with chain-of-custody requirements — the commercial vendors' procurement and compliance maturity remains a real moat through 2027.

15.2 Where moats are forming

If the parsing-quality moat is disappearing, the obvious question is: what is replacing it? Four categories are visible in the 2026 landscape and are likely to deepen through 2027.

15.2.1 Agentic orchestration platforms

The vendors that own the *workflow* around document understanding — not just the model that does extraction — have the strongest 2027 position. LlamaIndex with the LlamaParse + LlamaIndex stack is the canonical example. Landing AI ADE is building toward this with its expanded ontology and chunk-classification work.

The **platform tier is now three vendors deep, not one** (the May 2026 update — Chapter 7. details). Databricks Document Intelligence has `ai_parse_document`; Snowflake Cortex has `AI_PARSE_DOCUMENT`; Microsoft Fabric has Foundry Tools integration with Azure DI. The naming convergence (`ai_parse_document` / `AI_PARSE_DOCUMENT`) is a leading indicator — when two competing platform vendors ship products with effectively the same name and the same SQL-function shape, the category has crystallized faster than usual. By 2027 a fourth entrant (a hyperscaler-platform play, possibly from AWS via Bedrock Data Automation deepening into RedShift/SageMaker) is the expected addition.

The moat is integration depth, not parsing accuracy. A team that has built its agentic workflow on LlamaIndex's primitives cannot easily switch to Reducto for parsing without rebuilding the orchestration layer. The switching cost is the moat; the orchestration platform is the lock-in.

15.2.2 Compliance and grounding infrastructure

The 2027 procurement question that will increasingly determine vendor selection: *can you prove, to a regulator, where every extracted field came from?* The vendors that can answer this with a one-click audit trail will outsell vendors that have to engineer the answer per-customer.

Reducto's hybrid architecture with audit trails. Landing AI ADE's HIPAA + ZDR posture. Anterior AI's published fairness-evaluation methodology for healthcare prior-auth. These are not marketing differentiators; they are the table stakes for any vendor selling into healthcare, finance, or government in 2027.

The companies that retrofit compliance late will lose deals to companies that built compliance in from the start. The gap is widening, not narrowing, through 2026 and into 2027.

15.2.3 Observability for document agents

This category does not yet exist as a coherent product in 2026. By the end of 2027, it will be one of the most active areas of investment in the agentic-OCR ecosystem.

The need is concrete: production agentic OCR systems are black boxes from the operator's perspective. Why did the system escalate this document? Why did the reflection loop accept that wrong answer? What is the calibration trend over time? Which prompt variants are converging on this output? Current tooling — vendor dashboards, custom dashboards stitched together from API logs — answers these questions poorly.

The opportunity: a Datadog-equivalent observability layer for agentic document workflows. Extraction lineage. Calibration trending. Escalation-rate decomposition by field, document type, and OOD detector signal. Drift alerting. Reviewer-correction pattern analysis.

The pure-play vendors that try to be everything will struggle here; the specialized observability tools that integrate across vendor stacks will win. Expect the first credible such product by mid-2027 and meaningful market share by end of 2027.

15.2.4 Vertical specialization

Healthcare won 2026. The CMS prior-authorization rule, the public-reporting forcing function, and the early case studies (Anterior AI + Reducto reaching 99.24%) created a vertical where agentic OCR has clear, measurable, large ROI.

The vertical pattern will repeat. Legal contract review is partway through the same curve — clear ROI, large volume, regulatory pressure (less than healthcare, but rising). Financial services KYC and AML compliance are similar. Insurance claims adjudication. Government benefits administration. Each of these has a specific document distribution, a specific regulatory regime, and an ROI threshold that agentic OCR can clear.

Vendors that go *deep* on a vertical — not just adding “healthcare” to their feature list, but building the specific workflows, schemas, regulatory mappings, and HITL operations that the vertical requires — will outperform horizontal vendors in their chosen verticals. Expect a wave of vertical-specialist agentic-OCR vendors in 2027, some of them spinning out from the existing horizontal vendors, some of them new entrants targeting verticals the incumbents have not focused on.

15.3 Open research questions

Three problems are likely to dominate the research conversation in agentic OCR through 2027 and into 2028. None of them is at the model-architecture layer (the model layer is increasingly mature); all are at the system-operation layer.

Empirical failure-mode taxonomies from production deployments. The “Tiny Silent Hallucinations” paper [19] is the leading example of this genre — careful documentation of failure modes from real systems, with diagnostic and mitigation suggestions. The field needs much more of this. Most failure-mode discussion in 2026 is anecdotal or vendor-internal; what is needed is reproducible, cross-vendor, real-production characterization. Expect three to five major papers in this direction by end of 2027.

Long-form benchmarks beyond DABstep. DABstep [7] is the leading edge of multi-step reasoning evaluation, but its small N (450 tasks) and narrow distribution (Adyen’s operational workloads) limit its general usefulness. The next-generation benchmark will combine multi-document parsing with multi-step reasoning at larger scale, across diverse domains, with both extraction-success and reasoning-success measured separately. This is more engineering than it sounds; the benchmark that does this well will take a research group eighteen months to build and will reshape the conversation when it lands.

Cost-aware evaluation. No 2026 benchmark integrates accuracy and cost into a single metric. The procurement-relevant question — what is the cost-per-correct-extraction-with-provenance on a representative production distribution — is asked by every procurement process and answered by none of the public benchmarks. A benchmark that gets this right would be enormously useful and will probably be published in 2027 by either a research group with industry connections or an industry consortium with research credibility.

Honorable mentions: calibrated-confidence research (the OOD calibration problem remains genuinely open), reasoning-extraction separability (operationally important, conceptually under-explored), and the human-in-the-loop economics (under-studied because it requires production data that vendors guard carefully).

15.4 What to watch in 2027

Five concrete things to watch through 2027:

A new benchmark supersedes OmniDocBench. SCORE-Bench is the leading candidate; an alternative from a research group not yet active in 2026 is plausible. Whoever publishes the successor sets the procurement conversation for the next eighteen months.

Consolidation among classic IDP vendors. Hyperscience, ABBYY, Klippa, Rossum, UiPath Document Understanding, Docsumo, Nanonets — the field has more vendors than the market structure supports. Expect at least two acquisitions in 2027, possibly more if a hyperscaler decides to enter the IDP market by buying rather than building.

The first “agentic-OCR-as-foundation-model” play. A model trained from scratch with multi-pass agentic extraction as the training objective, not retrofitted from a general VLM. This is harder than it sounds and will probably take research effort rather than vendor effort to produce. Likely candidates: AllenAI (continuing the olmOCR line), an academic group, or possibly Mistral or another open-weight foundation-model vendor that wants document-AI as a strategic capability.

Hyperscaler agentic OCR catching up. Google, AWS, and Azure all have agentic-OCR work in progress in 2026. The hyperscaler product cycles are slow, but the resources behind them are enormous. By end of 2027, expect at least one of the three to have closed the agentic-feature gap with the AI-native specialists, at least on common workloads. The platform-integration advantage will then become decisive for enterprise procurement.

Open-source closing the orchestration gap. There is no production-grade open-source equivalent of LlamaParse Agentic or LandingAI ADE in 2026. The Docling framework is the closest, but it is a glue layer rather than a complete orchestration product. Expect this to change — possibly through a new open-source project, possibly through one of the existing frameworks (LangChain, LlamaIndex itself in its OSS form) maturing into the orchestration role. The end-state, by 2028, is a credible open-source path for end-to-end agentic OCR.

15.5 Final word

The 2024–2026 period was about proving that agentic OCR works. That work is largely done. The vendors exist; the benchmarks exist; the production deployments exist; the case studies exist. A buyer in 2026 who concludes that agentic OCR is a real category of technology with real applications is making the correct conclusion.

The 2027–2028 period is about making agentic OCR **reliable enough to trust without checking**. That is a different problem. It is the observability problem (can you tell what the system is doing?). It is the calibration problem (when the system says it is confident, is it actually right?). It is the HITL-economics problem (how do you reduce reviewer load by 90% without losing the 10% of cases that absolutely need human eyes?). It is the long-form-reasoning problem (when extraction is perfect, why does downstream behavior still fail?). All of these will be open in 2027, and the field's progress on them will determine whether agentic OCR remains a buzzword or becomes infrastructure.

The thesis of this book, restated for the final time: **reading is the ceiling on agent quality in 2026, and reading is mostly solved**. The next ceiling is **trust**.

That is the book to write next.

Named takes

In 2027, nobody will talk about parsing accuracy. They will talk about agent observability. That's the wave that's about to break.

Parsing quality will commoditize. Compliance, observability, HITL, and vertical specialization will not. Build vendor relationships and engineering capacity around the things that will not commoditize.

The 2026 question was "does agentic OCR work?" The answer is yes. The 2027 question is "can you trust it?" The answer is partial.

Appendices

A. Glossary

Short, sharp definitions of the load-bearing terms used in this book. Where a term has a contested or vendor-shaped definition in the wider field, the entry uses the book's working definition with a pointer to the chapter that develops it.

Agentic OCR

A document-processing system that is **goal-driven, multi-pass, self-correcting, tool-using, and grounded** (Chapter 3). All five attributes must be present. Three of five is "VLM with retries," not agentic OCR.

Calibration

The property that a model's reported confidence scores correspond to actual accuracy probabilities. A confidence of 0.9 from a well-calibrated model means roughly 90% accuracy on similar inputs. VLMs in 2026 are typically well-calibrated on their training distribution and uncalibrated on out-of-distribution data, which is most production traffic.

CER (Character Error Rate)

A classic OCR metric measuring per-character extraction accuracy. Right for plain-text OCR, wrong for structured extraction — penalizes harmless formatting differences and under-penalizes structural errors. See Chapter 11.

Confidence gating

The use of a model's reported confidence score as a threshold for triggering retry, escalation, or output. Common in 2026 production systems; structurally weak as the only gate because confidence scores are not calibrated probabilities. See Chapter 5.

DABstep

A 2025 benchmark from Adyen and Hugging Face testing multi-step reasoning over real-world data and documents. Top reasoning models score ~14.55% on hard tasks. The reality-check benchmark for the agentic OCR field. See Chapter 10.

Detection (in self-correction)

One of the three subsystems of self-correction — knowing that an extraction is wrong. The hardest of the three and the most common source of production failures. Requires an external oracle for reliability. See Chapter 5.

DocTags

An XML-like document representation format used by IBM Granite-Docling and the Docling framework. Preserves layout structure better than Markdown; lighter-weight than HTML for downstream consumption.

Escalation (in self-correction)

One of the three subsystems of self-correction — routing a document to a human reviewer when detection or repair cannot close the loop. Engineering-intensive; requires HITL queue, reviewer UI, and feedback loop. See Chapter 5.

Field-level F1

The default accuracy metric for structured extraction. Per-field precision and recall, harmonic mean. Use per-field rather than aggregate to surface which specific fields the system handles well or poorly. See Chapter 11.

Five-attribute scorecard

The procurement test for whether a system qualifies as agentic OCR. Score the system yes/partial/no on goal-driven, multi-pass, self-correcting, tool-using, and grounded. Five of five is the bar. See Chapter 3.

Golden set

A hand-labeled set of 200–500 documents from your actual production traffic, used to evaluate vendors, run regression tests, and predict production behavior. The single most important investment in agentic OCR. See Chapter 11.

Grounding

The property that every extracted field can be traced back to a region of the source document (page number, bounding box, often text span). Required for compliance, auditability, efficient HITL, and structural self-correction. See Chapter 3.

Hallucination (in agentic OCR)

An extraction that is plausible — type-correct, range-valid, format-compliant — and wrong. Rarer than in single-pass VLMs but harder to detect when it happens. See Chapter 14.

Hierarchical retrieval

A RAG pattern that retrieves at the section level first, then at the chunk level within retrieved sections. Produces more coherent generation context than flat top-k retrieval on structured documents. See Chapter 12.

HITL (Human-in-the-loop)

A workflow design in which human reviewers handle documents the agentic system escalates. Healthy 2026 deployments run 1–10% escalation rates with engineered reviewer UIs, feedback loops, and operational discipline. Not eliminable; volume reducible. See Chapter 14.

IDP (Intelligent Document Processing)

The 2010s category of template-driven document extraction, with vendors including ABBYY, Hyper-science, UiPath, Klippa, and Rossum. Strong on HITL and compliance; weaker on agentic capability without retrofit. See Chapter 7.

OmniDocBench

A 2025/2026 benchmark from OpenDataLab for document parsing and evaluation across 10 document types, 5 layout types, and 5 languages. Saturated at the top in 2026; the v1.7 hard subset (April 2026) restores discrimination. See Chapter 10.

OOD (Out-of-distribution)

A document that is sufficiently different from anything in a model’s training distribution that the model’s behavior is unreliable. Most production traffic includes OOD documents; agentic systems need OOD detection as a pre-extraction step. See Chapter 5.

ParseBench

A 2026 benchmark from LlamaIndex and Kaggle for end-to-end agentic OCR vendor comparison. ~2,000 pages and 167,000 test rules across five axes. Top result: LlamaParse Agentic at 84.9%. See Chapter 10.

Pattern A / B / C

The three reference architectures from Chapter 4. Pattern A is single-pass VLM; Pattern B is agentic loop with reflection (the 2026 default); Pattern C is hybrid CV+VLM (the high-stakes choice).

RD-TableBench

A 2025 benchmark from Reducto for adversarial table extraction. 1,000 PhD-annotated tables, Needleman-Wunsch cell alignment. Top result: Reducto at ~0.90 similarity, ~20pp ahead of hyperscalers. See Chapter 10.

Reflection loop

The plan-extract-critique-re-prompt-validate cycle that constitutes the core agentic OCR architecture (Pattern B). See Chapter 4 and Chapter 5.

Repair (in self-correction)

One of the three subsystems of self-correction — fixing a detected error, typically via re-extraction with revised instructions. The easiest of the three subsystems. See Chapter 5.

Schema validity

A metric measuring whether a system’s output parses against the downstream schema. Cheap, deterministic, surprisingly diagnostic. Often catches errors that confidence scores miss. See Chapter 11.

Shift-left thesis

The argument that moving structural and semantic understanding earlier in the pipeline (into parsing rather than downstream cleanup) produces lower cost-per-correct-extraction even when per-page cost increases. The Databricks 16% across-the-board agent gain is the canonical 2026 evidence. See Chapter 6.

Silent hallucination

A high-confidence wrong answer that the reflection loop accepts and the system ships. The dominant 2026 production failure mode. Documented at length in the OpenReview “Tiny Silent Hallucinations in Agentic AI” paper. See Chapter 5 and Chapter 14.

Single-pass VLM

A vision-language model that extracts structured output from a document image in one model call, with no reflection loop or specialized tool routing. The default architecture of 2022–2024. By Chapter 3’s definition, not agentic OCR. See Chapter 4 (Pattern A).

Tool-using

One of the five attributes from Chapter 3 — routine, orchestrated invocation of specialized tools (layout detectors, table extractors, KB lookups) during the extraction workflow. Tool-capable (the model has function-calling) is not the same as tool-using (the model routinely calls them).

VLM (Vision-Language Model)

A neural model that operates on image + text inputs and outputs. Modern agentic OCR systems use VLMs as the core extraction component; the agentic loop, tool routing, and HITL queue surround the VLM. See Chapter 4.

Wrapper-washing

A 2026 anti-pattern (Chapter 13). A classic-OCR pipeline with an LLM cleanup step, rebranded as “agentic OCR.” Fails the five-attribute test, typically on tool-using.

Zero Data Retention (ZDR)

A vendor commitment to not persist customer documents or extracted outputs beyond the duration of a single API call. Available from LandingAI ADE Team/Enterprise plans, Mistral Document AI self-hosted, certain Reducto configurations. Verify in eval rather than trusting the marketing claim. See Chapter 11.

Bibliography

B. Quick-Reference Cheatsheet

The book in one page. Print this, tape it to a wall, refer back when the marketing language starts to drift.

B.1 The five attributes

A system is **agentic OCR** if and only if it is:

1. **Goal-driven** — operates against a structured target (schema, extraction goal), not against the document.
2. **Multi-pass** — iterates with at least one self-review pass, not single-shot.
3. **Self-correcting** — has working detection, repair, and escalation subsystems (all three).
4. **Tool-using** — routinely invokes specialized tools (layout detector, table extractor, schema validator, KB lookup), not just function-calling-capable.
5. **Grounded** — every output field traces to a source region (page, bounding box).

Three of five is “VLM with retries.” Four of five is borderline. Five of five is the bar.

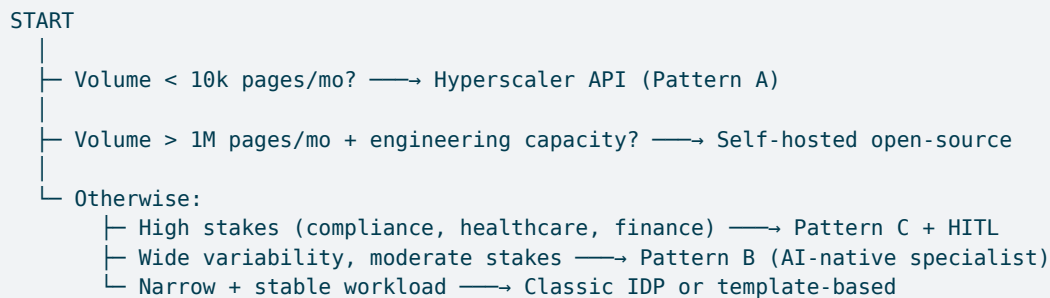
See Chapter 3 for the full definition; Chapter 13 for the anti-patterns that come from missing each attribute.

B.2 The three architecture patterns

Pattern	Description	Best for	Failure mode
A — Single-pass VLM	One model call; structured output	Low-stakes, low-variability workloads; commodity OCR	Tables, long documents, ungrounded output
B — Agentic loop with reflection	Plan → extract → critique → re-prompt → validate	The 2026 default; messy enterprise docs at moderate stakes	Silent hallucinations; retry thrash
C — Hybrid CV+VLM	Layout-first; route regions to specialized extractors	High-stakes, adversarial tables, regulated workloads	Engineering complexity; operational overhead

See Chapter 4 for the full treatment.

B.3 The decision tree in one diagram



See Chapter 9 for the full framework, including the “should I build this at all?” check in Section 9.5.

B.4 The cost rule of thumb

All-in cost-per-correct-extraction for true agentic OCR in 2026: \$0.05 – \$0.30 per page.

This includes parsing API spend, HITL amortized over typical 5% escalation, and eval/observability/vendor management.

Defensibility thresholds: - Downstream per-page value **below 1¢**: agentic OCR has already eaten your budget. Don't build. - Downstream per-page value **1–5¢**: marginal at best; wait for prices to drop. - Downstream per-page value **5–25¢**: economically defensible if the workload justifies the architecture. - Downstream per-page value **above 25¢**: comfortably above the threshold (healthcare prior-auth, KYC, contract review, clinical-note coding).

The objective function to minimize: **cost-per-correct-extraction-with-provenance**, not cost-per-page.

See Chapter 9 and Chapter 11.

B.5 Benchmarks at a glance

Benchmark	Year	Best use	Caveat
ParseBench	2026	Commercial vendor comparison (LlamaParse Agentic 84.9%)	Vendor-published
olmOCR-Bench	2025	Open-source model comparison (Chandra 83.1)	English-only
OmniDocBench v1.7	2026	Cross-validation (use hard subset)	Saturated at the top
RD-TableBench	2025	Table extraction (Reducto ~0.90)	Vendor-published
DABstep	2025	Reality check (~14.55% ceiling)	Small N
SCORE-Bench	2026	Generative-parser eval	New methodology
OfficeQA	2026	Shift-left thesis evidence (16% gain)	Databricks-maintained

The most important rule: benchmark rank ≠ production rank. Your golden set wins.

See Chapter 10.

B.6 Anti-pattern smell tests

A one-line check for each of the five dominant anti-patterns from Chapter 13:

1. **Wrapper test:** Does the system satisfy all five attributes? (*If not, it's wrapper-washing.*)
2. **Schema-flexibility test:** What does the system do with an unexpected field? (*If it drops it silently, schema rigidity.*)
3. **HITL test:** What happens to a confident-but-wrong output? (*If it ships, ignoring HITL.*)
4. **Calibration test:** Plot confidence vs. accuracy on your golden set — is the curve smooth? (*If not, you're over-relying on confidence.*)
5. **Decomposition test:** When a downstream answer is wrong, can you tell whether parsing or reasoning was at fault? (*If not, you're conflating them.*)

Print these. Use them before signing.

B.7 Six eval properties of a healthy 2026 deployment

From Chapter 11:

- **Golden set:** 200–500 docs from production, refreshed quarterly.
- **Metrics:** field-level F1, schema-validity, grounding accuracy (per-field, not aggregate).
- **Regression cadence:** weekly subset, monthly full, quarterly refresh.
- **HITL closed-loop:** corrections flow into golden set, prompt tuning, vendor selection.
- **Cost-per-correct-extraction:** tracked as the primary economic metric.
- **OOD slice:** held out and measured separately.

B.8 RAG + agentic OCR

The three concrete ways agentic OCR helps RAG (Chapter 12):

- **Structural metadata** → semantic chunking (chunks = sections, not fixed windows).
- **Table awareness** → tables as atomic chunks with cell-relationship metadata preserved.
- **Grounding** → real citations, audit infrastructure.

The diagnostic before upgrading the parser: hand-trace 50 wrong-answer queries. If parsing failures are under 30% of total, parsing is not your bottleneck. Don't spend \$500k upgrading the wrong layer.

B.9 The three named takes worth memorizing

1. *"If your agent can read perfectly but reason imperfectly, you have a model problem. If your agent reasons brilliantly on the wrong inputs, you have a document problem — and 2026 says the document problem is bigger."*
2. *"'Agentic' is the word that lets people charge 10× for a retry loop. The five-attribute test exists so you don't get fooled."*
3. *"Most teams pick a vendor before they pick a metric. This is backwards. Choose the metric that determines your downstream cost, then choose the vendor that minimizes that metric."*

B.10 What 2027 commoditizes (and doesn't)

Will commoditize: parsing quality, single-pass VLM extraction, basic table extraction, multilingual coverage.

Will not commoditize (yet): agentic orchestration platforms, compliance and grounding infrastructure, observability for document agents, vertical specialization.

Build your vendor relationships and your engineering investment around the categories that will not commoditize.

B.11 The thesis, restated

Reading is the ceiling on agent quality, and reading is mostly solved. The next ceiling is trust.

That is the entire book.

Bibliography

- [1] Databricks, “Why Frontier Agents Can’t Read Documents — and How We’re Fixing It.” [Online]. Available: <https://www.databricks.com/blog/why-frontier-agents-cant-read-documents-and-how-were-fixing-it>
- [2] Agentic AI Institute, “Agentic AI Enterprise Adoption 2026: 72% Production Proven.” [Online]. Available: <https://agenticaiinstitute.org/agentic-ai-enterprise-adoption-2026-governance-gap/>
- [3] Centers for Medicare and Medicaid Services, “CMS Interoperability and Prior Authorization Final Rule.” [Online]. Available: <https://www.cms.gov/>
- [4] LlamaIndex and Kaggle, “ParseBench: A Document Parsing Benchmark for AI Agents.” [Online]. Available: <https://www.llamaindex.ai/blog/parsebench>
- [5] AllenAI, “olmOCR-Bench: A Programmatic Benchmark for OCR Systems.” [Online]. Available: <https://github.com/allenai/olmocr>
- [6] Reducto AI, “Announcing RD-TableBench: An Open-Source Table Benchmark.” [Online]. Available: <https://reducto.ai/blog/rd-tablebench>
- [7] Adyen and Hugging Face, “DABstep: Data Agent Benchmark for Multi-step Reasoning,” in *arXiv preprint*, 2025. [Online]. Available: <https://arxiv.org/abs/2506.23719>
- [8] Google, “Tesseract OCR — open-sourced by Google in 2005.” [Online]. Available: <https://github.com/tesseract-ocr/tesseract>
- [9] ABBYY, “ABBYY Vantage.” [Online]. Available: <https://www.abbyy.com/vantage/>
- [10] Hyperscience, “Hyperscience Document Automation Platform.” [Online]. Available: <https://www.hyperscience.com/>
- [11] G. Kim, T. Hong, M. Yim, and others, “OCR-free Document Understanding Transformer,” in *European Conference on Computer Vision (ECCV)*, 2022. [Online]. Available: <https://arxiv.org/abs/2111.15664>
- [12] Y. Huang, T. Lv, L. Cui, and others, “LayoutLMv3: Pre-training for Document AI with Unified Text and Image Masking,” in *ACM Multimedia*, 2022. [Online]. Available: <https://arxiv.org/abs/2204.08387>
- [13] OpenAI, “GPT-4V(ision) System Card.” [Online]. Available: https://cdn.openai.com/papers/GPTV_System_Card.pdf
- [14] Various, “OCR-Agent: Agentic OCR with Capability and Memory Reflection.” [Online]. Available: <https://arxiv.org/abs/2602.21053>
- [15] Reducto AI, “Reducto’s Hybrid Architecture: Technical Deep Dive Into Agentic OCR and Multi-Pass Document Parsing.” [Online]. Available: <https://llms.reducto.ai/hybrid-architecture-agentic-ocr-deep-dive>
- [16] IBM, “IBM Granite-Docling: End-to-end Document Understanding.” [Online]. Available: <https://www.ibm.com/new/announcements/granite-docling-end-to-end-document-conversion>
- [17] PaddlePaddle, “PaddleOCR-VL and PP-StructureV3.” [Online]. Available: <https://github.com/PaddlePaddle/PaddleOCR>
- [18] Landing AI, “Agentic Document Extraction (ADE): Pricing and Features.” [Online]. Available: <https://docs.landing.ai/ade/ade-pricing>
- [19] Various, “Tiny Silent Hallucinations in Agentic AI: Hidden Failure Modes in Autonomous Systems.” [Online]. Available: <https://openreview.net/forum?id=1KxDazvI6L>
- [20] Mistral AI, “Introducing Mistral OCR 3.” [Online]. Available: <https://mistral.ai/news/mistral-ocr-3>
- [21] Snowflake, “Parsing documents with AI_PARSE_DOCUMENT.” [Online]. Available: <https://docs.snowflake.com/en/user-guide/snowflake-cortex/parse-document>
- [22] Microsoft, “Transform and Enrich Data with AI Functions in Microsoft Fabric.” [Online]. Available: <https://learn.microsoft.com/en-us/fabric/data-science/ai-functions/overview>
- [23] Hugging Face, “Supercharge your OCR Pipelines with Open Models.” [Online]. Available: <https://huggingface.co/blog/ocr-open-models>
- [24] OpenDataLab, “MonkeyOCR v1.5 Technical Report: Unlocking Robust Document Parsing for Complex Patterns.” [Online]. Available: <https://arxiv.org/abs/2511.10390>
- [25] AllenAI, “olmOCR-2: Open OCR with Reproducible Benchmarking.” [Online]. Available: <https://github.com/allenai/olmocr>

- [26] OpenDataLab, “OmniDocBench: A Comprehensive Benchmark for Document Parsing and Evaluation.” [Online]. Available: <https://github.com/pendatalab/OmniDocBench>
- [27] Unstructured.io, “Introducing SCORE-Bench: An Open Document Parsing Benchmark.” [Online]. Available: <https://unstructured.io/blog/introducing-score-bench-an-open-benchmark-for-document-parsing>
- [28] Seemore Data, “The Hidden Cost of Snowflake Cortex AI: A \$5K Single-Query Wake-Up Call.” [Online]. Available: <https://seemoredata.io/blog/snowflake-cortex-ai/>